BASE OF SINGULAR LAGRANGIAN FIBRATIONS MAY NOT BE QUOTIENTS OF PROJECTIVE SPACES

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Abstract. We disprove a conjecture of Liu, Liu, and Xu on the bases of Lagrangian fibrations: we construct a projective irreducible symplectic variety of dimension 12 with a Lagrangian fibration whose base is not isomorphic to the quotient of a projective space by any finite group. In contrast, the base of a Lagrangian fibration of a smooth projective irreducible symplectic variety is a projective space.

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1. Introduction

We work over the field of complex numbers $\mathbb{C}$.

Irreducible symplectic varieties are the symplectic factors in the singular Beauville–Bogomolov decomposition [GGK19, HP19, BGL22] of klt projective varieties with numerically trivial canonical class. We use the notion of irreducible symplectic variety in [LLX25, Definition 2.2(e)]. For a projective symplectic variety X with symplectic form σ, it asks that the algebra of global reflexive forms on X' is generated by the pullback of σ for any finite morphism $X' \to X$ from a normal variety X' that is étale in codimension one, and not only for X. Asking the condition only for X defines the cohomological irreducible symplectic varieties of [LLX25, Definition 2.2(f)], which form a larger class. The fibrations of an irreducible symplectic variety that we consider are Lagrangian fibrations, whose general fibers are Lagrangian subvarieties. By analogy with the case when the total space is smooth, Liu, Liu, and Xu proposed the following conjecture on the bases of Lagrangian fibrations.

Conjecture 1.1 ([LLX25, Conjecture 1.4])**.** *Let n be a positive integer, X a projective irreducible symplectic variety of dimension $2n$, and $f : X \to B$ a Lagrangian fibration. Then B is isomorphic to a quotient of $\mathbb{P}^n$.*

We refer the reader to Definitions 2.2 and 2.3 for the definitions. In Conjecture 1.1, a quotient of $\mathbb{P}^n$ means $\mathbb{P}^n/G$ for a finite subgroup $G \subset \mathrm{PGL}(n+1, \mathbb{C})$. This is consistent with the example of [LLX25, Theorem 1.2]: the base there is the weighted projective space $\mathbb{P}(1^{15}, 2^6, 3)$ (see Definition 2.8), where an exponent indicates how many times a weight is repeated, and $\mathbb{P}(1^{15}, 2^6, 3)$ is the quotient of $\mathbb{P}^{21}$ by a finite abelian group acting diagonally (cf. [Dol82, 1.2.2]).

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The bases $\mathbb{P}(1^{d+1},2)$ of the examples of [Car26, Theorem 5.2], where $1\leq d\leq 9$, are also of the form $\mathbb{P}^{d+1}/G$.

Here and below, until Theorem 1.2, $f : X \to B$ denotes a Lagrangian fibration as in Conjecture 1.1. We first consider the case when X is smooth. A smooth projective irreducible symplectic variety in the sense of Definition 2.2 is simply connected [GGK19, Corollary 13.3], hence a projective irreducible holomorphic symplectic manifold. When X is smooth, general fibers of f are abelian varieties [Mat99, Mat01], and any irreducible component of any fiber of f is Lagrangian [Mat00, Theorem 1].

When X is smooth, $B\cong\mathbb{P}^n$ by [OAI26, Theorem 1.1], and in particular Conjecture 1.1 holds. Before [OAI26], this was known in the following cases:

(1) When B is smooth [Hwa08, Theorem 1.2], [GL14, Theorem 1.1]. See also [BS26] for a Hodge-theoretic proof.
(2) When $n=2$: B is $\mathbb{P}^2$ or a normal Fano surface with one singular point, of type E_8, by [Ou19, Theorem 1.2], and $B\cong\mathbb{P}^2$ by [HX22, Theorem 0.1].
(3) When X is of $\mathrm{K3}^{[n]}$, generalized Kummer, OG6, or OG10 type [DHMV24, Theorem 1.3] and the references there (see [OAI26, Section 1] for the deduction of $B\cong\mathbb{P}^n$ from [DHMV24, Theorem 1.3]).

See also [MX26, Theorem 1.3], which shows that, for smooth X, B is smooth if it has only isolated quotient singularities, and see [SY22, Theorem 0.4(a)] for the intersection cohomology of B.

We now turn to the case when X is singular, where much less is known. [Mat15, Theorem 1.10] proved that $B\cong\mathbb{P}^n$ when moreover X is $\mathbb{Q}$-factorial terminal and B is smooth. Cohomological evidence for Conjecture 1.1 was established in [FSY22, Theorem 0.1(c)] (see [LLX24, p. 4]): if moreover the second Betti number of X is at least 5, then the intersection cohomology of B with complex coefficients is isomorphic to $H^*(\mathbb{P}^n,\mathbb{C})$.

We now explain where the argument for smooth X breaks down when X is singular. The proof of [OAI26, Theorem 1.1] has two stages [OAI26, Section 1.1]: it first rules out the singularities of B that are not quotient singularities, and then it eliminates the stabilizers of the quotient singularities of B. The second stage is [OAI26, Proposition 11.1]: under the assumptions of [OAI26, Theorem 1.1], if B has only quotient singularities and there exists a finite surjective morphism from an abelian variety onto B, then B is smooth.

One place where the smoothness of X enters is the following. For a quotient singularity $b\in B$, there exist an analytic open neighborhood $V\subset B$ of b and a finite Galois cover $U\to V$, étale in codimension one, with U smooth. Pulling it back to the open subset $f^{-1}(V)$ of X and normalizing gives a finite cover of $f^{-1}(V)$ that is étale in codimension one, as f is equidimensional [Mat00, Theorem 1], hence étale by purity of the branch locus, since X is smooth (see the proof of [OAI26, Lemma 11.2]). An argument of Kollár (see the footnote to [LLX24, Remark 1.3(b)]), showing that the total space in [LLX25, Theorem 1.2] is singular, also uses purity of the branch locus on a smooth total space.

When X is singular, a finite cover of X that is étale in codimension one need not be étale. Thus the argument by purity of the branch locus is no longer available, and the stabilizers can indeed survive: the bases of [LLX25, Theorem 1.2] and [Car26, Theorem 5.2] are singular. Therefore, for singular X one can only expect that $B\cong\mathbb{P}^n/G$ for a finite subgroup $G\subset\mathrm{PGL}(n+1,\mathbb{C})$, and, when B has only quotient singularities, the remaining question is whether the surviving stabilizers always come from an action of a finite group on $\mathbb{P}^n$.

Whether $B\cong\mathbb{P}^n/G$ cannot be decided at a single quotient singularity of B: for any finite subgroup $K\subset\mathrm{GL}(n,\mathbb{C})$ acting on the last n homogeneous coordinates of $\mathbb{P}^n$, the open subset $\{z_0\neq 0\}\cong\mathbb{C}^n$ is K-stable and K acts on it linearly, so the germ of $\mathbb{P}^n/K$ at the image of $[1:0:\cdots:0]$ is the quotient singularity $\mathbb{C}^n/K$. Hence every quotient singularity of dimension n occurs as a germ of a quotient of $\mathbb{P}^n$, and when B has only quotient singularities, an obstruction

to Conjecture 1.1 must be global. The goal of this paper is to show that the answer is negative: Conjecture 1.1 does not hold for $n = 6$.

Our main result is the following.

Theorem 1.2. *There exist an abelian variety Y of dimension 12, a symplectic form σ_Y on Y, a finite group $\widetilde{H}$ of automorphisms of Y preserving σ_Y, and a surjective morphism $g : Y/\widetilde{H} \to B$ onto a normal projective variety B, such that the following hold.*

(1) For any projective $\mathbb{Q}$-factorial terminalization $\mu : X \to Y/\widetilde{H}$ (see Definition 2.4), the pullback by μ of the symplectic form on $Y/\widetilde{H}$ induced by σ_Y extends to a symplectic form σ_X on X, (X, σ_X) is a projective irreducible symplectic variety of dimension 12, and $g \circ \mu : X \to B$ is a Lagrangian fibration of (X, σ_X).

(2) B is not isomorphic to $\mathbb{P}^6/G$ for any finite subgroup $G \subset \mathrm{PGL}(7, \mathbb{C})$.

The quotient $Y/\widetilde{H}$ is a symplectic variety (Construction-Proposition 5.1), hence has canonical singularities [BGMM25, Proposition 3.3]. Projective $\mathbb{Q}$-factorial terminalizations of $Y/\widetilde{H}$ exist by [BCHM10, Corollary 1.4.3] (see also [BGMM25, Definition 3.4]). Thus Theorem 1.2 shows that Conjecture 1.1 fails for $n = 6$.

We now describe Y, σ_Y, $\widetilde{H}$, and g. Let $E \subset \mathbb{P}^2$ be the Hesse cubic curve

$$x^3 + y^3 + z^3 - 3(1 + \sqrt{3})xyz = 0,$$

which is an elliptic curve with origin $o := [0 : 1 : -1]$. The curve E has an automorphism τ of order 4 with $\tau(o) = o$ (Construction-Proposition 4.1), so E has j-invariant 1728 [Har77, Chapter IV, Corollary 4.7]. Let Q be the root lattice of type E_6 with its positive definite bilinear form $(\cdot, \cdot)$, let W be the Weyl group of Q, and let $A := Q \otimes_{\mathbb{Z}} E$, an abelian variety of dimension 6 on which $w \in W$ acts by $w \otimes \mathrm{id}_E$. The quotient A/W is isomorphic to the weighted projective space $\mathbb{P}(1, 1, 1, 2, 2, 2, 3)$ (see [BS06, Section 2.3, Corollary] and Theorem 4.4). The study of such quotients goes back to [Loo76] (see [Rai23, Section 1]). Let $T \subset A$ be the group of W-invariant points of A, which is isomorphic to $E[3]$ (Construction-Proposition 4.3(2)), and let H be the group of automorphisms of A generated by W, the translations by the points of T, and the automorphism τ_* of A induced by τ. Then H is finite, $H/W \cong E[3] \rtimes \mathbb{Z}/4\mathbb{Z}$, and $|H/W| = 36$. Let $B := A/H$. Let $Y := A \times A$, and let $\widetilde{H}$ be the group of automorphisms of Y generated by

$$(x, y) \to (wx, wy), \quad (x, y) \to (x + t, y), \quad (x, y) \to (\tau_* x, \tau_*^{-1} y)$$

for $w \in W$ and $t \in T$. Let σ_Y be the translation-invariant symplectic form on Y pairing the first and the second factor of $Y = A \times A$ by $(\cdot, \cdot)$, and let $g : Y/\widetilde{H} \to B$ be the morphism induced by the first projection $Y \to A$ (Construction-Propositions 4.3 and 5.1). The variety X in Theorem 1.2 is a terminalization of the quotient of a smooth symplectic variety by a finite group preserving the symplectic form, as in [Fuj83, Men22, BGMM25]. The quotient $Y/\widetilde{H}$ itself is not irreducible symplectic in the sense of Definition 2.2: since $\widetilde{H}$ preserves σ_Y, the quotient morphism $Y \to Y/\widetilde{H}$ is étale outside a closed subset of codimension at least 2 (the fixed locus of any non-trivial element of $\widetilde{H}$ has codimension at least 2, see Lemma 5.4(1)(2)), while Y carries 12 linearly independent holomorphic 1-forms.

Remark 1.3. Let B be the variety in Construction-Proposition 4.3. Then there exists a finite surjective morphism $\mathbb{P}^6 \to B$.

Proof. The coordinate power map

$$p : \mathbb{P}^6 \to \mathbb{P}(1, 1, 1, 2, 2, 2, 3), \quad [z_0 : \cdots : z_6] \to [z_0 : z_1 : z_2 : z_3^2 : z_4^2 : z_5^2 : z_6^3]$$

is the quotient morphism of $\mathbb{P}^6$ by the group $\mu_2^3 \times \mu_3$, whose factors act by multiplication on z_3, z_4, z_5, and z_6 respectively [Dol82, 1.2.2]. Hence it is finite and surjective. The composition of p with the isomorphism $\mathbb{P}(1, 1, 1, 2, 2, 2, 3) \cong A/W$ of Theorem 4.4 and the quotient morphism $A/W \to A/H = B$ is a finite surjective morphism $\mathbb{P}^6 \to B$. □

Thus B is a finite image of $\mathbb{P}^6$, but not a quotient of $\mathbb{P}^6$ by a finite group (Theorem 1.2(2)).

Sketch of the proof. Theorem 1.2(1) follows from Propositions 5.7 and 5.8, and Theorem 1.2(2) is Theorem 4.8. Proposition 5.7 follows from Proposition 5.5, by which the terminalization kills the period lattice $\Lambda_Y \cong \mathbb{Z}^{24}$ of Y (Construction 5.3) and the diagonal copy of W in $\pi_1(X_{\mathrm{reg}})$, and from Lemma 5.6. Suppose that $B \cong \mathbb{P}^6/G$ for a finite subgroup $G \subset \mathrm{PGL}(7,\mathbb{C})$. We derive a contradiction in three steps.

(1) Since the elements of H fixing a prime divisor of A pointwise generate W (Proposition 4.5), the morphism $\mathbb{P}^6 \to \mathbb{P}^6/G \cong B$ factors as

$$\mathbb{P}^6 \to \mathbb{P}^6/J \cong A/W \to A/H = B$$

for a normal subgroup $J \subset G$, compatibly with an isomorphism $G/J \cong H/W$ (Proposition 3.3).

(2) Since $A/W \cong \mathbb{P}(1,1,1,2,2,2,3)$ is well-formed (Theorem 4.4), J is the image of a finite complex reflection group normalized by a finite lift of G (Lemma 3.5). Thus the 3-dimensional space M_2 of generators of degree 2 of $\mathbb{P}(1,1,1,2,2,2,3)$ (Definition-Lemma 2.10) splits into subspaces of dimension at most 2 permuted by H/W (Lemma 3.7).

(3) The group H/W acts primitively on M_2 (Proposition 4.7), which contradicts Step (2).

The contradiction uses the fact that an irreducible finite complex reflection group has at most two basic invariants of each degree [ST54, Table VII], while $\dim M_2 = 3$. The obstruction depends only on the action of H/W on M_2, so we never determine G or J. Theorem 4.4 and Lemma 5.6 follow directly from known results, and the main new work is in Steps (2) and (3) and in Proposition 5.5.

(a) Reflection groups and generators of degree 2. The presentation $B \cong \mathbb{P}^6/G$ is a priori unrelated to $A \to B$. Proposition 3.3 relates G to H: as $A/W \to B$ is étale in codimension one, purity of the branch locus on $\mathbb{P}^6$ and the simple connectedness of $\mathbb{P}^6$ lift $\mathbb{P}^6 \to B$ to A/W.

[ST54, Theorem 5.1 and Table VII] describes the rings of invariants of finite complex reflection groups, whereas J is a subgroup of $\mathrm{PGL}(7,\mathbb{C})$. Moreover, if J is the image of a finite complex reflection group whose basic invariants have degrees $d_0,\dots,d_6$, then $\mathbb{P}^6/J \cong \mathbb{P}(d_0,\dots,d_6)$, which need not be well-formed (Definition 2.8), so the d_i need not be the weights of $A/W \cong \mathbb{P}(1,1,1,2,2,2,3)$. Since $\mathbb{P}^6/J$ is isomorphic to the well-formed weighted projective space $\mathbb{P}(1,1,1,2,2,2,3)$, it has no non-trivial finite cover étale in codimension one (Lemma 3.4). Hence J is generated by non-identity elements fixing a hyperplane of $\mathbb{P}^6$ pointwise, which lift uniquely to reflections (Lemma 3.5).

Lemma 3.7 shows that, after passing to a well-formed weighted projective space isomorphic to $\mathbb{P}^6/J$, the generators of each degree $m \geq 2$ still split into blocks of dimension at most 2 permuted by G. The lemma says nothing about degree 1: the invariants of degree 1 coming from vectors fixed by the reflection group can span a space of any dimension that need not split into subspaces of dimension at most 2 permuted by G. Since $\dim M_3 = 1$, only M_2 can carry an obstruction, and as $\dim M_2 = 3$, the obstruction is primitivity.

(b) The action on M_2. [Rai23, Theorem 2.5 and Section 4.1] provides only an abstract isomorphism $A/W \cong \mathbb{P}(1,1,1,2,2,2,3)$, and the weights do not determine the action of H/W on M_2, since $\mathrm{Aut}(\mathbb{P}(1,1,1,2,2,2,3))$ contains $\mathrm{GL}(3,\mathbb{C})$ acting on the coordinates of degree 2. We use the root system E_6 since [Rai23, Section 5.2] then provides a model in which the action can be computed: A/W parametrizes the cubic surfaces containing E as an anti-canonical curve, given by equations $F + x_0q_2 + x_0^2q_1 + x_0^3q_0 = 0$ in $\mathbb{P}^3$, where $E = \{F = 0\} \subset \mathbb{P}^2 = \{x_0 = 0\}$ and $q_j \in \mathbb{C}[x,y,z]$ is homogeneous of degree j, modulo translations of x,y,z and weighted scaling, and the coordinates of degree 2 are the coefficients of q_1. By Definition-Proposition 4.6 and [Rai23, p. 45], $H/W \cong E[3] \rtimes \langle\tau\rangle$ acts on the model through its projective action on $\mathbb{P}^2 \supset E$, hence on M_2 through a finite lift to $\mathrm{GL}(3,\mathbb{C})$ twisted by a character (Proposition 4.7).

The translations by $E[3]$ alone would act by monomial matrices, permuting three lines in M_2, which is allowed in Step (2). This is why $j(E) = 1728$: the automorphism τ of order 4 acts on

$E[3] \cong \mathbb{F}_3^2$ with $\tau^2 = -1$, and since -1 is not a square in $\mathbb{F}_3$, τ has no eigenvector in $E[3]$, which forces primitivity (Lemma 4.2).

(c) A symplectic form preserved by the group. The group H preserves no non-zero holomorphic 2-form on A, since the derivative of τ_* at the origin is a scalar whose square is -1. Thus, as on the cotangent bundle of A, we let $\widetilde{H}$ act on the second factor of $Y = A \times A$ through the contragredients of the linear parts with respect to $(\cdot,\cdot)$, so that τ_*^{-1} replaces τ_* and the translations act trivially on the second factor. Then $\widetilde{H}$ preserves σ_Y, the first projection induces g, and σ_Y vanishes on the fibers $\{a\} \times A$ (Construction-Proposition 5.1 and Proposition 5.8).

(d) The period lattice. By purity of the branch locus, the finite covers of X étale in codimension one are controlled by $\pi_1(X_{\mathrm{reg}})$. [BGMM25, Proposition 8.1] computes $\pi_1(X_{\mathrm{reg}})$ when the symplectic manifold is simply connected, which Y is not, and π_1 of the open subset $U \subset Y/\widetilde{H}$ of Construction 5.3 is an extension of $\widetilde{H}$ by Λ_Y (Lemma 5.4). For a root α and a period ω of E, the translations by $(\alpha \otimes \omega, 0)$ and $(0, \alpha \otimes \omega)$ are products of two affine lifts of the diagonal reflection (s_α, s_α) to the universal cover of Y, each fixing an affine subspace of codimension 2. Near the image of a general point of such a subspace, $Y/\widetilde{H}$ is a smooth germ times a Du Val surface singularity, with simply connected inverse image in X (Lemma 2.5(2), based on [BGMM25, Definition 3.6 and Proposition 3.7]). Thus the lifts are trivial in $\pi_1(X_{\mathrm{reg}})$, and so are Λ_Y and the diagonal copy of W (Proposition 5.5), as for the singular Kummer surface [BGMM25, Remark 8.2].

Finally, by Proposition 5.5, for any finite morphism $X' \to X$ from a normal variety that is étale in codimension one, X' is birational to the quotient of Y by a subgroup of $\widetilde{H}$ containing the diagonal copy of W. By a theorem of Steinberg [Ste68, Section 14] (see also [Bou02, Chapter V, §2, Exercise 3(d)] and [Hu25, Theorem 1.2]), the holomorphic forms on Y invariant under the diagonal copy of W form the 7-dimensional subalgebra $\mathbb{C}[\sigma_Y]$ of the 2^{12}-dimensional algebra of holomorphic forms on Y (Lemma 5.6). This implies Proposition 5.7.

Structure of the paper. In Section 2 we recall preliminary material and fix notation. In particular, we introduce irreducible symplectic varieties, Lagrangian fibrations, and the spaces of generators of weighted projective spaces, and we prove local properties of terminalizations of symplectic varieties (Lemma 2.5). In Section 3 we compare the quotient of a normal quasi-projective variety by a finite group with the quotient by the subgroup generated by the elements fixing a prime divisor pointwise, and we study weighted projective spaces which are quotients of projective spaces by finite groups. In Section 4 we construct B and prove Theorem 4.8, which is Theorem 1.2(2). In Section 5 we construct $Y/\widetilde{H}$ and g, and we complete the proof of Theorem 1.2.

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2. Preliminaries

We adopt the standard notation and definitions for the MMP from [Sho92, KM98, BCHM10] and use them freely.

2.1. Symplectic varieties.

Definition 2.1. Let X be a normal variety, X_{reg} the smooth locus of X, and $i : X_{\mathrm{reg}} \to X$ the inclusion. For any non-negative integer k, the sheaf of *reflexive k-forms* on X is $\Omega_X^{[k]} := i_*\Omega^k_{X_{\mathrm{reg}}}$. A finite surjective morphism $q : X' \to X$ from a normal variety X' is called *quasi-étale* if there exists a closed subset $T \subset X$ with $\operatorname{codim}_X T \geq 2$ such that q is étale over $X \backslash T$. For such q

and any reflexive k-form η on X, the *reflexive pullback* $q^{[*]}\eta$ is the reflexive k-form on X' whose restriction to $q^{-1}(X_{\mathrm{reg}}\backslash T)$ is $q^*\eta$.

Quasi-étale morphisms are defined in [LLX25, Definition 2.1] for any morphism between normal varieties. We only use the finite surjective case, which is the case in [LLX25, Definition 2.2(e)].

Definition 2.2 ([LLX25, Definition 2.2])**.** Let X be a normal quasi-projective variety. A *symplectic form* on X is a closed holomorphic 2-form on X_{reg} that is non-degenerate at any point of X_{reg}. Let σ be a symplectic form on X. We say that (X,σ) is a *symplectic variety* if for some (equivalently, any) resolution $h : X'' \to X$, the 2-form $h^*\sigma$ on $h^{-1}(X_{\mathrm{reg}})$ extends to a holomorphic 2-form on X''. We say that a symplectic variety (X,σ) is an *irreducible symplectic variety* if X is projective and for any finite quasi-étale morphism $q : X' \to X$ from a normal variety X',

$$\bigoplus_{k\geq 0} H^0(X', \Omega^{[k]}_{X'}) = \mathbb{C}[q^{[*]}\sigma]$$

as graded algebras, where the product on the left-hand side is induced by the exterior product of forms on X'_{reg}.

For a normal quasi-projective variety, Definition 2.2 coincides with [BGMM25, Definition 3.2], in which a symplectic variety is a normal variety with rational singularities and a symplectic form on its smooth locus. Indeed, if (X,σ) is a symplectic variety, then X has Gorenstein canonical singularities by [LLX25, Lemma 2.3], hence rational singularities [KM98, Theorem 5.22]. Conversely, if X has rational singularities and σ is a symplectic form on X, then $h^*\sigma$ extends to a holomorphic 2-form on any resolution h of X by [BGMM25, Section 3.1], the paragraph after Definition 3.2. Thus we may cite results of [BGMM25] on symplectic varieties for symplectic varieties in the sense of Definition 2.2.

Definition 2.3. Let (X,σ) be a symplectic variety and $X_{\mathrm{sing}} := X\backslash X_{\mathrm{reg}}$. A closed subvariety $F \subset X$ is called *Lagrangian* if $\dim F = \frac{1}{2}\dim X$, $F \not\subset X_{\mathrm{sing}}$, and the pullback of σ to the smooth locus of $F \cap X_{\mathrm{reg}}$ is zero. A projective surjective morphism $f : X \to B$ with connected fibers onto a normal quasi-projective variety B is called a *Lagrangian fibration* if general fibers of f are Lagrangian subvarieties of X.

The definition of Lagrangian fibrations is [LLX25, Definition 2.6]. Lagrangian subvarieties are not defined in [LLX25], and we use the usual definition. The condition $F \not\subset X_{\mathrm{sing}}$ excludes the subvarieties of X_{sing}, for which the condition on σ is empty.

Definition 2.4 ([BGMM25, Definition 3.4])**.** Let Z be a variety with canonical singularities. A *terminalization* of Z is a proper birational morphism $\mu : X \to Z$ such that X has terminal singularities and $K_X = \mu^*K_Z$. A terminalization $\mu : X \to Z$ is called a *projective $\mathbb{Q}$-factorial terminalization* if μ is projective and X is $\mathbb{Q}$-factorial.

If Z is quasi-projective, then projective $\mathbb{Q}$-factorial terminalizations of Z exist by [BCHM10, Corollary 1.4.3] (cf. [BGMM25, Definition 3.4]).

We use Lemma 2.5 in Section 5: item (1) is used in Lemma 5.4 and Proposition 5.8, item (2) is used in Proposition 5.5 to kill the local fundamental groups at the points where Z is locally the product of a smooth germ and a Du Val surface singularity, and item (3) gives the symplectic form σ_X on a projective terminalization.

Lemma 2.5. *Let (Z,σ_Z) be a symplectic variety and $\mu : X \to Z$ a terminalization. Then the following hold.*

(1) μ is an isomorphism over Z_{reg}.
(2) Let $z \in Z$ be a point such that the analytic germ $(Z \ni z)$ is isomorphic to

$$(\mathbb{C}^{\dim Z-2} \times (\mathbb{C}^2/G_z) \ni (0,\bar{0}))$$

for a finite subgroup $G_z \subset \mathrm{SL}(2,\mathbb{C})$, where $\bar{0}$ is the image of the origin of $\mathbb{C}^2$. Then there exists an analytic open neighborhood N_z of z such that $\mu^{-1}(N_z) \to N_z$ is isomorphic to the restriction of the product of $\mathrm{id}_{\mathbb{C}^{\dim Z-2}}$ and the minimal resolution of $\mathbb{C}^2/G_z$, and $\mu^{-1}(N_z)$ is a simply connected complex manifold.

(3) *If μ is projective, then the 2-form $\mu^*\sigma_Z$ on $\mu^{-1}(Z_{\mathrm{reg}})$ extends to a symplectic form σ_X on X, and (X,σ_X) is a symplectic variety.*

Proof. Let Z° be the largest open subset of Z consisting of points z such that either Z is smooth at z, or the formal completion of Z at z is isomorphic to the product of the formal completion of a smooth variety at a point and the formal completion of a canonical surface singularity. By [BGMM25, Definition 3.6, the paragraph after it, and Proposition 3.7], $\mu^{-1}(Z^\circ) \to Z^\circ$ is isomorphic over Z° to the resolution μ° of Z° obtained by repeatedly blowing up the singular locus.

(1) A blow-up is an isomorphism over the complement of the center. Thus for each of the blow-ups whose composite is μ°, the center, which is the singular locus of the variety blown up, is contained in the inverse image of Z_{sing}, and μ° is an isomorphism over $Z_{\mathrm{reg}} \subset Z^\circ$. This implies (1).

(2) If $G_z = \{1\}$, then $z \in Z_{\mathrm{reg}}$, and (2) follows from (1), with N_z the image of a small open ball. Thus we may assume that $G_z \neq \{1\}$. Let $f : S \to \mathbb{C}^2/G_z$ be the minimal resolution.

The formal completion of Z at z is the completion of the analytic local ring of Z at z [Ser56, Proposition 3]. Thus the formal completion of Z at z is isomorphic to the product of the formal completion of $\mathbb{C}^{\dim Z-2}$ at 0 and the formal completion of $\mathbb{C}^2/G_z$ at $\bar{0}$. Since $G_z \subset \mathrm{SL}(2,\mathbb{C})$, $\bar{0} \in \mathbb{C}^2/G_z$ is a Du Val singularity (see [KM98, Remark 4.21(1)]), so it is canonical [KM98, Theorem 4.20]. Hence $z \in Z^\circ$ (the set of points satisfying the condition in the definition of Z° is open, being the union of the strata of dimension $\geq \dim Z - 2$ of [Kal06, Theorem 2.3], as in [BGMM25, Section 3.2], the paragraph after Definition 3.6). Since a finite group preserves a Hermitian inner product, G_z is conjugate in $\mathrm{GL}(2,\mathbb{C})$ to a subgroup of $\mathrm{SU}(2)$, and conjugation does not change the germ $(\mathbb{C}^2/G_z \ni \bar{0})$. Thus we may assume $G_z \subset \mathrm{SU}(2)$, so that each ball $\mathbb{B}_\delta$ below is G_z-stable and $U_\delta = \mathbb{B}_\delta/G_z$. For any $\epsilon > 0$, let $\mathbb{B}_\epsilon \subset \mathbb{C}^2$ be the open ball of radius ϵ centered at the origin. Let U_ϵ be the image of $\mathbb{B}_\epsilon$ in $\mathbb{C}^2/G_z$. Then there exist an open ball $D \subset \mathbb{C}^{\dim Z-2}$ centered at 0, a real number $\epsilon > 0$, and an analytic open neighborhood $N_z \subset Z^\circ$ of z, such that $N_z \cong D \times U_\epsilon$ and z corresponds to $(0,\bar{0})$.

Any element of $G_z\backslash\{1\}$ has eigenvalues λ and λ^{-1} for some $\lambda \neq 1$, so G_z acts freely on $\mathbb{C}^2\backslash\{0\}$. Thus the singular locus of $D \times U_\epsilon$ is $D \times \{\bar{0}\}$, and the reduced ideal of $D \times \{\bar{0}\}$ is the pullback of the reduced ideal of $\bar{0}$ along the flat projection $D \times U_\epsilon \to U_\epsilon$. Since blowing up commutes with flat base change, the blow-up of $D \times U_\epsilon$ along the singular locus is D times the blow-up of U_ϵ along $\bar{0}$. Repeating this, the restriction of μ° over $N_z \cong D \times U_\epsilon$ is $\mathrm{id}_D \times g|_{g^{-1}(U_\epsilon)}$, where $g : S' \to \mathbb{C}^2/G_z$ is the resolution obtained by repeatedly blowing up the singular locus. Here the process for $\mathbb{C}^2/G_z$ terminates since the process for N_z terminates and the singular locus of each blow-up is contained in the inverse image of $\bar{0}$. Let C be a g-exceptional curve. Since $K_X = \mu^*K_Z$ and $K_{D\times g^{-1}(U_\epsilon)}$ is the pullback of $K_{g^{-1}(U_\epsilon)}$,

$$(K_{S'} \cdot C) = (K_X \cdot (\{0\} \times C)) = (\mu^*K_Z \cdot (\{0\} \times C)) = 0.$$

Thus no g-exceptional curve is a (-1)-curve, so g is the minimal resolution (see [BHPV04, Chapter III, Theorem 6.2 and the definition before it]), that is, $g = f$. Let $V := f^{-1}(U_\epsilon)$. Then $\mu^{-1}(N_z) \to N_z$ is isomorphic to $\mathrm{id}_D \times f|_V : D \times V \to D \times U_\epsilon$, and $\mu^{-1}(N_z)$ is a complex manifold.

Since D is contractible, it remains to show that V is simply connected. By [KM98, Theorems 4.7 and 4.22], $f^{-1}(\bar{0})$ is a union of smooth rational curves, any two of which intersect transversally in at most one point, and the dual graph of $f^{-1}(\bar{0})$ is a tree. By van Kampen's theorem and induction on the number of curves, $f^{-1}(\bar{0})$ is connected and simply connected. For any $0 < \delta \leq \epsilon$, f induces an isomorphism

$$f^{-1}(U_\delta)\backslash f^{-1}(\bar{0}) \cong (\mathbb{B}_\delta\backslash\{0\})/G_z.$$

Since $\mathbb{B}_\delta\backslash\{0\}$ is simply connected and G_z acts freely on it, the inclusion

$$f^{-1}(U_\delta)\backslash f^{-1}(\bar{0}) \subset V\backslash f^{-1}(\bar{0})$$

induces an isomorphism of fundamental groups, both groups being G_z. Since $V\backslash f^{-1}(\bar{0})$ is connected and dense in V, V is connected. Since $f^{-1}(\bar{0})$ is a proper analytic subset of the complex manifold V, the homomorphism $\pi_1(V\backslash f^{-1}(\bar{0})) \to \pi_1(V)$ is surjective [Kol95, Proposition 2.10.1], so $\pi_1(V)$ is finite.

Let $p : V' \to V$ be the universal cover and k the degree of p. By the surjectivity above, $p^{-1}(f^{-1}(U_\delta)\backslash f^{-1}(\bar{0}))$ is connected for any $0 < \delta \leq \epsilon$. Since $f^{-1}(\bar{0})$ is simply connected, $p^{-1}(f^{-1}(\bar{0}))$ is the disjoint union of k compact subsets $C_1, \dots, C_k$, each mapped homeomorphically onto $f^{-1}(\bar{0})$ by p. Let $N_1, \dots, N_k$ be pairwise disjoint open neighborhoods of $C_1, \dots, C_k$ in V'. Since $f \circ p$ is proper over U_ϵ, the compact subsets $(f \circ p)^{-1}(\overline{U_\delta})$, $0 < \delta < \epsilon$, have intersection $p^{-1}(f^{-1}(\bar{0}))$, so there exists $0 < \delta < \epsilon$ such that $(f \circ p)^{-1}(U_\delta) \subset N_1 \cup \cdots \cup N_k$. The connected set $(f \circ p)^{-1}(U_\delta)\backslash p^{-1}(f^{-1}(\bar{0}))$ is contained in N_j for some j. On the other hand, each C_i is a nowhere dense analytic subset of the 2-dimensional complex manifold V', and $(f \circ p)^{-1}(U_\delta)$ is an open neighborhood of C_i, so C_i is contained in the closure of $(f \circ p)^{-1}(U_\delta)\backslash p^{-1}(f^{-1}(\bar{0}))$. Thus $N_i \cap N_j \neq \emptyset$ for any i, so $i = j$ for any i, and $k = 1$. Hence V is simply connected, and so is $\mu^{-1}(N_z) \cong D \times V$.

(3) This is [BGMM25, Proposition 3.17], and we show that the symplectic form on X extends $\mu^*\sigma_Z$. Assume that μ is projective, and write $\dim Z = 2n$. By (1), $\mu^{-1}(Z_{\mathrm{reg}})$ is a dense open subset of X_{reg}, and $\mu^*\sigma_Z$ is a symplectic form on $\mu^{-1}(Z_{\mathrm{reg}})$. In particular, an extension of $\mu^*\sigma_Z$ to X_{reg} is unique if it exists. Let $h : X'' \to X$ be a resolution which is an isomorphism over X_{reg}. Then $\mu \circ h$ is a resolution of Z, and by Definition 2.2, $(\mu \circ h)^*\sigma_Z$ extends to a holomorphic 2-form σ'' on X''. The restriction of σ'' to $h^{-1}(X_{\mathrm{reg}}) \cong X_{\mathrm{reg}}$ is a holomorphic 2-form σ_X on X_{reg} extending $\mu^*\sigma_Z$, and σ_X is closed since $d\sigma_X = 0$ on the dense open subset $\mu^{-1}(Z_{\mathrm{reg}})$.

Let K_Z and K_X be the divisors of σ_Z^n and σ_X^n. Since σ_Z^n has no zero on Z_{reg} and Z_{sing} has codimension at least 2 in Z, $K_Z = 0$. Since μ is a terminalization, $K_X = \mu^*K_Z = 0$ (this is an equality of divisors since $\sigma_X^n = \mu^*\sigma_Z^n$ as rational forms, as in [KM98, Definition 2.25]). Thus σ_X^n has no zero on X_{reg}, and σ_X is non-degenerate at any point of X_{reg}. Hence σ_X is a symplectic form on X. Moreover, X is normal since X has terminal singularities, and X is quasi-projective since μ is projective and Z is quasi-projective. Since $h^*\sigma_X = \sigma''|_{h^{-1}(X_{\mathrm{reg}})}$ extends to the holomorphic 2-form σ'' on X'', (X, σ_X) is a symplectic variety. □

2.2. **Reflection groups.**

Definition 2.6. Let V be a complex vector space with $0 < \dim V < +\infty$. An element $g \in \mathrm{GL}(V)$ of finite order is called a *reflection* if the subspace $\{v \in V \mid gv = v\}$ is a hyperplane of V. A *finite complex reflection group* on V is a finite subgroup of $\mathrm{GL}(V)$ generated by reflections.

Definition 2.7. Let V be a complex vector space with $0 < \dim V < +\infty$ and $G \subset \mathrm{GL}(V)$ a subgroup. A *system of imprimitivity* of G is a decomposition $V = V_1 \oplus \cdots \oplus V_k$ into non-zero subspaces with $k \geq 2$, such that for any $g \in G$ and any $1 \leq j \leq k$, there exists $1 \leq j' \leq k$ with $g(V_j) = V_{j'}$. We say that G is *primitive* if G acts irreducibly on V and G has no system of imprimitivity.

2.3. **Weighted projective spaces.**

Definition 2.8. Let d and $w_0, \dots, w_d$ be positive integers. The *weighted projective space* $\mathbb{P}(w_0, \dots, w_d)$ is $\operatorname{Proj} \mathbb{C}[x_0, \dots, x_d]$, where $\deg x_i = w_i$ for any i. We say that $\mathbb{P}(w_0, \dots, w_d)$ is *well-formed* if the greatest common divisor of any d of the integers $w_0, \dots, w_d$ is 1.

Definition-Proposition 2.9. Let $\mathbb{P} := \mathbb{P}(w_0, \dots, w_d)$ be a well-formed weighted projective space. The graded ring $R := \bigoplus_{m \geq 0} H^0(\mathbb{P}, \mathcal{O}_{\mathbb{P}}(m))$ is called the *Cox ring* of $\mathbb{P}$. Then the following hold.

(1) The class group of $\mathbb{P}$ is isomorphic to $\mathbb{Z}$ and generated by the class of $\mathcal{O}_{\mathbb{P}}(1)$, and R is isomorphic to $\mathbb{C}[x_0, \dots, x_d]$ with $\deg x_i = w_i$.
(2) Let $\mathbb{P}'$ be a well-formed weighted projective space with Cox ring R'. Then any isomorphism $\mathbb{P} \to \mathbb{P}'$ is induced by a graded isomorphism $R' \to R$, which is unique up to composition with an automorphism of R acting on R_m by multiplication by c^m for any $m \geq 0$, for some $c \in \mathbb{C}^*$.

Proof. (1) This is [PS17, Lemma A.2.1].

(2) Let $V \subset \mathbb{P}$ be the open subset of points at which the weights of the non-vanishing coordinates have greatest common divisor 1. Then $\mathcal{O}_{\mathbb{P}}(1)|_V$ is invertible and the multiplication maps induce isomorphisms

$$(\mathcal{O}_{\mathbb{P}}(1)|_V)^{\otimes m} \cong \mathcal{O}_{\mathbb{P}}(m)|_V$$

for any integer m [Dol82, 1.5.5], $\operatorname{codim}_{\mathbb{P}}(\mathbb{P} \backslash V) \geq 2$ since, by well-formedness, any prime divides at most $d-1$ of the weights, and $\mathcal{O}_{\mathbb{P}}(m)$ is reflexive (cf. [PS17, Section A.2]). Hence $\mathcal{O}_{\mathbb{P}}(m)$ is the reflexive hull of $\mathcal{O}_{\mathbb{P}}(1)^{\otimes m}$, and the multiplication of R is induced by the tensor product. Similarly, $\mathcal{O}_{\mathbb{P}'}(m)$ is the reflexive hull of $\mathcal{O}_{\mathbb{P}'}(1)^{\otimes m}$, and the multiplication of R' is induced by the tensor product. In particular, for any integer m, the class of $\mathcal{O}_{\mathbb{P}}(m)$ in $\operatorname{Cl}(\mathbb{P})$ is m times the class of $\mathcal{O}_{\mathbb{P}}(1)$, and similarly for $\mathbb{P}'$.

Let $\phi : \mathbb{P} \to \mathbb{P}'$ be an isomorphism. By (1), the pullback ϕ^* induces an isomorphism

$$\operatorname{Cl}(\mathbb{P}') \cong \mathbb{Z} \to \operatorname{Cl}(\mathbb{P}) \cong \mathbb{Z},$$

which maps the class of $\mathcal{O}_{\mathbb{P}'}(1)$ to plus or minus the class of $\mathcal{O}_{\mathbb{P}}(1)$. For a sufficiently divisible positive integer N, $\mathcal{O}_{\mathbb{P}}(N)$ and $\mathcal{O}_{\mathbb{P}'}(N)$ are ample line bundles (cf. [Gro61, Proposition 4.6.18]), ϕ^* preserves ampleness, and $\mathcal{O}_{\mathbb{P}}(-N)$ is not ample. Thus the sign is plus, and there exists an isomorphism

$$\psi_1 : \phi^* \mathcal{O}_{\mathbb{P}'}(1) \to \mathcal{O}_{\mathbb{P}}(1),$$

unique up to multiplication by an element of $H^0(\mathbb{P}, \mathcal{O}_{\mathbb{P}}^*) = \mathbb{C}^*$. Since $\mathcal{O}_{\mathbb{P}}(m)$ and $\mathcal{O}_{\mathbb{P}'}(m)$ are the reflexive hulls of $\mathcal{O}_{\mathbb{P}}(1)^{\otimes m}$ and $\mathcal{O}_{\mathbb{P}'}(1)^{\otimes m}$, the reflexive hulls of the tensor powers of ψ_1 are isomorphisms

$$\psi_m : \phi^* \mathcal{O}_{\mathbb{P}'}(m) \to \mathcal{O}_{\mathbb{P}}(m)$$

for any $m \geq 0$, compatible with the multiplications. Let $\alpha : R' \to R$ be the map defined by $\alpha(s) := \psi_m(\phi^* s)$ for any $s \in R'_m$. Since the isomorphisms ψ_m are compatible with the multiplications and ϕ^* is bijective on global sections, α is a graded isomorphism. Since each ψ_m is $\mathcal{O}_{\mathbb{P}}$-linear, for any non-negative integer m and any $s, t \in R'_m$ with $s \neq 0$,

$$\frac{\alpha(t)}{\alpha(s)} = \phi^* \left(\frac{t}{s} \right)$$

as rational functions on $\mathbb{P}$. Thus α induces ϕ.

Let $\alpha' : R' \to R$ be another graded isomorphism inducing ϕ, and $\beta := \alpha' \circ \alpha^{-1}$. Since Proj is contravariant, β is a graded automorphism of R inducing $\phi^{-1} \circ \phi = \mathrm{id}_{\mathbb{P}}$. Thus β is the identity on the degree zero part of any homogeneous localization of R, and so β induces an $\mathcal{O}_{\mathbb{P}}$-linear automorphism of $\mathcal{O}_{\mathbb{P}}(m)$ for any $m \geq 0$. Since $\mathcal{O}_{\mathbb{P}}(m)$ is a reflexive sheaf of rank one on the normal projective variety $\mathbb{P}$, any such automorphism is the multiplication by an element of $H^0(\mathbb{P}, \mathcal{O}_{\mathbb{P}}^*) = \mathbb{C}^*$. Hence for any $m \geq 0$ with $R_m \neq 0$, β acts on $R_m = H^0(\mathbb{P}, \mathcal{O}_{\mathbb{P}}(m))$ by multiplication by some $c_m \in \mathbb{C}^*$. By (1), fix homogeneous generators $x_0, \dots, x_d$ of R with $\deg x_i = w_i$. Then $\beta(x_i) = c_{w_i} x_i$ for any i. For any i and j, both $x_i^{w_j}$ and $x_j^{w_i}$ are non-zero elements of $R_{w_i w_j}$, so

$$c_{w_i}^{w_j} = c_{w_i w_j} = c_{w_j}^{w_i}.$$

Since $\mathbb{P}$ is well-formed, the greatest common divisor of $w_0, \dots, w_d$ is 1. Let $n_0, \dots, n_d$ be integers such that $\sum_i n_i w_i = 1$, and $c := \prod_i c_{w_i}^{n_i}$. Then for any j,

$$c^{w_j} = \prod_i c_{w_i}^{n_i w_j} = \prod_i c_{w_j}^{n_i w_i} = c_{w_j}.$$

Thus $\beta(x_i) = c^{w_i}x_i$ for any i, and β acts on R_m by multiplication by c^m for any $m \geq 0$. Since $\alpha' = \beta \circ \alpha$, this proves the uniqueness. □

Definition-Lemma 2.10. Let $\mathbb{P}$ and $\mathbb{P}'$ be well-formed weighted projective spaces with Cox rings R and R' (Definition-Proposition 2.9), and let $\phi : \mathbb{P} \to \mathbb{P}'$ be an isomorphism. Let $R_+ \subset R$ be the ideal generated by homogeneous elements of positive degree. For any positive integer m, the degree m part $M_m := (R_+/R_+^2)_m$ of R_+/R_+^2 is called the *space of generators of degree m* of $\mathbb{P}$. Let $R'_+ \subset R'$ be the ideal generated by homogeneous elements of positive degree, and let $M'_m := (R'_+/(R'_+)^2)_m$ for any positive integer m. Then for any positive integer m, the following hold.

(1) Any graded isomorphism $\alpha : R \to R'$ inducing ϕ^{-1} induces a linear isomorphism $\alpha_m : M_m \to M'_m$, and the class $[\phi]_m$ of α_m modulo $\mathbb{C}^*$ does not depend on the choice of α.
(2) For any isomorphism $\phi' : \mathbb{P}' \to \mathbb{P}''$ to a well-formed weighted projective space $\mathbb{P}''$, $[\phi' \circ \phi]_m = [\phi']_m \circ [\phi]_m$. In particular, $g \to [g]_m$ is a group homomorphism $\mathrm{Aut}(\mathbb{P}) \to \mathrm{PGL}(M_m)$, and
$$[\phi \circ g \circ \phi^{-1}]_m = [\phi]_m \circ [g]_m \circ [\phi]_m^{-1}$$
for any $g \in \mathrm{Aut}(\mathbb{P})$.

For a finite subgroup $G \subset \mathrm{Aut}(\mathbb{P})$ and a positive integer m with $M_m \neq 0$, we say that G acts *irreducibly* (resp. *primitively*) on M_m if the inverse image of $\{[g]_m \mid g \in G\}$ in $\mathrm{SL}(M_m)$ acts irreducibly on M_m (resp. is primitive). Moreover, for any finite subgroup $G \subset \mathrm{Aut}(\mathbb{P})$, any positive integer m with $M_m \neq 0$, and any subgroup $\widehat{G} \subset \mathrm{GL}(M_m)$ whose image in $\mathrm{PGL}(M_m)$ is $\{[g]_m \mid g \in G\}$, G acts irreducibly (resp. primitively) on M_m if and only if $\widehat{G}$ acts irreducibly on M_m (resp. is primitive).

Proof. (1) A graded isomorphism $\alpha : R \to R'$ maps R_+ onto R'_+ and R_+^2 onto $(R'_+)^2$, so α induces a linear isomorphism $\alpha_m : M_m \to M'_m$. By Definition-Proposition 2.9(2) applied to $\phi^{-1} : \mathbb{P}' \to \mathbb{P}$, a graded isomorphism $R \to R'$ inducing ϕ^{-1} exists, and it is unique up to composition with an automorphism γ of R' acting on R'_k by multiplication by c^k for any $k \geq 0$, for some $c \in \mathbb{C}^*$. Since γ preserves R'_+ and $(R'_+)^2$ and acts on M'_m by multiplication by c^m, $(\gamma \circ \alpha)_m = c^m \alpha_m$. This implies (1).

(2) Let $\alpha : R \to R'$ be a graded isomorphism inducing ϕ^{-1}, and $\alpha' : R' \to R''$ a graded isomorphism inducing ϕ'^{-1}, where R'' is the Cox ring of $\mathbb{P}''$. Since Proj is contravariant, $\alpha' \circ \alpha$ induces $\phi^{-1} \circ \phi'^{-1} = (\phi' \circ \phi)^{-1}$. Since $(\alpha' \circ \alpha)_m = \alpha'_m \circ \alpha_m$, we have $[\phi' \circ \phi]_m = [\phi']_m \circ [\phi]_m$. We use graded isomorphisms inducing ϕ^{-1} rather than ϕ in (1), so that the order of the composition is not reversed. Since id_R induces $\mathrm{id}_{\mathbb{P}}$, $[\mathrm{id}_{\mathbb{P}}]_m$ is the class of id_{M_m}. Thus $g \to [g]_m$ is a group homomorphism $\mathrm{Aut}(\mathbb{P}) \to \mathrm{PGL}(M_m)$, $[\phi^{-1}]_m = [\phi]_m^{-1}$, and
$$[\phi \circ g \circ \phi^{-1}]_m = [\phi]_m \circ [g]_m \circ [\phi]_m^{-1}$$
for any $g \in \mathrm{Aut}(\mathbb{P})$.

Finally, let $\widehat{G}$ be as in the last sentence of the statement. Any element of $\widehat{G}$ is a scalar multiple of an element of the inverse image of $\{[g]_m \mid g \in G\}$ in $\mathrm{SL}(M_m)$, and any element of the inverse image is a scalar multiple of an element of $\widehat{G}$. Since scalar multiplications preserve any subspace of M_m, a subspace of M_m is $\widehat{G}$-invariant if and only if it is invariant under the inverse image, and a decomposition of M_m is a system of imprimitivity of $\widehat{G}$ if and only if it is a system of imprimitivity of the inverse image. This proves the last sentence of the statement. □

Remark 2.11. The graded lifts of an automorphism of $\mathbb{P}$ are unique only up to $\mathbb{C}^*$ (Definition-Proposition 2.9(2)), and they need not preserve the span of a fixed set of generators of degree m. For example, for weighted coordinates $x_0, \ldots, x_6$ of degrees $1, 1, 1, 2, 2, 2, 3$ on $\mathbb{P}(1, 1, 1, 2, 2, 2, 3)$, the automorphism given by $x_3 \to x_3 + x_0^2$ and fixing the other coordinates does not preserve $\mathbb{C}x_3 \oplus \mathbb{C}x_4 \oplus \mathbb{C}x_5$, while any graded automorphism preserves R_+^2. Thus the projective representation on M_m is intrinsic, and $[\phi]_m$ transports it between isomorphic weighted projective spaces. We

use primitivity rather than irreducibility since Section 3 only yields decompositions of M_m into subspaces of dimension at most 2 permuted by the group, and some irreducible actions permute such subspaces.

3. Quotients by reflections

In this section, we prove two general results on finite quotients, which are used in Section 4 to exclude any isomorphism $B \cong \mathbb{P}^6/G$, where both the finite subgroup $G \subset \mathrm{PGL}(7,\mathbb{C})$ and the branch divisor of $\mathbb{P}^6 \to B$ are arbitrary.

3.1. **Quasi-étale quotients.** In this subsection, we show that an isomorphism between a finite quotient of X and a finite quotient of $\mathbb{P}^d$ descends to the quotient of X by the subgroup generated by the elements fixing a prime divisor pointwise (Proposition 3.3).

Set-up 3.1. Let X, H, and H_0 be as follows.

(1) X is a normal quasi-projective variety, and $H \subset \mathrm{Aut}(X)$ is a finite subgroup.
(2) $H_0 \subset H$ is the subgroup generated by the elements $h \in H$ such that the fixed locus of h contains a prime divisor on X.

The quotient morphism $X \to X/H$ is ramified in codimension one only along prime divisors fixed pointwise by non-identity elements of H, and such elements generate H_0. Thus the quotient morphism $X/H_0 \to X/H$ has no ramification in codimension one (Lemma 3.2).

Lemma 3.2. *Notation and conditions as in Set-up 3.1. Then H_0 is a normal subgroup of H, the induced action of H/H_0 on X/H_0 is faithful, and the quotient morphism $X/H_0 \to X/H$ is quasi-étale.*

Proof. For any finite subgroup $H' \subset \mathrm{Aut}(X)$, the quotient X/H' exists, the quotient morphism $X \to X/H'$ is finite and surjective, the fibers of $X \to X/H'$ are the H'-orbits, $\mathcal{O}_{X/H'}$ is the sheaf of H'-invariant sections of the direct image of $\mathcal{O}_X$, the pair consisting of X/H' and $X \to X/H'$ is unique up to isomorphism, and $X \to X/H'$ is étale over the image of any H'-invariant open subset of X on which H' acts freely [Mum74, §7, Theorem on p. 66 and Remark on p. 69]. In particular, X/H' is normal, since the ring of invariants of an integrally closed domain under a finite group is integrally closed. Moreover, X/H' is again quasi-projective [Har92, Lecture 10, p. 127], so for any finite subgroup $H'' \subset \mathrm{Aut}(X/H')$, the quotient $(X/H')/H''$ exists and has the properties above with X and H' replaced by X/H' and H'' [Mum74, §7, Theorem on p. 66 and Remark on p. 69]. Let $\pi_0 : X \to X/H_0$ be the quotient morphism. For any $h \in H$, let

$$\mathrm{Fix}(h) := \{x \in X \mid h(x) = x\}$$

be the fixed locus of h. Since X is separated, $\mathrm{Fix}(h)$ is closed in X.

Let $h \in H$ and $g \in H$, and let S be a prime divisor on X with $S \subset \mathrm{Fix}(h)$. Then $g(S)$ is a prime divisor on X and $g(S) \subset \mathrm{Fix}(ghg^{-1})$. Thus conjugation by g permutes the generators of H_0, and H_0 is a normal subgroup of H. For any $h \in H$ and $u \in H_0$, we have

$$\pi_0 \circ h \circ u = \pi_0 \circ (huh^{-1}) \circ h = \pi_0 \circ h.$$

Hence $\pi_0 \circ h$ is H_0-invariant, and there exists a unique automorphism k of X/H_0 such that $k \circ \pi_0 = \pi_0 \circ h$. The induced action of H/H_0 on X/H_0 is the action in which hH_0 acts by k.

We show that the action of H/H_0 on X/H_0 is faithful. Let $h \in H$ such that hH_0 acts trivially on X/H_0. Then $\pi_0(h(x)) = \pi_0(x)$ for any closed point $x \in X$, so $h(x) = u(x)$ for some $u \in H_0$. Thus

$$X = \bigcup_{u \in H_0} \mathrm{Fix}(u^{-1}h).$$

Since X is irreducible, $X = \mathrm{Fix}(u^{-1}h)$ for some $u \in H_0$. Thus $u^{-1}h = \mathrm{id}_X$, and $h = u \in H_0$.

We may therefore regard H/H_0 as a finite subgroup of $\mathrm{Aut}(X/H_0)$. By the first paragraph of this proof, applied with $H' = H_0$ and $H'' = H/H_0$, the quotient of X/H_0 by H/H_0 exists. The

composition of π_0 with the quotient morphism of X/H_0 by H/H_0 is a topological quotient of X by H, and the invariant sections satisfy

$$\left((\pi_{0*}\mathcal{O}_X)^{H_0}\right)^{H/H_0} = (\pi_{0*}\mathcal{O}_X)^H.$$

By the uniqueness of the quotient, the quotient of X/H_0 by H/H_0 is X/H, and the quotient morphism $q : X/H_0 \to X/H$ is the morphism in the lemma. In particular, q is finite and surjective, and X/H_0 is normal.

It remains to show that q is quasi-étale. For any $k \in H/H_0$, let $\mathrm{Fix}(k) \subset X/H_0$ be the fixed locus of k. Let $k \in H/H_0$ be a non-identity element, and let $h \in H$ be a representative of k. Suppose that $\mathrm{Fix}(k)$ contains a prime divisor F on X/H_0. Since π_0 is finite and surjective, there exists an irreducible component S of $\pi_0^{-1}(F)$ with $\pi_0(S) = F$, and S is a prime divisor on X. For any closed point $x \in S$, we have $\pi_0(h(x)) = k(\pi_0(x)) = \pi_0(x)$, so $h(x) = u(x)$ for some $u \in H_0$. Thus

$$S \subset \bigcup_{u \in H_0} \mathrm{Fix}(u^{-1}h).$$

Since S is irreducible, $S \subset \mathrm{Fix}(u^{-1}h)$ for some $u \in H_0$. By the definition of H_0, $u^{-1}h \in H_0$. Thus $h \in H_0$ and k is the identity, a contradiction. Therefore $\mathrm{codim}_{X/H_0} \mathrm{Fix}(k) \geq 2$ for any non-identity element $k \in H/H_0$. Let

$$T_0 := \bigcup_{k \in H/H_0, k \neq 1} \mathrm{Fix}(k).$$

Then T_0 is closed, $\mathrm{codim}_{X/H_0} T_0 \geq 2$, and T_0 is H/H_0-invariant, since $k'(\mathrm{Fix}(k)) = \mathrm{Fix}(k'kk'^{-1})$ for any $k, k' \in H/H_0$. The group H/H_0 acts freely on $(X/H_0)\backslash T_0$. Since the fibers of q are the H/H_0-orbits and T_0 is H/H_0-invariant, $q^{-1}((X/H)\backslash q(T_0)) = (X/H_0)\backslash T_0$, so q is étale over $(X/H)\backslash q(T_0)$. Since q is finite, $q(T_0)$ is closed and $\dim q(T_0) \leq \dim T_0 \leq \dim X - 2$. Hence q is quasi-étale. □

By purity of the branch locus and the simple connectedness of $\mathbb{P}^d$, the morphism $\mathbb{P}^d \to X/H$ induced by an isomorphism $\mathbb{P}^d/G \cong X/H$ lifts to the quasi-étale cover $X/H_0 \to X/H$ of Lemma 3.2, and Proposition 3.3 shows that the actions of G and H/H_0 then correspond to each other.

Proposition 3.3. *Notation and conditions as in Set-up 3.1. Let d be a positive integer, $G \subset \mathrm{PGL}(d+1, \mathbb{C})$ a finite subgroup, and $\phi : \mathbb{P}^d/G \to X/H$ an isomorphism. Let $q : X/H_0 \to X/H$ be the quotient morphism. Then there exist a normal subgroup $J \subset G$, an isomorphism $\psi : \mathbb{P}^d/J \to X/H_0$, and a group isomorphism $\rho : G/J \to H/H_0$, such that $\psi \circ \gamma = \rho(\gamma) \circ \psi$ for any $\gamma \in G/J$ and $q \circ \psi = \phi \circ p$, where $p : \mathbb{P}^d/J \to \mathbb{P}^d/G$ is the quotient morphism, and G/J and H/H_0 act on $\mathbb{P}^d/J$ and X/H_0 by the actions induced by G and H (see Lemma 3.2).*

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we construct a finite surjective morphism $s : \mathbb{P}^d \to X/H_0$ such that $q \circ s = \phi \circ \pi_G$, where $\pi_G : \mathbb{P}^d \to \mathbb{P}^d/G$ is the quotient morphism.

By Lemma 3.2 and [Mum74, §7, Theorem on p. 66 and Remark on p. 69], q is finite and surjective, X/H_0 is normal, and the fibers of q are the H/H_0-orbits, since the fibers of $X \to X/H$ are the H-orbits. Since $X/H \cong \mathbb{P}^d/G$ is projective of dimension d and q is finite and surjective, X/H_0 is a complete normal variety of dimension d.

Let X'' be an irreducible component of $\mathbb{P}^d \times_{X/H} (X/H_0)$ which dominates $\mathbb{P}^d$, with the reduced structure, where the fiber product is taken with respect to $\phi \circ \pi_G$ and q. Such a component exists, since the projection $\mathbb{P}^d \times_{X/H} (X/H_0) \to \mathbb{P}^d$ is a base change of q, hence finite and surjective. Let X' be the normalization of X'', and let $a : X' \to \mathbb{P}^d$ and $b : X' \to X/H_0$ be the induced

morphisms. Then a is finite and surjective, b is finite, since the projection to X/H_0 is a base change of the finite morphism $\phi \circ \pi_G$, and

$$q \circ b = \phi \circ \pi_G \circ a.$$

By Lemma 3.2, there exists a closed subset $T \subset X/H$ with $\operatorname{codim}_{X/H} T \geq 2$ such that q is étale over $(X/H)\backslash T$. Let $T' := (\phi \circ \pi_G)^{-1}(T)$. Since $\phi \circ \pi_G$ is finite, $\operatorname{codim}_{\mathbb{P}^d} T' \geq 2$. Over $\mathbb{P}^d \backslash T'$, the projection $\mathbb{P}^d \times_{X/H} (X/H_0) \to \mathbb{P}^d$ is étale, so the preimage of $\mathbb{P}^d \backslash T'$ in $\mathbb{P}^d \times_{X/H} (X/H_0)$ is smooth, and the irreducible components of the preimage are pairwise disjoint. Thus the preimage of $\mathbb{P}^d \backslash T'$ in X'' is an open and closed subset of the preimage in $\mathbb{P}^d \times_{X/H} (X/H_0)$, which is smooth and étale over $\mathbb{P}^d \backslash T'$. Hence the normalization $X' \to X''$ is an isomorphism over $\mathbb{P}^d \backslash T'$, and a is étale over $\mathbb{P}^d \backslash T'$. Since a is finite, $\operatorname{codim}_{X'} a^{-1}(T') \geq 2$. The set of points of X' where a is not étale is contained in $a^{-1}(T') \neq X'$. Since X' is normal and $\mathbb{P}^d$ is smooth, purity of the branch locus [Gro71, Exposé X, Théorème 3.1] implies that a is étale. Since X' is irreducible and $\mathbb{P}^d$ is simply connected [Gro71, Exposé XI, Proposition 1.1], the finite étale morphism a is an isomorphism. Let $s := b \circ a^{-1}$. Then s is finite and $q \circ s = \phi \circ \pi_G$. Since $s(\mathbb{P}^d)$ is closed of dimension d and X/H_0 is irreducible of dimension d, s is surjective.

Step 2. In this step we construct a surjective group homomorphism $\tilde{\rho} : G \to H/H_0$ such that $s \circ g = \tilde{\rho}(g) \circ s$ for any $g \in G$.

Let $g \in G$. Then

$$q \circ s \circ g = \phi \circ \pi_G \circ g = \phi \circ \pi_G = q \circ s.$$

Since the fibers of q are the H/H_0-orbits, for any closed point $y \in \mathbb{P}^d$ there exists $k \in H/H_0$ such that $s(gy) = k(s(y))$. Thus

$$\mathbb{P}^d = \bigcup_{k \in H/H_0} \{y \in \mathbb{P}^d \mid s(gy) = k(s(y))\}.$$

Since X/H_0 is separated, each set in the union is closed. Since $\mathbb{P}^d$ is irreducible, there exists $k \in H/H_0$ such that $s \circ g = k \circ s$. If $k' \in H/H_0$ also satisfies $s \circ g = k' \circ s$, then k and k' coincide on $s(\mathbb{P}^d) = X/H_0$, so $k = k'$, since H/H_0 acts faithfully on X/H_0 (Lemma 3.2). Let $\tilde{\rho}(g) := k$. For any $g, g' \in G$, we have

$$s \circ gg' = \tilde{\rho}(g) \circ s \circ g' = \tilde{\rho}(g)\tilde{\rho}(g') \circ s,$$

so $\tilde{\rho}(gg') = \tilde{\rho}(g)\tilde{\rho}(g')$ by uniqueness. Thus $\tilde{\rho}$ is a group homomorphism.

We show that $\tilde{\rho}$ is surjective. For any $k \in H/H_0$, let $\operatorname{Fix}(k) \subset X/H_0$ be the fixed locus of k. Since H/H_0 acts faithfully on X/H_0, $\operatorname{Fix}(k)$ is a proper closed subset of X/H_0 for any non-identity element k. Since X/H_0 is irreducible, there exists a closed point $z \in X/H_0$ which is not contained in $\operatorname{Fix}(k)$ for any non-identity element $k \in H/H_0$. Pick $y \in s^{-1}(z)$. Let $k \in H/H_0$, and pick $y' \in s^{-1}(k(z))$. Then

$$\phi(\pi_G(y')) = q(k(z)) = q(z) = \phi(\pi_G(y)).$$

Since ϕ is an isomorphism, $\pi_G(y') = \pi_G(y)$. Since the fibers of π_G are the G-orbits [Mum74, §7, Theorem on p. 66 and Remark on p. 69], $y' = gy$ for some $g \in G$. Then

$$\tilde{\rho}(g)(z) = \tilde{\rho}(g)(s(y)) = s(gy) = s(y') = k(z),$$

so $z \in \operatorname{Fix}(k^{-1}\tilde{\rho}(g))$. By the choice of z, $\tilde{\rho}(g) = k$.

Step 3. In this step we construct J, ψ, and ρ, and conclude the proof.

Let $J := \operatorname{Ker} \tilde{\rho}$. Then J is a normal subgroup of G, and $\tilde{\rho}$ induces a group isomorphism $\rho : G/J \to H/H_0$. Let $\pi_J : \mathbb{P}^d \to \mathbb{P}^d/J$ be the quotient morphism. For any $\gamma = gJ \in G/J$, the automorphism γ of $\mathbb{P}^d/J$ satisfies $\gamma \circ \pi_J = \pi_J \circ g$, and $\pi_G = p \circ \pi_J$ by the uniqueness of the quotient [Mum74, §7, Theorem on p. 66 and Remark on p. 69] (the quotient of $\mathbb{P}^d/J$ by G/J exists by [Mum74, §7, Theorem on p. 66], as $\mathbb{P}^d$ is covered by G-stable affine open subsets, whose images are G/J-stable affine open subsets covering $\mathbb{P}^d/J$). By **Step 2**, $s \circ j = s$ for any

$j \in J$. Since $\mathbb{P}^d/J$ is the topological quotient of $\mathbb{P}^d$ by J and $\mathcal{O}_{\mathbb{P}^d/J}$ is the sheaf of J-invariant sections of $\pi_{J*}\mathcal{O}_{\mathbb{P}^d}$ [Mum74, §7, Theorem on p. 66 and Remark on p. 69], there exists a morphism $\psi : \mathbb{P}^d/J \to X/H_0$ such that

$$s = \psi \circ \pi_J.$$

Since s is surjective, ψ is surjective.

We show that ψ is finite. Let $U \subset X/H_0$ be an affine open subset. Since s is finite, $s^{-1}(U)$ is a J-invariant affine open subset of $\mathbb{P}^d$, and $\psi^{-1}(U) = \pi_J(s^{-1}(U))$ is the quotient of $s^{-1}(U)$ by J. By [Mum74, §7, Theorem on p. 66 and Remark on p. 69], $\psi^{-1}(U)$ is affine with coordinate ring $\mathcal{O}(s^{-1}(U))^J$. Since $\mathcal{O}(s^{-1}(U))$ is a finite $\mathcal{O}(U)$-module and $\mathcal{O}(U)$ is Noetherian, the submodule $\mathcal{O}(s^{-1}(U))^J$ is a finite $\mathcal{O}(U)$-module. Thus ψ is finite.

We show that $\deg \psi = 1$. The varieties $\mathbb{P}^d$, $\mathbb{P}^d/J$, $\mathbb{P}^d/G$, and X/H_0 are complete of dimension d. The groups G and J act faithfully on $\mathbb{P}^d$, and H/H_0 acts faithfully on X/H_0 (Lemma 3.2). A finite group acting faithfully on a variety acts with trivial stabilizers on the complement of the union of the fixed loci of the non-identity elements, which is a dense open subset. Since the fibers of the quotient morphisms are the orbits and the quotient morphisms are separable [Mum74, §7, Theorem on p. 66 and Remark on p. 69], we have

$$\deg \pi_G = |G|, \quad \deg \pi_J = |J|, \quad \deg q = |H/H_0|$$

by [Mum74, §6, Application 3, pp. 62–63]. Since $q \circ \psi \circ \pi_J = q \circ s = \phi \circ \pi_G$, ϕ is an isomorphism, and $\tilde{\rho}$ is surjective with kernel J, we have

$$|H/H_0| \cdot \deg \psi \cdot |J| = |G| = |J| \cdot |H/H_0|.$$

Thus $\deg \psi = 1$, and ψ is a finite birational morphism. Let $U \subset X/H_0$ be an affine open subset. Since ψ is finite, surjective, and birational, $\psi^{-1}(U)$ is affine, and $\mathcal{O}(\psi^{-1}(U))$ is a domain that is a finite $\mathcal{O}(U)$-module and the inclusion $\mathcal{O}(U) \subset \mathcal{O}(\psi^{-1}(U))$ induces an isomorphism of fraction fields. Since X/H_0 is normal, $\mathcal{O}(U)$ is integrally closed. Thus $\mathcal{O}(\psi^{-1}(U)) = \mathcal{O}(U)$, and therefore ψ is an isomorphism.

Finally, for any $\gamma = gJ \in G/J$, we have

$$\psi \circ \gamma \circ \pi_J = \psi \circ \pi_J \circ g = s \circ g = \tilde{\rho}(g) \circ s = \rho(\gamma) \circ \psi \circ \pi_J$$

and

$$q \circ \psi \circ \pi_J = q \circ s = \phi \circ \pi_G = \phi \circ p \circ \pi_J.$$

Since π_J is surjective, $\psi \circ \gamma = \rho(\gamma) \circ \psi$ and $q \circ \psi = \phi \circ p$. □

3.2. **Finite quotients of projective spaces.** In this subsection, we study finite quotients $\mathbb{P}^d/J$ which are isomorphic to well-formed weighted projective spaces. In the situation of Set-up 3.1, if X/H_0 is a well-formed weighted projective space (as for A/W in Section 4) and $X/H \cong \mathbb{P}^d/G$ for a finite subgroup $G \subset \mathrm{PGL}(d+1, \mathbb{C})$, then Proposition 3.3 provides a normal subgroup J of G with $\mathbb{P}^d/J \cong X/H_0$. We show that J is the image of a finite complex reflection group (Lemma 3.5), and that the generators of degree at least two of the Cox ring of $\mathbb{P}^d/J$ split into subspaces of dimension at most two permuted by G (Lemma 3.7).

Lemma 3.4. *Let $\mathbb{P} := \mathbb{P}(w_0, \dots, w_d)$ be a well-formed weighted projective space, and let $q : X' \to \mathbb{P}$ be a finite quasi-étale morphism from a normal variety X'. Then q is an isomorphism.*

Proof. Write $\mathbb{P} = \operatorname{Proj} \mathbb{C}[x]$, where $\mathbb{C}[x] := \mathbb{C}[x_0, \dots, x_d]$ with $\deg x_i = w_i$, and let $\mathbb{C}^*$ act on $\mathbb{C}^{d+1} = \operatorname{Spec} \mathbb{C}[x]$ by

$$t \cdot (z_0, \dots, z_d) = (t^{w_0} z_0, \dots, t^{w_d} z_d).$$

Let $h : \mathbb{C}^{d+1} \setminus \{0\} \to \mathbb{P}$ be the morphism induced by the inclusions $\mathbb{C}[x]_{(f)} \subset \mathbb{C}[x]_f$, where f runs over the homogeneous elements of positive degree. Then h is $\mathbb{C}^*$-invariant, and $h^{-1}(D_+(f))$ is the open subset $D(f) \subset \mathbb{C}^{d+1}$ where $f \neq 0$. Moreover, $\mathbb{P}$ is normal, because an element c of the fraction field of $\mathbb{C}[x]_{(f)}$ which is integral over $\mathbb{C}[x]_{(f)}$ belongs to the normal domain $\mathbb{C}[x]_f$ and is

homogeneous of degree 0, since $cb = a$ for some non-zero homogeneous elements a, b of degree 0, so $c \in \mathbb{C}[x]_{(f)}$.

A point $z \in \mathbb{C}^{d+1}\backslash\{0\}$ has trivial stabilizer in $\mathbb{C}^*$ if and only if the integers w_i with $z_i \neq 0$ have greatest common divisor 1. Let $\Omega \subset \mathbb{C}^{d+1}\backslash\{0\}$ be the subset of points with trivial stabilizer. Then Ω is the union of the open subsets $D(f_I)$, where I runs over the subsets of $\{0, \dots, d\}$ with $\gcd(w_i \mid i \in I) = 1$ and $f_I := \prod_{i\in I} x_i$. Moreover, $\mathbb{C}^{d+1}\backslash\Omega$ is the union of the linear subspaces

$$\Lambda_p := \{z \in \mathbb{C}^{d+1} \mid z_i = 0 \text{ for any } i \text{ with } p \nmid w_i\},$$

where p runs over the primes, and $\Lambda_p = \{0\}$ unless p divides $w_0 \cdots w_d$. Since $\mathbb{P}$ is well-formed, for any prime p there exist at least two indices i with $p \nmid w_i$, so $\operatorname{codim} \Lambda_p \geq 2$. Thus Ω is the complement in $\mathbb{C}^{d+1}$ of finitely many linear subspaces of real codimension at least 4. A loop in Ω bounds a disk in $\mathbb{C}^{d+1}$, which can be perturbed relative to the boundary to be transverse to, hence disjoint from, the subspaces Λ_p. Thus Ω is simply connected. Similarly, a path in $\mathbb{C}^{d+1}$ joining two points z, z' of Ω can be perturbed, with z and z' fixed, to be disjoint from the subspaces Λ_p, so Ω is connected.

Let I be a subset of $\{0, \dots, d\}$ with $\gcd(w_i \mid i \in I) = 1$. By Bézout's identity, there exist integers a_i $(i \in I)$ with $\sum_{i\in I} a_i w_i = 1$. Then $\mu_I := \prod_{i\in I} x_i^{a_i} \in \mathbb{C}[x]_{f_I}$ is homogeneous of degree 1, so $\mu_I(t \cdot z) = t\mu_I(z)$. The homomorphism $\mathbb{C}[x]_{(f_I)} \to \mathbb{C}[x]_{f_I}/(\mu_I - 1)$ is an isomorphism, with inverse induced by $u \to u\mu_I^{-e}$ for homogeneous u of degree e. Thus h maps the closed subset $\{\mu_I = 1\}$ of $D(f_I)$ isomorphically onto $D_+(f_I)$, and

$$D(f_I) \to \mathbb{C}^* \times D_+(f_I), \quad z \to (\mu_I(z), h(z))$$

is a $\mathbb{C}^*$-equivariant isomorphism. Let $\mathbb{P}^\circ \subset \mathbb{P}$ be the union of the open subsets $D_+(f_I)$. Then $h^{-1}(\mathbb{P}^\circ) = \Omega$, and $h|_\Omega : \Omega \to \mathbb{P}^\circ$ is a principal $\mathbb{C}^*$-bundle, locally trivial in the Zariski topology. In particular, $\mathbb{P}^\circ$ is smooth and connected. By the long exact sequence of homotopy groups of $h|_\Omega$ [Hat02, Theorem 4.41 and Proposition 4.48], $\pi_1(\Omega) = 0$ and the connectedness of $\mathbb{C}^*$ imply that $\pi_1(\mathbb{P}^\circ) = 0$ in the analytic topology.

We show that $\operatorname{codim}_{\mathbb{P}}(\mathbb{P}\backslash\mathbb{P}^\circ) \geq 2$. Let $\xi \in \mathbb{P}$ be a point, and let I be the set of indices i with $\xi \in D_+(x_i)$. If $\gcd(w_i \mid i \in I) = 1$, then $\xi \in D_+(f_I) \subset \mathbb{P}^\circ$. Otherwise, a prime p divides w_i for any $i \in I$, so ξ belongs to the closed subset $\{x_i = 0 \text{ for any } i \text{ with } p \nmid w_i\}$ of $\mathbb{P}$. The closed subset $\{x_i = 0 \text{ for any } i \text{ with } p \nmid w_i\}$ is Proj of a polynomial ring in at most $d-1$ variables, so the dimension of $\{x_i = 0 \text{ for any } i \text{ with } p \nmid w_i\}$ is at most $d-2$. Since only finitely many primes p divide $w_0 \cdots w_d$, $\operatorname{codim}_{\mathbb{P}}(\mathbb{P}\backslash\mathbb{P}^\circ) \geq 2$.

Let $T \subset \mathbb{P}$ be a closed subset with $\operatorname{codim}_{\mathbb{P}} T \geq 2$ such that q is étale over $\mathbb{P}\backslash T$. Let $X'^\circ := q^{-1}(\mathbb{P}^\circ)$, a non-empty open subset of the normal variety X'. The morphism $q|_{X'^\circ} : X'^\circ \to \mathbb{P}^\circ$ is finite and surjective, $\mathbb{P}^\circ$ is smooth, and $q|_{X'^\circ}$ is étale outside $q^{-1}(T)$, whose codimension in X'° is at least 2 as q is finite. By purity of the branch locus [Gro71, Exposé X, Théorème 3.1], the set of points of X'° where q is not étale is empty or of pure codimension 1. Thus $q|_{X'^\circ}$ is finite étale. Since $\mathbb{P}^\circ$ is simply connected, the finite topological covering of $\mathbb{P}^\circ$ in the analytic topology induced by $q|_{X'^\circ}$ is trivial. By the uniqueness in the Riemann existence theorem [Gro71, Exposé XII, Théorème 5.1], X'° is isomorphic over $\mathbb{P}^\circ$ to a disjoint union of copies of $\mathbb{P}^\circ$. Since X'° is irreducible, $q|_{X'^\circ}$ is an isomorphism. Thus q is finite and birational. Since q is finite and birational, $q_*\mathcal{O}_{X'}$ is a sheaf of $\mathcal{O}_{\mathbb{P}}$-algebras contained in the sheaf of rational functions on $\mathbb{P}$ and integral over $\mathcal{O}_{\mathbb{P}}$. Since $\mathbb{P}$ is normal, we have $q_*\mathcal{O}_{X'} = \mathcal{O}_{\mathbb{P}}$. Therefore, q is an isomorphism. □

The theory of [ST54] considers linear groups generated by reflections, whereas J is a subgroup of $\mathrm{PGL}(d+1, \mathbb{C})$, which is not a priori generated by elements fixing a hyperplane of $\mathbb{P}^d$ pointwise, and the inverse images in $\mathrm{SL}(d+1, \mathbb{C})$ of such elements are not reflections, since the determinant of a reflection is not 1. Lemma 3.5 provides a finite complex reflection group $\widehat{J}$ lifting J and a finite group $\widehat{G}$ normalizing $\widehat{J}$. Lemma 3.4 is used only to show that J is generated by elements fixing a hyperplane of $\mathbb{P}^d$ pointwise.

Lemma 3.5. *Let $d \geq 2$ be an integer, and let $J \subset G \subset \mathrm{PGL}(d+1,\mathbb{C})$ be finite subgroups. Assume that J is normal in G and $\mathbb{P}^d/J$ is isomorphic to a well-formed weighted projective space. Then there exist a finite complex reflection group $\widehat{J} \subset \mathrm{GL}(d+1,\mathbb{C})$ whose image in $\mathrm{PGL}(d+1,\mathbb{C})$ is J, and a finite subgroup $\widehat{G} \subset \mathrm{GL}(d+1,\mathbb{C})$ whose image in $\mathrm{PGL}(d+1,\mathbb{C})$ is G, with $\widehat{J}$ normal in $\widehat{G}$.*

Proof. Let $J_{\mathrm{ref}} \subset J$ be the subgroup generated by the elements of J whose fixed locus contains a prime divisor on $\mathbb{P}^d$. By Lemma 3.2, applied to $X = \mathbb{P}^d$ and $H = J$, the induced action of J/J_{ref} on $\mathbb{P}^d/J_{\mathrm{ref}}$ is faithful, and the quotient morphism $\mathbb{P}^d/J_{\mathrm{ref}} \to \mathbb{P}^d/J$ is quasi-étale. Since $\mathbb{P}^d/J$ is isomorphic to a well-formed weighted projective space, the quotient morphism is an isomorphism by Lemma 3.4. Since the action of J/J_{ref} on $\mathbb{P}^d/J_{\mathrm{ref}}$ is faithful, the degree of the quotient morphism is $|J/J_{\mathrm{ref}}|$. Thus $J = J_{\mathrm{ref}}$.

Let $j \in J$ be an element of order n, and let $g \in \mathrm{GL}(d+1,\mathbb{C})$ be a lift of j. Since g^n is a scalar matrix, g is diagonalizable, and the fixed locus of j is the union of the projectivizations of the eigenspaces of g. Hence, if $j \neq 1$, the fixed locus of j contains a prime divisor if and only if j fixes a hyperplane of $\mathbb{P}^d$ pointwise, that is, g has an eigenspace Π with $\dim \Pi = d$. Assume that $j \neq 1$ and g has such an eigenspace Π, with eigenvalue λ. Let $\hat{j} := \lambda^{-1} g$. Then $\hat{j}|_\Pi = \mathrm{id}_\Pi$, and $\hat{j}^n$ is a scalar matrix which is the identity on Π, so $\hat{j}^n = 1$. Thus $\hat{j}$ is diagonalizable, with eigenvalue 1 on Π and one more eigenvalue ϵ_j with $\epsilon_j^n = 1$. Since $j \neq 1$, $\hat{j} \neq 1$, so $\epsilon_j \neq 1$. Hence $\hat{j}$ is a reflection with $(\det \hat{j})^{|J|} = \epsilon_j^{|J|} = 1$. Any other lift of j is $c\hat{j}$ for some $c \in \mathbb{C}^* \backslash \{1\}$. The eigenvalue c of $c\hat{j}$ has multiplicity d, so $\dim\{v \in \mathbb{C}^{d+1} \mid c\hat{j}v = v\} \leq 1 < d$, and $c\hat{j}$ is not a reflection. Thus $\hat{j}$ is the unique lift of j which is a reflection.

Let $\widehat{J} \subset \mathrm{GL}(d+1,\mathbb{C})$ be the subgroup generated by the reflections $\hat{j}$, where j runs over the elements $j \neq 1$ of J fixing a hyperplane of $\mathbb{P}^d$ pointwise. Since $J = J_{\mathrm{ref}}$, the image of $\widehat{J}$ in $\mathrm{PGL}(d+1,\mathbb{C})$ is J. Any element $k \in \widehat{J}$ satisfies $(\det k)^{|J|} = 1$, so any element of the kernel of $\widehat{J} \to J$ is a scalar matrix $c \cdot 1$ with $c^{(d+1)|J|} = 1$. Thus the kernel of $\widehat{J} \to J$ is finite, and $\widehat{J}$ is a finite complex reflection group.

Let $G_1 \subset \mathrm{SL}(d+1,\mathbb{C})$ be the inverse image of G. Then G_1 maps onto G, and G_1 is finite, as the kernel of $G_1 \to G$ consists of the scalar matrices $c \cdot 1$ with $c^{d+1} = 1$. Let $g \in G_1$ with image $[g] \in G$, and let $j \neq 1$ be an element of J fixing a hyperplane of $\mathbb{P}^d$ pointwise. Since J is normal in G, $j' := [g]j[g]^{-1}$ is an element of J with $j' \neq 1$ fixing a hyperplane of $\mathbb{P}^d$ pointwise, and $g\hat{j}g^{-1}$ is a reflection lifting j'. By the uniqueness above, $g\hat{j}g^{-1} = \hat{j}' \in \widehat{J}$. Thus G_1 normalizes $\widehat{J}$, and $\widehat{G} := G_1\widehat{J}$ is a finite subgroup of $\mathrm{GL}(d+1,\mathbb{C})$ containing $\widehat{J}$ as a normal subgroup. Since G_1 maps onto G and $\widehat{J}$ maps onto $J \subset G$, the image of $\widehat{G}$ in $\mathrm{PGL}(d+1,\mathbb{C})$ is G. □

Remark 3.6. The degrees of the ring of invariants $\mathbb{C}[x_0,\ldots,x_d]^{\widehat{J}}$ described in [ST54] need not be well-formed, for example if $d \geq 2$ and $\widehat{J}$ consists of the diagonal matrices $\mathrm{diag}(1,\pm 1,\ldots,\pm 1)$, in which case the ring of invariants is $\mathbb{C}[x_0,x_1^2,\ldots,x_d^2]$, with degrees $(1,2,\ldots,2)$. Passing to the second Veronese subring [Gro61, Proposition 2.4.7(i)] gives $\mathbb{P}^d/J \cong \mathbb{P}^d$, whose Cox ring has all weights equal to 1, so the ring of invariants is in general not the Cox ring of $\mathbb{P}^d/J$. Lemma 3.7 shows that, in degrees $m \geq 2$, the decomposition of the generators survives the passage to well-formed weights (**Step 3** of the proof of Lemma 3.7) and carries over to any isomorphic model, whereas in degree 1 the decomposition can fail, since the generators of degree 1 coming from the vectors fixed by $\widehat{J}$ span a $\widehat{G}$-stable space of arbitrarily large dimension that need not have such a decomposition. For example, if $d = 2$, $\widehat{J} = \{1\}$, and $\widehat{G} \subset \mathrm{SO}(3)$ is the rotation group of a regular icosahedron, then $\mathbb{P}^d/J = \mathbb{P}^2$, the space of generators of degree 1 is the space of linear forms, and $\widehat{G}$ acts on it primitively (Definition 2.7), as $\widehat{G} \cong A_5$ is simple and acts irreducibly on $\mathbb{C}^3$.

Lemma 3.7. *Let d be a positive integer, $\widehat{J} \subset \mathrm{GL}(d+1,\mathbb{C})$ a finite complex reflection group, and $\widehat{G} \subset \mathrm{GL}(d+1,\mathbb{C})$ a finite subgroup containing $\widehat{J}$ as a normal subgroup. Let J and G be the images of $\widehat{J}$ and $\widehat{G}$ in $\mathrm{PGL}(d+1,\mathbb{C})$. Then the following hold.*

(1) $\mathbb{P}^d/J$ is isomorphic to a well-formed weighted projective space.

(2) Let $\mathbb{P}$ be a well-formed weighted projective space, $\phi : \mathbb{P}^d/J \to \mathbb{P}$ an isomorphism, and $m \geq 2$ an integer with $M_m \neq 0$, where M_m is the space of generators of degree m of $\mathbb{P}$. Then there exists a decomposition $M_m = M_{m,1} \oplus \cdots \oplus M_{m,s}$ with $\dim M_{m,j} \in \{1,2\}$ for any j, such that for any $g \in G$ and any $1 \leq j \leq s$, we have $[\phi \circ \bar{g} \circ \phi^{-1}]_m(M_{m,j}) = M_{m,j'}$ for some $1 \leq j' \leq s$, where $\bar{g}$ is the automorphism of $\mathbb{P}^d/J$ induced by g.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we show that $\operatorname{Proj} S \cong \mathbb{P}^d/J$, where $S := \mathbb{C}[x_0,\dots,x_d]^{\widehat{J}}$.

Here $x_0,\dots,x_d$ are the coordinates of $\mathbb{C}^{d+1}$, and for $g \in \mathrm{GL}(d+1,\mathbb{C})$ and $f \in \mathbb{C}[x_0,\dots,x_d]$ we write $g^*f := f \circ g$. Thus S is the ring of polynomials f with $g^*f = f$ for any $g \in \widehat{J}$, and $g^*S = S$ for any $g \in \widehat{G}$, since $\widehat{J}$ is normal in $\widehat{G}$. After a linear change of coordinates we may assume that $\widehat{J}$ is unitary (cf. [ST54, p. 274]), so by [ST54, Theorem 5.1], S is a polynomial ring in $d+1$ homogeneous elements. Any x_i is a root of the monic polynomial $\prod_{g\in\widehat{J}}(T - g^*x_i) \in S[T]$, whose coefficients other than the leading one are homogeneous of positive degree. Evaluating at a common zero $v \in \mathbb{C}^{d+1}$ of the homogeneous elements of positive degree of S, we get $x_i(v)^{|\widehat{J}|} = 0$ for any i, so $v = 0$. Hence the open subsets $D_+(f) \subset \mathbb{P}^d$, where f runs over the homogeneous elements of positive degree of S, cover $\mathbb{P}^d$. Any such $D_+(f)$ is a J-stable affine open subset with coordinate ring $\mathbb{C}[x_0,\dots,x_d]_{(f)}$, on which $\widehat{J}$ acts through J, and $(\mathbb{C}[x_0,\dots,x_d]_{(f)})^{\widehat{J}} = S_{(f)}$. By [Gro71, Exposé V, Proposition 1.1], $D_+(f)/J \cong \operatorname{Spec} S_{(f)}$. Gluing the isomorphisms $D_+(f)/J \cong \operatorname{Spec} S_{(f)}$ over the J-stable open subsets $D_+(f)$, we obtain an isomorphism $\mathbb{P}^d/J \cong \operatorname{Proj} S$ induced by the inclusion $S \subset \mathbb{C}[x_0,\dots,x_d]$. For $g \in \widehat{G}$, we also write $\bar{g}$ for the automorphism of $\mathbb{P}^d/J$ induced by the image of g in G. Since the automorphism of $\mathbb{P}^d$ induced by g^* is the image of g, the automorphism of $\operatorname{Proj} S$ induced by g^* corresponds to $\bar{g}$.

Step 2. In this step we choose homogeneous generators of S in subspaces of dimension at most two permuted by $\widehat{G}$.

By complete reducibility [Ser77, Section 1.3, Theorem 1 and Section 1.4, Theorem 2],

$$\mathbb{C}^{d+1} = L_0 \oplus L_1 \oplus \cdots \oplus L_t,$$

where L_0 is the subspace of vectors fixed by $\widehat{J}$, and $\widehat{J}$ acts irreducibly and non-trivially on L_k for $1 \leq k \leq t$. For a reflection $r \in \widehat{J}$, $\operatorname{rank}(\mathrm{id}_{\mathbb{C}^{d+1}} - r) = 1$, so r acts non-trivially on exactly one L_k, and $k \geq 1$. For $1 \leq k \leq t$, let $\widehat{J}_k \subset \widehat{J}$ be the subgroup generated by the reflections acting non-trivially on L_k. Then $\widehat{J}_k$ acts trivially on L_l for $l \neq k$ and as a finite complex reflection group on L_k. Since the reflections generate $\widehat{J}$, we have $\widehat{J} = \widehat{J}_1 \times \cdots \times \widehat{J}_t$ (see also [ST54, p. 275]), and $\widehat{J}_k$ is the subgroup of elements of $\widehat{J}$ acting trivially on $\bigoplus_{l\neq k} L_l$. Thus the kernel of the action of $\widehat{J}$ on L_k is $\prod_{l\neq k}\widehat{J}_l$ for $1 \leq k \leq t$, and the kernel of the action on L_0 is $\widehat{J}$. Since $\widehat{J}_l \neq \{1\}$ for $1 \leq l \leq t$ (as $\widehat{J}$ acts non-trivially on L_l and is generated by reflections), the kernels of the actions of $\widehat{J}$ on $L_0, L_1, \dots, L_t$ are pairwise different. Hence $L_1,\dots,L_t$ are pairwise non-isomorphic non-trivial irreducible $\widehat{J}$-modules, and L_k is the isotypic component of $\mathbb{C}^{d+1}$ of type L_k. Since $\widehat{J}$ is normal in $\widehat{G}$, for $g \in \widehat{G}$ and $1 \leq k \leq t$, $g(L_k)$ is a non-trivial irreducible $\widehat{J}$-submodule of $\mathbb{C}^{d+1}$, so $g(L_k)$ is the isotypic component L_l containing $g(L_k)$ for some $1 \leq l \leq t$. Similarly, $g(L_0) = L_0$. Thus $\widehat{G}$ preserves L_0 and permutes $L_1,\dots,L_t$.

For $0 \leq k \leq t$, let $\mathbb{C}[L_k] \subset \mathbb{C}[x_0,\dots,x_d]$ be the subring generated by the linear forms vanishing on $\bigoplus_{l\neq k} L_l$, and for $1 \leq k \leq t$, let $S_k := \mathbb{C}[L_k]^{\widehat{J}_k}$. Then

$$\mathbb{C}[x_0,\dots,x_d] = \mathbb{C}[L_0] \otimes \mathbb{C}[L_1] \otimes \cdots \otimes \mathbb{C}[L_t],$$

and $\widehat{J}_k$ acts trivially on $\mathbb{C}[L_l]$ for $l \neq k$. Since the Reynolds operator of $\widehat{J}$ is the tensor product of the Reynolds operators of $\widehat{J}_1, \dots, \widehat{J}_t$, we obtain

$$S = \mathbb{C}[L_0] \otimes S_1 \otimes \cdots \otimes S_t.$$

By [ST54, Theorem 5.1], S_k is a polynomial ring in $\dim L_k$ homogeneous elements, and by [ST54, Table VII, p. 301], for any integer e, at most two of them have degree e. For $g \in \widehat{G}$ and $1 \leq k, l \leq t$ with $g(L_l) = L_k$, we have $g^*\mathbb{C}[L_k] = \mathbb{C}[L_l]$ and $g\widehat{J}_l g^{-1} = \widehat{J}_k$, so $g^* S_k = S_l$. Moreover, $g^*\mathbb{C}[L_0] = \mathbb{C}[L_0]$ for any $g \in \widehat{G}$.

For $1 \leq k \leq t$, let $S_{k,+} \subset S_k$ be the ideal of elements of positive degree, and let $\widehat{G}_k \subset \widehat{G}$ be the stabilizer of L_k. Fix a set of representatives of the $\widehat{G}$-orbits on $\{1, \dots, t\}$. For a representative k and $e \geq 1$, the maps g^* for $g \in \widehat{G}_k$ preserve $(S_k)_e$ and $(S_{k,+}^2)_e$, so by [Ser77, Section 1.3, Theorem 1], there exists a complement $C_{k,e}$ of $(S_{k,+}^2)_e$ in $(S_k)_e$ with $g^* C_{k,e} = C_{k,e}$ for any $g \in \widehat{G}_k$. For l in the $\widehat{G}$-orbit of a representative k and $e \geq 1$, let $C_{l,e} := g^* C_{k,e}$ for some $g \in \widehat{G}$ with $g(L_l) = L_k$. Since $(g'g)^* = g^* \circ g'^*$ for $g, g' \in \widehat{G}$, $C_{l,e}$ does not depend on g, and

$$g^* C_{k,e} = C_{l,e} \quad \text{for any } g \in \widehat{G} \text{ and } 1 \leq k, l \leq t \text{ with } g(L_l) = L_k.$$

Since S_k is a polynomial ring in homogeneous elements, $\dim C_{k,e}$ is the number of generators of degree e of S_k, so $\dim C_{k,e} \leq 2$ and $\sum_{e \geq 1} \dim C_{k,e} = \dim L_k$. By induction on the degree, $\bigoplus_{e \geq 1} C_{k,e}$ generates S_k. Let $C_0 \subset \mathbb{C}[L_0]$ be the space of linear forms. Then $g^* C_0 = C_0$ for any $g \in \widehat{G}$.

Choose a basis of C_0 and a basis of any non-zero $C_{k,e}$. By the decomposition of S above, the union $f_0, \dots, f_d$ of the bases of C_0 and of the non-zero $C_{k,e}$ consists of $\dim L_0 + \sum_{k=1}^t \dim L_k = d+1$ homogeneous elements generating S. Since $\mathbb{C}[x_0, \dots, x_d]$ is integral over S, $\dim S = d+1$, so $f_0, \dots, f_d$ are algebraically independent and $S = \mathbb{C}[f_0, \dots, f_d]$. Thus the generators $f_0, \dots, f_d$ form bases of subspaces of S of the following two kinds: the subspace C_0 of generators of degree 1, stable under g^* for any $g \in \widehat{G}$, and the non-zero subspaces $C_{k,e}$, of dimension 1 or 2, each consisting of homogeneous elements of one degree and permuted by g^* for any $g \in \widehat{G}$.

Step 3. In this step we replace S by a well-formed weighted polynomial ring $S' \subset S$ with $\operatorname{Proj} S' \cong \operatorname{Proj} S$, and we prove (1).

For a graded ring S'' and a positive integer n, let $S''^{(n)} := \bigoplus_{e \geq 0} S''_{ne}$, graded by $(S''^{(n)})_e := S''_{ne}$. By [Gro61, Proposition 2.4.7(i)], the inclusion $S''^{(n)} \subset S''$ induces an isomorphism $\operatorname{Proj} S'' \cong \operatorname{Proj} S''^{(n)}$. Let c be the greatest common divisor of the degrees of $f_0, \dots, f_d$. Then $S = S^{(c)}$, so $\operatorname{Proj} S$ does not change if we divide the degrees by c.

Next, assume that the weights $c_0, \dots, c_d$ of $f_0, \dots, f_d$ have greatest common divisor 1 and are not well-formed. Then there exist a prime p and an index i with $p \mid c_l$ for any $l \neq i$ and $p \nmid c_i$. A monomial in $f_0, \dots, f_d$ has weight divisible by p if and only if the exponent of f_i is divisible by p, so $S^{(p)} = \mathbb{C}[f_i^p, f_l \ (l \neq i)]$, and we divide the weights of $S^{(p)}$ by p: f_i^p has weight c_i, and f_l has weight c_l/p for $l \neq i$. The new weights have greatest common divisor 1, since a prime dividing all of them divides $c_0, \dots, c_d$, and the sum of the new weights is smaller than $c_0 + \cdots + c_d$. We repeat the passage from a weighted polynomial ring S'' whose weights have greatest common divisor 1 and are not well-formed to $S''^{(p)}$ with weights divided by p until the weights are well-formed.

During the procedure, g^* acts on the current ring by graded automorphisms for any $g \in \widehat{G}$, and the current generators form bases of subspaces of both kinds of **Step 2**: a subspace of generators of weight 1 stable under any g^*, coming from C_0, and subspaces of dimension 1 or 2, each consisting of generators of one weight and permuted by the maps g^*, coming from the $C_{k,e}$. We check by induction on the number of operations that the current generators form bases of subspaces of both kinds. The property holds after the division by c, since $c = 1$ if $C_0 \neq 0$. At an operation with a prime p and an index i such that $p \mid c_l$ for any $l \neq i$ and $p \nmid c_i$, where $c_0, \dots, c_d$ are the current weights, f_i is the only generator of weight c_i, so the subspace of the

current decomposition containing f_i is $\mathbb{C}f_i$, which is stable under any g^*. Replacing $\mathbb{C}f_i$ with $\mathbb{C}f_i^p$ keeps both kinds, and the other subspaces do not change. Moreover, if $c_l = 1$, then $l = i$ as $p \nmid 1$, and f_i^p has weight $c_i = 1$. Thus the generators coming from C_0 have weight 1 throughout.

At the end, we obtain a graded ring $S' \subset S$ which is a polynomial ring in homogeneous elements $f'_0, \dots, f'_d$ with well-formed weights $a_0, \dots, a_d$, and an isomorphism $\psi : \mathbb{P}^d/J \to \mathbb{P}' := \operatorname{Proj} S'$ induced by the inclusion $S' \subset S$ and **Step 1**. Since $\mathbb{P}' \cong \mathbb{P}(a_0, \dots, a_d)$, this proves (1). For $g \in \widehat{G}$, g^* is a graded automorphism of S', and by **Step 1**, the automorphism of $\mathbb{P}'$ induced by g^* is $\psi \circ \bar{g} \circ \psi^{-1}$. Moreover, for any $m \geq 2$, the generators f'_i of weight m form bases of subspaces $C'_{m,1}, \dots, C'_{m,s}$ of S'_m, with $\dim C'_{m,j} \in \{1, 2\}$ for any j, which are permuted by g^* for any $g \in \widehat{G}$.

Step 4. In this step we prove (2).

Let R' be the Cox ring of $\mathbb{P}'$, and let M'_m be the space of generators of degree m of $\mathbb{P}'$. The natural map $S'_m \to H^0(\mathbb{P}', \mathcal{O}_{\mathbb{P}'}(m))$, sending a homogeneous element to the section it defines, is an isomorphism for any m. It is the canonical isomorphism of [PS17, Lemma A.2.1] in Definition-Proposition 2.9(1), and we use it to identify S' with R'. For $g \in \widehat{G}$, the graded automorphism $(g^{-1})^*$ of R' induces $(\psi \circ \bar{g} \circ \psi^{-1})^{-1}$. Hence, by Definition-Lemma 2.10(1), $[\psi \circ \bar{g} \circ \psi^{-1}]_m$ is the class of the linear map induced by $(g^{-1})^*$ on M'_m. Let $m \geq 2$. Since S' is a polynomial ring in $f'_0, \dots, f'_d$, the images in M'_m of the generators f'_i of weight m form a basis of M'_m. Thus

$$M'_m = M'_{m,1} \oplus \cdots \oplus M'_{m,s},$$

where $M'_{m,j}$ is the image of $C'_{m,j}$ in M'_m, and for any $g \in \widehat{G}$ and any $1 \leq j \leq s$,

$$[\psi \circ \bar{g} \circ \psi^{-1}]_m(M'_{m,j}) = M'_{m,j'}$$

for some $1 \leq j' \leq s$, as $(g^{-1})^*$ permutes $C'_{m,1}, \dots, C'_{m,s}$.

Now let $\mathbb{P}$ be a well-formed weighted projective space, $\phi : \mathbb{P}^d/J \to \mathbb{P}$ an isomorphism, and $m \geq 2$ an integer with $M_m \neq 0$. Since $[\phi \circ \psi^{-1}]_m$ is the class modulo $\mathbb{C}^*$ of a linear isomorphism $M'_m \to M_m$, the subspaces

$$M_{m,j} := [\phi \circ \psi^{-1}]_m(M'_{m,j}), \quad 1 \leq j \leq s,$$

are well defined, $M_m = M_{m,1} \oplus \cdots \oplus M_{m,s}$, and $\dim M_{m,j} \in \{1, 2\}$ for any j. Let $g \in G$. Since the image of $\widehat{G}$ in $\mathrm{PGL}(d+1, \mathbb{C})$ is G, g is the image of an element of $\widehat{G}$, which we also denote by g. Since

$$\phi \circ \bar{g} \circ \phi^{-1} = (\phi \circ \psi^{-1}) \circ (\psi \circ \bar{g} \circ \psi^{-1}) \circ (\phi \circ \psi^{-1})^{-1},$$

Definition-Lemma 2.10(2) implies that

$$[\phi \circ \bar{g} \circ \phi^{-1}]_m = [\phi \circ \psi^{-1}]_m \circ [\psi \circ \bar{g} \circ \psi^{-1}]_m \circ [\phi \circ \psi^{-1}]_m^{-1}.$$

Thus

$$[\phi \circ \bar{g} \circ \phi^{-1}]_m(M_{m,j}) = [\phi \circ \psi^{-1}]_m(M'_{m,j'}) = M_{m,j'}$$

for some $1 \leq j' \leq s$, which proves (2). □

4. A quotient of an abelian sixfold

In this section, we construct the base $B = A/H$ of the fibration in the main theorem, and we prove Theorem 4.8 at the end of Subsection 4.3. The variety B depends only on the pair (A, H), and the symplectic variety over B is constructed in Section 5.

4.1. **An elliptic curve and a group of order 36.** In this subsection, we construct an elliptic curve $E \subset \mathbb{P}^2$ and a group Γ of order 36 of automorphisms of E, and we show that the linear groups lifting Γ act primitively on $\mathbb{C}^3$ (Lemma 4.2). As explained in the sketch of the proof in the introduction, the translations by $E[3]$ alone act through monomial matrices, so we also need an automorphism of E of order 4 fixing o, which exists only when the j-invariant is 1728 [Har77, Chapter IV, Corollary 4.7].

We use the Hesse form since there the translations by $E[3]$ are given by the monomial matrices g_1, g_2 below, and for the parameter $1+\sqrt{3}$, the matrix g_3 below also preserves E, with $g_3|_E = \tau$ (Construction-Proposition 4.1(2)).

Construction-Proposition 4.1. Let $\zeta := \exp(2\pi i/3)$. Let $E \subset \mathbb{P}^2$ be the plane cubic curve

$$x^3 + y^3 + z^3 - 3(1+\sqrt{3})xyz = 0,$$

let $o := [0:1:-1]$, and let $g_1, g_2, g_3 \in \mathrm{PGL}(3,\mathbb{C})$ be the classes of the matrices

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & \zeta & 0 \\ 0 & 0 & \zeta^2 \end{pmatrix}, \quad \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \quad \begin{pmatrix} 1 & 1 & 1 \\ 1 & \zeta & \zeta^2 \\ 1 & \zeta^2 & \zeta \end{pmatrix}$$

respectively. Then the following hold.

(1) E is smooth, $o \in E$, and $\mathcal{O}_E(1) \cong \mathcal{O}_E(3o)$. In particular, E is an elliptic curve with origin o.
(2) The automorphisms g_1, g_2, and g_3 of $\mathbb{P}^2$ preserve E. The restrictions of g_1 and g_2 to E generate the group of translations of E by the 3-torsion points $E[3]$, and $\tau := g_3|_E$ is an automorphism of E of order 4 such that $\tau(o) = o$ and $\tau^2 = -\mathrm{id}_E$. In particular, τ is a group automorphism of E.
(3) Let $\Gamma \subset \mathrm{Aut}(E)$ be the subgroup generated by τ and the translations by $E[3]$. Then $\Gamma \cong E[3] \rtimes \langle\tau\rangle$, $|\Gamma| = 36$, restriction to E induces an isomorphism from the subgroup $\langle g_1, g_2, g_3\rangle$ of $\mathrm{PGL}(3,\mathbb{C})$ onto Γ, and the only subgroups of $E[3]$ invariant under τ are 0 and $E[3]$.

Proof. Let $s_0 := 1+\sqrt{3}$, and for $s \in \mathbb{C}$, let

$$F_s := x^3 + y^3 + z^3 - 3sxyz.$$

Then $E = \{F_{s_0} = 0\}$. In this proof, we also write g_1, g_2, and g_3 for the three matrices in the statement, and for a cubic form F, we write $F \circ g_3$ for the cubic form $F(g_3(x,y,z)^{\mathrm{T}})$.

(1) Let $[x:y:z]$ be a singular point of $\{F_s = 0\}$. Then

$$x^2 = syz, \quad y^2 = sxz, \quad z^2 = sxy.$$

If one of x, y, and z is 0, then the three equations imply $x = y = z = 0$, which is impossible. Otherwise, multiplying the three equations, we obtain $x^2y^2z^2 = s^3x^2y^2z^2$, hence $s^3 = 1$. Since $s_0 > 1$ is a real number, $s_0^3 \neq 1$, so E is smooth. We have $F_{s_0}(0,1,-1) = 0$, so $o \in E$. Since the gradient of F_{s_0} at $(0,1,-1)$ is $(3s_0, 3, 3)$, the tangent line of E at o is $\{s_0x + y + z = 0\}$. A direct calculation shows

$$F_{s_0}(x, y, -s_0x - y) = (1 - s_0^3)x^3.$$

Since $1 - s_0^3 \neq 0$, the divisor of the linear form $s_0x + y + z$ on E is $3o$. Thus $\mathcal{O}_E(1) \cong \mathcal{O}_E(3o)$.

(2) The substitutions $(x,y,z) \to (x, \zeta y, \zeta^2 z)$ and $(x,y,z) \to (z,x,y)$ preserve F_{s_0}, so g_1 and g_2 preserve E. As matrices, $g_2g_1 = \zeta^2 g_1 g_2$, and $g_1^r g_2^{r'}$ is a scalar matrix only if $r \equiv r' \equiv 0 \bmod 3$. Thus the subgroup $\Gamma_0 \subset \mathrm{PGL}(3,\mathbb{C})$ generated by g_1 and g_2 is isomorphic to $(\mathbb{Z}/3\mathbb{Z})^2$.

We show that no non-identity element of Γ_0 has a fixed point on E. For $r \not\equiv 0 \bmod 3$, the fixed points of g_1^r in $\mathbb{P}^2$ are $[1:0:0]$, $[0:1:0]$, and $[0:0:1]$, and F_{s_0} takes the value 1 at each of them. Since $g_1^r g_2^2$ is the inverse of $g_1^{-r} g_2$ in $\mathrm{PGL}(3,\mathbb{C})$, it remains to consider $g_1^r g_2$. Let $v = (x,y,z) \neq 0$ be an eigenvector of $g_1^r g_2$ with eigenvalue μ, that is,

$$(z, \zeta^r x, \zeta^{2r} y) = \mu(x,y,z).$$

Then x, y, and z are non-zero, $\mu^3 = 1$, $x^3 = y^3 = z^3$, and $xyz = \zeta^r x^3$. Hence

$$F_{s_0}(x, y, z) = 3x^3(1 - s_0\zeta^r) \neq 0,$$

because $(s_0\zeta^r)^3 = s_0^3 \neq 1$. Thus $[x : y : z] \notin E$.

Write $E = \mathbb{C}/\Lambda$ for a lattice $\Lambda \subset \mathbb{C}$, with o the image of 0. Then any automorphism of E is of the form $v \to cv + b$, where $c \in \mathbb{C}^*$, $c\Lambda = \Lambda$, and $b \in E$ [Har77, Chapter IV, Lemma 4.9 and Proposition 4.18]. If $c \neq 1$, then $(1 - c)v = b$ has a solution $v \in E$, which is a fixed point. Thus the restriction to E of any non-identity element of Γ_0 is the translation by a non-zero point of $E[3]$, because Γ_0 has exponent 3. In particular, restriction to E is injective on Γ_0. Since $|\Gamma_0| = 9 = |E[3]|$, the restrictions of g_1 and g_2 generate the group of translations by $E[3]$ (see also [AD09, Section 4]).

Let S be the matrix of $(x, y, z) \to (x, z, y)$. A direct calculation shows $g_3^2 = 3S$, so g_3 has order 4 in $\mathrm{PGL}(3, \mathbb{C})$. Since $g_3(0, 1, -1)^{\mathrm{T}} = i\sqrt{3}(0, 1, -1)^{\mathrm{T}}$, g_3 fixes o. A direct calculation shows

$$(x^3 + y^3 + z^3) \circ g_3 = 3(x^3 + y^3 + z^3) + 18xyz, \quad (xyz) \circ g_3 = x^3 + y^3 + z^3 - 3xyz,$$

hence for any $s \neq 1$,

$$F_s \circ g_3 = 3(1 - s)F_{(s+2)/(s-1)}.$$

Since $s_0^2 - 2s_0 - 2 = 0$, we have $(s_0 + 2)/(s_0 - 1) = s_0$, so $F_{s_0} \circ g_3 = 3(1 - s_0)F_{s_0}$. Thus g_3 preserves E.

Let $\tau := g_3|_E$. Then $\tau(o) = o$ and $\tau^4 = \mathrm{id}_E$. The fixed locus of S in $\mathbb{P}^2$ is the union of the line $\{y = z\}$ and the point o, which does not contain E, so $\tau^2 = S|_E \neq \mathrm{id}_E$. Hence τ has order 4. Since $\tau(o) = o$, τ is of the form $v \to cv$ with $c^4 = 1$ and $c^2 \neq 1$, so $c^2 = -1$ and $\tau^2 = -\mathrm{id}_E$. In particular, τ is a group automorphism of E.

(3) For $u \in E[3]$, let t_u be the translation of E by u. Since τ is a group automorphism of E by (2), $\tau t_u \tau^{-1} = t_{\tau(u)}$. As matrices, a direct calculation shows

$$g_3 g_1 = g_2^{-1} g_3, \quad g_3 g_2 = g_1 g_3.$$

Thus Γ_0 is normal in $\langle g_1, g_2, g_3\rangle$, and any element of $\langle g_1, g_2, g_3\rangle$ is of the form γg_3^k with $\gamma \in \Gamma_0$ and $0 \leq k \leq 3$. By (2), the restriction of γg_3^k to E is $t_u\tau^k$ for some $u \in E[3]$, and $u = 0$ only if $\gamma = 1$. Since $t_u\tau^k(o) = u$ and τ has order 4 by (2), the elements $t_u\tau^k$ with $u \in E[3]$ and $0 \leq k \leq 3$ are pairwise distinct. Hence restriction to E is an isomorphism from $\langle g_1, g_2, g_3\rangle$ onto Γ, $|\Gamma| = 36$, and $\Gamma \cong E[3] \rtimes \langle\tau\rangle$.

Under the isomorphism $\Gamma_0 \cong E[3]$ given by restriction to E, the action of τ on $E[3]$ corresponds to conjugation by g_3 on Γ_0. In the basis g_1, g_2 of $\Gamma_0 \cong \mathbb{F}_3^2$, conjugation by g_3 has the matrix

$$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix},$$

whose characteristic polynomial $X^2 + 1$ is irreducible over $\mathbb{F}_3$, because -1 is not a square modulo 3. A τ-invariant subgroup of $E[3]$ of order 3 would be spanned by an eigenvector over $\mathbb{F}_3$ of the matrix of conjugation by g_3. Thus the only τ-invariant subgroups of $E[3]$ are 0 and $E[3]$. □

Lemma 4.2. *Let $\widehat{\Gamma} \subset \mathrm{GL}(3, \mathbb{C})$ be a finite subgroup whose image in $\mathrm{PGL}(3, \mathbb{C})$ is $\langle g_1, g_2, g_3\rangle$. Then for any homomorphism $\lambda : \widehat{\Gamma} \to \mathbb{C}^*$, the subgroups*

$$\{\lambda(\widehat{\gamma})\widehat{\gamma} \mid \widehat{\gamma} \in \widehat{\Gamma}\} \quad \textit{and} \quad \{\lambda(\widehat{\gamma})(\widehat{\gamma}^{-1})^{\mathrm{T}} \mid \widehat{\gamma} \in \widehat{\Gamma}\}$$

of $\mathrm{GL}(3, \mathbb{C})$ are primitive.

Proof. The image of

$$\{\lambda(\widehat{\gamma})\widehat{\gamma} \mid \widehat{\gamma} \in \widehat{\Gamma}\}$$

in $\mathrm{PGL}(3, \mathbb{C})$ is $\langle g_1, g_2, g_3\rangle$. The image of

$$\{\lambda(\widehat{\gamma})(\widehat{\gamma}^{-1})^{\mathrm{T}} \mid \widehat{\gamma} \in \widehat{\Gamma}\}$$

is the image of $\langle g_1, g_2, g_3\rangle$ under the automorphism of $\mathrm{PGL}(3,\mathbb{C})$ induced by $M \to (M^{-1})^{\mathrm{T}}$. Since the matrix of g_1 is diagonal, the matrix of g_2 is a permutation matrix, and the matrix of g_3 is symmetric, we have

$$(g_1^{-1})^{\mathrm{T}} = g_1^{-1}, \quad (g_2^{-1})^{\mathrm{T}} = g_2, \quad (g_3^{-1})^{\mathrm{T}} = g_3^{-1},$$

so the image of $\langle g_1, g_2, g_3\rangle$ under $M \to (M^{-1})^{\mathrm{T}}$ is $\langle g_1, g_2, g_3\rangle$ as well. Hence it suffices to show that any subgroup of $\mathrm{GL}(3,\mathbb{C})$ whose image in $\mathrm{PGL}(3,\mathbb{C})$ is $\langle g_1, g_2, g_3\rangle$ is primitive (Definition 2.7).

Let $\widehat{\Gamma}' \subset \mathrm{GL}(3,\mathbb{C})$ be a subgroup whose image in $\mathrm{PGL}(3,\mathbb{C})$ is $\langle g_1, g_2, g_3\rangle$. Then $\widehat{\Gamma}'$ contains $c_1 g_1$ and $c_2 g_2$ for some $c_1, c_2 \in \mathbb{C}^*$, where g_1 and g_2 also denote the matrices in Construction-Proposition 4.1. The eigenvalues 1, ζ, and ζ^2 of g_1 are distinct, so the eigenlines of $c_1 g_1$ are the three coordinate lines, and $c_2 g_2$ permutes them cyclically. Hence a $\widehat{\Gamma}'$-invariant subspace of $\mathbb{C}^3$ is a sum of coordinate lines invariant under $c_2 g_2$, that is, 0 or $\mathbb{C}^3$. Thus $\widehat{\Gamma}'$ acts irreducibly on $\mathbb{C}^3$.

Suppose that $\mathbb{C}^3 = V_1 \oplus \cdots \oplus V_k$ is a system of imprimitivity of $\widehat{\Gamma}'$. For any $1 \le j \le k$, the sum of the $V_{j'}$ in the $\widehat{\Gamma}'$-orbit of V_j is $\widehat{\Gamma}'$-invariant, so $\widehat{\Gamma}'$ acts transitively on $\{V_1, \dots, V_k\}$, and $\dim V_1 = \cdots = \dim V_k$. Since $k \ge 2$ and $\sum_j \dim V_j = 3$, we have $k = 3$ and $\dim V_j = 1$ for any j. Since scalar matrices preserve each V_j, the action of $\widehat{\Gamma}'$ on $\{V_1, V_2, V_3\}$ induces a homomorphism $\psi : \langle g_1, g_2, g_3\rangle \to \mathfrak{S}_3$ to the symmetric group $\mathfrak{S}_3$. Let $\Gamma_0 \subset \mathrm{PGL}(3,\mathbb{C})$ be the subgroup generated by g_1 and g_2, and let $\Gamma_1 := \Gamma_0 \cap \operatorname{Ker}\psi$. By Construction-Proposition 4.1(2)(3), restriction to E maps Γ_0 isomorphically onto the group of translations by $E[3]$, so $|\Gamma_0| = 9$. Since $|\mathfrak{S}_3| = 6$, the image of Γ_0 in $\mathfrak{S}_3$ has order dividing 3, so $\Gamma_1 \ne 1$. Under restriction to E, conjugation by g_3 becomes conjugation by τ, and $\tau t_u \tau^{-1} = t_{\tau(u)}$ for any $u \in E[3]$, where t_u is the translation of E by u. Since restriction to E is injective on $\langle g_1, g_2, g_3\rangle$ (Construction-Proposition 4.1(3)) and maps Γ_0 onto the group of translations by $E[3]$, we obtain $g_3 \Gamma_0 g_3^{-1} = \Gamma_0$. Since $\operatorname{Ker}\psi$ is normal in $\langle g_1, g_2, g_3\rangle$ and $\Gamma_1 = \Gamma_0 \cap \operatorname{Ker}\psi$, we have $g_3 \Gamma_1 g_3^{-1} = \Gamma_1$. Hence the points $u \in E[3]$ whose translations are restrictions of elements of Γ_1 form a non-zero τ-invariant subgroup of $E[3]$, which is $E[3]$ by Construction-Proposition 4.1(3). Thus $\Gamma_1 = \Gamma_0$, so $c_1 g_1$ and $c_2 g_2$ preserve V_1. Then V_1 is an eigenline of $c_1 g_1$, that is, a coordinate line, and $c_2 g_2$ preserves it, which is impossible. Thus $\widehat{\Gamma}'$ is primitive. □

4.2. **The abelian sixfold.** In this subsection, we construct the abelian sixfold A, the group H, and the quotient $B = A/H$, and we determine the elements of H whose fixed locus contains a prime divisor.

Construction-Proposition 4.3. Let $I_{1,6} := \mathbb{Z}h \oplus \bigoplus_{i=1}^{6} \mathbb{Z}e_i$ be the lattice with the symmetric bilinear form $x \cdot y$ determined by $h \cdot h = 1$, $e_i \cdot e_i = -1$, and $h \cdot e_i = e_i \cdot e_j = 0$ for $i \ne j$. Let $\kappa := -3h + \sum_{i=1}^{6} e_i$, $Q := \{x \in I_{1,6} \mid x \cdot \kappa = 0\}$, and $P := I_{1,6}/\mathbb{Z}\kappa$. For $x, y \in Q$, let $(x, y) := -x \cdot y$. Let W be the group of automorphisms of $I_{1,6}$ generated by the reflections $s_\alpha(x) := x + (x \cdot \alpha)\alpha$ for $\alpha \in Q$ with $(\alpha, \alpha) = 2$ (each s_α fixes κ since $\alpha \cdot \kappa = 0$, so W acts on Q and on P), and let $\delta : P \to \mathbb{Z}/3\mathbb{Z}$ be the homomorphism $[x] \to -x \cdot \kappa \bmod 3$. Let $A := \operatorname{Hom}_{\mathbb{Z}}(P, E)$, on which $w \in W$ acts by $\chi \to \chi \circ w^{-1}$. For $u \in E[3]$, let $\iota_u \in A$ be the homomorphism $p \to \delta(p)u$. Let τ_* be the automorphism $\chi \to \tau \circ \chi$ of A, let $H \subset \operatorname{Aut}(A)$ be the subgroup generated by W, the translations by ι_u for $u \in E[3]$, and τ_*, and let $B := A/H$. Then the following hold.

(1) $(Q, (\cdot,\cdot))$ is the root lattice of type E_6 with Weyl group W, and the form $x \cdot y$ induces a W-equivariant isomorphism $P \cong \operatorname{Hom}_{\mathbb{Z}}(Q, \mathbb{Z})$. The map $Q \otimes_{\mathbb{Z}} E \to A$ sending $q \otimes u$ to the homomorphism $[x] \to (x \cdot q)u$ is an isomorphism of algebraic groups, and it is W-equivariant when $w \in W$ acts on $Q \otimes_{\mathbb{Z}} E$ by $w \otimes 1$. In particular, A is an abelian variety of dimension 6.

(2) δ is a well-defined surjective W-invariant homomorphism, and $u \to \iota_u$ is an isomorphism from $E[3]$ onto the group of W-invariant points of A.

(3) There exists a unique injective homomorphism $\Gamma \to H$ that sends the translation of E by $u \in E[3]$ to the translation of A by ι_u and sends τ to τ_*. The image of Γ commutes with W and intersects W trivially. In particular, H is finite, the induced homomorphism

$W \times \Gamma \to H$ is an isomorphism, $H/W \cong \Gamma$, and B is a normal projective variety of dimension 6.

Proof. (1) By [Dol12, Lemma 8.2.6 and Section 8.2.3], $(Q, (\cdot,\cdot))$ is the root lattice of type E_6, the roots are the vectors $\alpha \in Q$ with $(\alpha,\alpha) = 2$, and

$$h - e_1 - e_2 - e_3, \quad e_1 - e_2, \quad e_2 - e_3, \quad e_3 - e_4, \quad e_4 - e_5, \quad e_5 - e_6$$

are simple roots. For any root α, s_α preserves $x \cdot y$ because $\alpha \cdot \alpha = -2$, and $s_\alpha(\kappa) = \kappa$ because $\alpha \cdot \kappa = 0$. Hence W preserves $x \cdot y$, fixes κ, and preserves Q, and the restriction of s_α to Q is the reflection $x \to x - (x,\alpha)\alpha$. Let $w \in W$ act trivially on Q. For any $x \in I_{1,6}$, we have

$$3x = (3x - (x \cdot \kappa)\kappa) + (x \cdot \kappa)\kappa \quad \text{and} \quad 3x - (x \cdot \kappa)\kappa \in Q,$$

because $\kappa \cdot \kappa = 3$. Thus $w(3x) = 3x$, and $wx = x$ because $I_{1,6}$ is torsion-free. Hence restriction to Q is injective on W, and W is the Weyl group of $(Q, (\cdot,\cdot))$.

The form $x \cdot y$ on $I_{1,6}$ is unimodular. Since Q is the kernel of $x \to x \cdot \kappa$, the group $I_{1,6}/Q$ is torsion-free, so Q is a direct summand of $I_{1,6}$. Hence any homomorphism $Q \to \mathbb{Z}$ extends to a homomorphism $I_{1,6} \to \mathbb{Z}$, which is of the form $y \to x \cdot y$ for some $x \in I_{1,6}$ by unimodularity. Thus the homomorphism $I_{1,6} \to \mathrm{Hom}_{\mathbb{Z}}(Q,\mathbb{Z})$ sending x to $q \to x \cdot q$ is surjective. The kernel is $\{x \in I_{1,6} \mid x \cdot Q = 0\}$, which is contained in $\mathbb{Q}\kappa$ because $\kappa \cdot \kappa \neq 0$, and which equals $\mathbb{Z}\kappa$ because κ is primitive in $I_{1,6}$. Thus $P \cong \mathrm{Hom}_{\mathbb{Z}}(Q,\mathbb{Z})$. Since W preserves $x \cdot y$ and fixes κ, we have $(wx) \cdot q = x \cdot (w^{-1}q)$ for any $w \in W$, $x \in I_{1,6}$, and $q \in Q$, so the isomorphism $P \cong \mathrm{Hom}_{\mathbb{Z}}(Q,\mathbb{Z})$ is W-equivariant when W acts on $\mathrm{Hom}_{\mathbb{Z}}(Q,\mathbb{Z})$ by $f \to f \circ w^{-1}$.

Since $\kappa \cdot q = 0$ for $q \in Q$, the homomorphism $[x] \to (x \cdot q)u$ is well-defined. Let $q_1, \dots, q_6$ be a basis of Q, and let $p_1, \dots, p_6$ be the basis of P such that $p_i \cdot q_j$ is 1 if $i = j$ and 0 otherwise. Evaluation at $p_1, \dots, p_6$ is a bijection $A \to E^6$, and a change of basis of P induces an automorphism of E^6 given by an integer matrix. We view A as an abelian variety through the bijection $A \to E^6$. Under the isomorphisms $Q \otimes_{\mathbb{Z}} E \cong E^6$ given by $q_1, \dots, q_6$ and $A \cong E^6$ given by $p_1, \dots, p_6$, the map $Q \otimes_{\mathbb{Z}} E \to A$ in the statement is the identity of E^6. Thus it is an isomorphism of algebraic groups, and $\dim A = 6$. For $w \in W$ and the image χ of $q \otimes u$, we have

$$(w\chi)([x]) = \chi([w^{-1}x]) = ((w^{-1}x) \cdot q)u = (x \cdot wq)u,$$

so $w\chi$ is the image of $wq \otimes u$. Thus the map is W-equivariant.

(2) Since $\kappa \cdot \kappa = 3$, the class $-x \cdot \kappa \bmod 3$ depends only on $[x] \in P$, so δ is well-defined. Since $-\kappa \cdot e_1 = 1$, δ is surjective. Since W preserves $x \cdot y$ and fixes κ, $\delta([wx]) = -(wx) \cdot (w\kappa) \bmod 3 = \delta([x])$, so δ is W-invariant. For $u \in E[3]$, $\delta(p)u$ is well-defined, so ι_u is a homomorphism $P \to E$, and $u \to \iota_u$ is a homomorphism. For $w \in W$, we have $(w\iota_u)(p) = \delta(w^{-1}p)u = \delta(p)u$, so ι_u is W-invariant. Since $\iota_u([e_1]) = u$, the map $u \to \iota_u$ is injective. Conversely, let $\chi \in A$ be W-invariant, and let $\alpha \in Q$ with $(\alpha,\alpha) = 2$. For any $x \in I_{1,6}$, since s_α is an involution, we have

$$\chi([x]) = \chi([s_\alpha x]) = \chi([x]) + (x \cdot \alpha)\chi([\alpha]),$$

so $(x \cdot \alpha)\chi([\alpha]) = 0$. If $\alpha = m\beta$ for some $\beta \in I_{1,6}$ and a positive integer m, then m^2 divides $\alpha \cdot \alpha = -2$, so $m = 1$. Since $x \cdot y$ is unimodular and α is primitive in $I_{1,6}$, there exists $x \in I_{1,6}$ with $x \cdot \alpha = 1$, and hence $\chi([\alpha]) = 0$. By (1), Q is generated by the vectors α with $(\alpha,\alpha) = 2$, so χ vanishes on the image of Q in P. The image of Q in P is $\mathrm{Ker}\,\delta$: if $x \cdot \kappa \in 3\mathbb{Z}$, then $x - \frac{x \cdot \kappa}{3}\kappa \in Q$ because $\kappa \cdot \kappa = 3$. Since $\delta([e_1]) = 1$ and χ vanishes on $\mathrm{Ker}\,\delta$, we obtain $3\chi([e_1]) = \chi([3e_1]) = 0$ and $\chi(p) = \delta(p)\chi([e_1])$ for any $p \in P$. Thus $\chi = \iota_u$ for $u := \chi([e_1]) \in E[3]$.

(3) For $u \in E[3]$, let t_u be the translation of E by u. By Construction-Proposition 4.1(3), any element of Γ can be written uniquely as $t_u\tau^k$ with $u \in E[3]$ and $0 \le k \le 3$. For $\gamma = t_u\tau^k \in \Gamma$, let $\gamma_A \in \mathrm{Aut}(A)$ be the automorphism

$$\chi \to \tau^k \circ \chi + \iota_u.$$

For $u' \in E[3]$ and $0 \le k' \le 3$, since $\tau^{k'}$ is a group automorphism of E (Construction-Proposition 4.1(2)), we have

$$\tau^{k'} \circ \iota_u = \iota_{\tau^{k'}(u)}, \qquad t_{u'}\tau^{k'} t_u \tau^k = t_{u'+\tau^{k'}(u)} \tau^{k+k'}.$$

Thus $\gamma \to \gamma_A$ is a homomorphism $\Gamma \to \mathrm{Aut}(A)$. The image of t_u is the translation by ι_u, and the image of τ is τ_*, so $\gamma_A \in H$ for any $\gamma \in \Gamma$. Since Γ is generated by τ and the t_u, such a homomorphism is unique. If $\gamma_A = \mathrm{id}_A$ for $\gamma = t_u\tau^k$, then evaluating at $\chi = 0$ we obtain $\iota_u = 0$, so $u = 0$ by (2). Hence $\tau^k \circ \chi = \chi$ for any $\chi \in A$, and taking χ with $\chi(p_1) = v$ for arbitrary $v \in E$, we obtain $\tau^k = \mathrm{id}_E$. Thus the homomorphism is injective. Since ι_u is W-invariant, for $w \in W$ we have

$$w(\tau^k \circ \chi + \iota_u) = \tau^k \circ (\chi \circ w^{-1}) + \iota_u,$$

so the image of Γ commutes with W.

We show that $W \to \mathrm{Aut}(A)$ is injective. Under the isomorphism in (1), τ_* corresponds to $1 \otimes \tau$, and the tangent space of A at 0 is $Q \otimes_{\mathbb{Z}} T_o E \cong Q \otimes_{\mathbb{Z}} \mathbb{C}$. The differential of $w \in W$ is $w \otimes 1$, and the differential of τ_* is the multiplication by the differential $c \in \mathbb{C}^*$ of τ at o, where $c^2 = -1$ because $\tau^2 = -\mathrm{id}_E$. If $w \in W$ acts trivially on A, then the differential $w \otimes 1$ of w at 0 is the identity of $Q \otimes_{\mathbb{Z}} \mathbb{C}$, so w acts trivially on Q, and $w = 1$ by (1).

Now suppose that $\gamma_A \in W$ for $\gamma = t_u\tau^k$. Since elements of W fix $0 \in A$ and $\gamma_A(0) = \iota_u$, we have $\iota_u = 0$, so $u = 0$, and $\gamma_A = \tau_*^k$. Suppose that $\tau_*^k = w$ for some $w \in W$ and $1 \le k \le 3$. If k is odd, then $w \otimes 1 = c^k$ on $Q \otimes_{\mathbb{Z}} \mathbb{C}$ with $c^k \in \{i, -i\}$, which is impossible because $w \otimes 1$ maps $Q \otimes_{\mathbb{Z}} \mathbb{R} \ne 0$ into itself. If $k = 2$, then $w \otimes 1 = -1$, so w is an involution whose eigenvalue -1 has multiplicity 6 on $Q \otimes_{\mathbb{Z}} \mathbb{C}$, which is impossible because the eigenvalue -1 of any involution in W on $Q \otimes_{\mathbb{Z}} \mathbb{C}$ has multiplicity at most 4 [BFS21, Section 3.2, (3.4)]. Thus $k = 0$, and the image of Γ intersects W trivially.

Since the image of Γ commutes with W and intersects W trivially, the map $W \times \Gamma \to H$ sending (w, γ) to $w\gamma_A$ is an injective homomorphism. It is surjective because H is generated by W and the image of Γ. Thus $H \cong W \times \Gamma$ is finite, and $H/W \cong \Gamma$. By [Mum74, §7, Theorem on p. 66 and Remark on p. 69], the quotient morphism $\pi : A \to B$ exists, π is finite and surjective, and $\mathcal{O}_B = (\pi_* \mathcal{O}_A)^H$. Since A is normal and the ring of invariants of an integrally closed domain under a finite group is integrally closed, B is normal. Since π is finite and surjective, $\dim B = \dim A = 6$. By [Har92, Lecture 10, pp. 126–127], B is projective. □

We write A as $\mathrm{Hom}_{\mathbb{Z}}(P, E)$ instead of $Q \otimes_{\mathbb{Z}} E$ because $I_{1,6}$ is the Picard lattice of a marked cubic surface with canonical class κ, and in Subsection 4.3, we compare A with the model of [Rai23, Section 5.2] via the smooth rational surfaces there, whose anti-canonical models are cubic surfaces, by restricting line bundles to an anti-canonical curve E (Definition-Proposition 4.6), where δ is the degree on E modulo 3.

Theorem 4.4 ([Rai23, Theorem 2.5 and Section 4.1])**.** *The quotient A/W is isomorphic to the weighted projective space $\mathbb{P}(1, 1, 1, 2, 2, 2, 3)$.*

Proof. By Construction-Proposition 4.3(1), A is W-equivariantly isomorphic to $Q \otimes_{\mathbb{Z}} E$ with W acting by $w \otimes 1$, where Q is the root lattice of type E_6 and W is the Weyl group. The group W acting on $Q \otimes_{\mathbb{Z}} E$ is an irreducible crystallographic reflection group in the sense of [Rai23, Section 2], since for simply laced root systems the crystallographic lattice is the root lattice tensored with E [Rai23, Section 4.1]. By [Rai23, Theorem 2.5], $(Q \otimes_{\mathbb{Z}} E)/W$ is a weighted projective space, and by the row E_6 of the table in [Rai23, Section 4.1], the weights are $1, 1, 1, 2, 2, 2, 3$. □

That A/W is a weighted projective space was proved in [BS06, Section 2.3, Corollary]. See also [Loo76] and [Rai23, Section 1]. Since any six of the weights $1, 1, 1, 2, 2, 2, 3$ include a weight 1, the weighted projective space $\mathbb{P}(1, 1, 1, 2, 2, 2, 3)$ is well-formed (Definition 2.8). Only the generators of degree 2 can lead to a contradiction with Lemma 3.7(2): the lemma says nothing about the generators of degree 1, and there exists only one generator of degree 3. The weight 2 appears

three times, one more than the bound 2 on the dimensions of the blocks in Lemma 3.7(2), and Proposition 4.7 shows that H/W acts primitively on the three-dimensional space of generators of degree 2.

To use Section 3, we need that the subgroup of H generated by the elements whose fixed locus contains a prime divisor is exactly W, so that the intermediate quotient is the weighted projective space of Theorem 4.4. The case of linear part $-w$ with $w \in W$ is excluded because an involution in W has eigenvalue -1 of multiplicity at most 4 [BFS21, Section 3.2, (3.4)].

Proposition 4.5. *Let $\eta \in H$ with $\eta \neq \mathrm{id}_A$. Then the fixed locus of η contains a prime divisor on A if and only if $\eta = s_\alpha$ for some $\alpha \in Q$ with $(\alpha, \alpha) = 2$. In particular, the subgroup of H generated by the elements whose fixed locus contains a prime divisor on A is W, the induced action of H/W on A/W is faithful, and the quotient morphism $A/W \to B$ is quasi-étale.*

Proof. For a group automorphism ℓ of A and $b \in A$, the fixed locus of $x \to \ell(x) + b$ is empty or a coset of $\mathrm{Ker}(1-\ell)$: if x_0 is a fixed point, then x is a fixed point if and only if $(1-\ell)(x - x_0) = 0$. Moreover, $\dim \mathrm{Ker}(1-\ell)$ is the nullity of $1 - d\ell$ on the tangent space T_0A: the tangent space at 0 of the group scheme $\mathrm{Ker}(1-\ell)$ is the kernel of $1 - d\ell$, and $\mathrm{Ker}(1-\ell)$ is smooth because any group scheme over a field of characteristic 0 is smooth [Mum74, §11, Theorem on p. 101].

Under the isomorphism of Construction-Proposition 4.3(1), we have

$$T_0A = Q \otimes_{\mathbb{Z}} T_oE \cong Q \otimes_{\mathbb{Z}} \mathbb{C},$$

the differential of $w \in W$ is $w \otimes 1$, and the differential of τ_* is the multiplication by the differential $c \in \mathbb{C}^*$ of τ at o, where $c^2 = -1$ because $\tau^2 = -\mathrm{id}_E$. By Construction-Proposition 4.3(3), we may write

$$\eta(x) = \tau_*^k(w(x)) + \iota_u$$

with $0 \leq k \leq 3$, $w \in W$, and $u \in E[3]$, and the differential of $\tau_*^k w$ is $c^k(w \otimes 1)$.

Suppose that the fixed locus of η contains a prime divisor on A. Then the fixed locus is non-empty, and $\dim \mathrm{Ker}(1 - \tau_*^k w) \geq 5$. If $\dim \mathrm{Ker}(1 - \tau_*^k w) = 6$, then $\tau_*^k w = \mathrm{id}_A$, so η is the translation by ι_u, and since η has a fixed point, $\iota_u = 0$ and $\eta = \mathrm{id}_A$, which is not the case. Hence the nullity of $1 - c^k(w \otimes 1)$ is 5, that is,

$$\mathrm{rank}(1 - c^k(w \otimes 1)) = 1.$$

Since w has finite order, $w \otimes 1$ is diagonalizable on $Q \otimes_{\mathbb{Z}} \mathbb{C}$. We consider three cases.

Case 1. k is odd. Then $c^{-k} \in \{i, -i\}$ is an eigenvalue of $w \otimes 1$ of multiplicity 5. Since w preserves $Q \otimes_{\mathbb{Z}} \mathbb{R}$, the complex conjugate $-c^{-k}$ is also an eigenvalue of multiplicity 5, which is impossible because $5 + 5 > 6$.

Case 2. $k = 2$. Then $c^k = -1$, and -1 is an eigenvalue of $w \otimes 1$ of multiplicity 5. The remaining eigenvalue is real, because the non-real eigenvalues of $w \otimes 1$ come in complex conjugate pairs with equal multiplicities. Since w has finite order, the remaining eigenvalue is 1 (it is ± 1, and it is not -1, since the multiplicity of -1 is exactly 5). Thus w is an involution whose eigenvalue -1 has multiplicity 5, which is impossible because the eigenvalue -1 of any involution in W has multiplicity at most 4 [BFS21, Section 3.2, (3.4)].

Case 3. $k = 0$. Then $\mathrm{rank}(1 - w \otimes 1) = 1$, so exactly one eigenvalue of $w \otimes 1$ is not 1, and the eigenvalue different from 1 is $\det w \in \{1, -1\}$, hence -1. Thus w is an involution whose eigenvalue -1 has multiplicity 1. By [BFS21, Section 3.1], an involution in W whose eigenvalue -1 has multiplicity 1 is a reflection of W, that is, s_α for a pair of opposite roots $\pm\alpha$. Hence $w = s_\alpha$ for some $\alpha \in Q$ with $(\alpha, \alpha) = 2$.

It remains to show that $\iota_u = 0$ in Case 3. Since $\eta(x) = s_\alpha(x) + \iota_u$ has a fixed point x_0, we have $\iota_u = (1 - s_\alpha)(x_0)$. For $q \in Q$ and $v \in E$, we have

$$(1 - s_\alpha)(q \otimes v) = (q, \alpha)\alpha \otimes v,$$

so ι_u is the image of $\alpha \otimes v$ for some $v \in E$. Since the root system of type E_6 is irreducible and all its roots α satisfy $(\alpha, \alpha) = 2$, by [Bou02, Chapter VI, §1, no. 3, Proposition 11], there exists $w' \in W$ such that $w'(\alpha)$ is a simple root α_1. Since the Dynkin diagram of E_6 is connected, there exists a simple root β_1 with $(\alpha_1, \beta_1) = -1$. Let $\beta := w'^{-1}(\beta_1)$. Then $(\alpha, \beta) = -1$, so $s_\beta(\alpha) = \alpha + \beta$. Since ι_u is W-invariant (Construction-Proposition 4.3(2)), we have

$$\alpha \otimes v = s_\beta(\alpha \otimes v) = \alpha \otimes v + \beta \otimes v,$$

so $\beta \otimes v = 0$. The simple roots form a basis of Q [Bou02, Chapter VI, §1, no. 9], so $\beta = w'^{-1}(\beta_1)$ is a member of a basis of Q, and $v \to \beta \otimes v$ is injective. Thus $v = 0$, $\iota_u = 0$, and $u = 0$ by Construction-Proposition 4.3(2). Hence $\eta = s_\alpha$.

Conversely, the fixed locus of s_α is $\operatorname{Ker}(1 - s_\alpha)$, and $\dim \operatorname{Ker}(1 - s_\alpha) = 5$ because $1 - s_\alpha \otimes 1$ has rank 1 on $Q \otimes_{\mathbb{Z}} \mathbb{C}$. Thus the fixed locus of s_α contains a prime divisor on A.

Since W is generated by the reflections s_α, the subgroup of H generated by the elements whose fixed locus contains a prime divisor on A is W. Since A is projective and H is finite (Construction-Proposition 4.3(3)), the subgroup H_0 of Set-up 3.1, applied to $X = A$ and the group H, is W. By Lemma 3.2, the induced action of H/W on A/W is faithful, and the quotient morphism $A/W \to A/H = B$ is quasi-étale. □

4.3. **Generators of degree two.** In this subsection, we compute the action of the residual group $H/W \cong \Gamma$ on the space M_2 of generators of degree 2 of $A/W \cong \mathbb{P}(1, 1, 1, 2, 2, 2, 3)$, and prove that H/W acts primitively on M_2. Primitivity contradicts the decomposition into blocks of Lemma 3.7, and we prove Theorem 4.8 at the end of this subsection.

Definition-Proposition 4.6 rewrites the action of H on A as an action on homomorphisms from P to a quotient of $\operatorname{Pic}(E)$, which we compare with the action in [Rai23, p. 45] in the proof of Proposition 4.7.

Definition-Proposition 4.6. Let $\overline{\operatorname{Pic}}(E) := \operatorname{Pic}(E)/\mathbb{Z}[\mathcal{O}_E(3o)]$, let $\deg : \overline{\operatorname{Pic}}(E) \to \mathbb{Z}/3\mathbb{Z}$ be the homomorphism induced by the degree, and let $\operatorname{Hom}_\delta(P, \overline{\operatorname{Pic}}(E))$ be the set of homomorphisms $\rho : P \to \overline{\operatorname{Pic}}(E)$ such that $\deg \circ \rho = \delta$. Let $\rho_0 \in \operatorname{Hom}_\delta(P, \overline{\operatorname{Pic}}(E))$ be the homomorphism $p \to \delta(p)[\mathcal{O}_E(o)]$, and identify E with $\operatorname{Pic}^0(E)$ by $u \to [\mathcal{O}_E(u)] - [\mathcal{O}_E(o)]$. Then the following hold.

(1) The map $A \to \operatorname{Hom}_\delta(P, \overline{\operatorname{Pic}}(E))$, $\chi \to \rho_0 + \chi$, is bijective.
(2) Any $\gamma \in \Gamma$ satisfies $\gamma^* \mathcal{O}_E(3o) \cong \mathcal{O}_E(3o)$. Under the bijection in (1), the element $w \in W$ acts by $\rho \to \rho \circ w^{-1}$, and $\gamma \in \Gamma$, viewed as an element of H by Construction-Proposition 4.3(3), acts by $\rho \to (\gamma^{-1})^* \circ \rho$.

Proof. The degree homomorphism $\operatorname{Pic}(E) \to \mathbb{Z}$ sends $[\mathcal{O}_E(3o)]$ to 3, so it induces

$$\deg : \overline{\operatorname{Pic}}(E) \to \mathbb{Z}/3\mathbb{Z}.$$

Since $\deg \mathcal{O}_E(3o)^{\otimes k} = 3k$ for any $k \in \mathbb{Z}$, the homomorphism $\operatorname{Pic}^0(E) \to \overline{\operatorname{Pic}}(E)$ is injective, and the image of $\operatorname{Pic}^0(E)$ is the kernel of deg: if $\deg L = 3k$ for a line bundle L on E, then $[L] - k[\mathcal{O}_E(3o)] \in \operatorname{Pic}^0(E)$. We regard $\operatorname{Pic}^0(E)$ as a subgroup of $\overline{\operatorname{Pic}}(E)$ via the injection. Since $3[\mathcal{O}_E(o)] = [\mathcal{O}_E(3o)] = 0$ in $\overline{\operatorname{Pic}}(E)$, ρ_0 is a well-defined homomorphism, and $\deg \circ \rho_0 = \delta$ because $\deg \mathcal{O}_E(o) = 1$. The map $u \to [\mathcal{O}_E(u)] - [\mathcal{O}_E(o)]$ is a bijection $E \to \operatorname{Pic}^0(E)$ [Har77, Chapter IV, Example 1.3.7], and it is an isomorphism of groups for the group law of E with origin o. Thus $A = \operatorname{Hom}_{\mathbb{Z}}(P, \operatorname{Pic}^0(E))$.

(1) For any $\chi \in A$, $\deg \circ(\rho_0 + \chi) = \delta$ because $\deg \circ \chi = 0$. Conversely, let $\rho : P \to \overline{\operatorname{Pic}}(E)$ be a homomorphism with $\deg \circ \rho = \delta$. Then $\deg \circ(\rho - \rho_0) = 0$, so $\rho - \rho_0$ is a homomorphism $P \to \operatorname{Pic}^0(E)$, that is, an element of A. Thus $\chi \to \rho_0 + \chi$ is bijective.

(2) Let $w \in W$. By Construction-Proposition 4.3(2), $\delta \circ w^{-1} = \delta$. Hence $\rho \to \rho \circ w^{-1}$ preserves $\operatorname{Hom}_\delta(P, \overline{\operatorname{Pic}}(E))$, $\rho_0 \circ w^{-1} = \rho_0$, and

$$(\rho_0 + \chi) \circ w^{-1} = \rho_0 + \chi \circ w^{-1}$$

for any $\chi \in A$. The right-hand side corresponds to the image of χ under w.

Let $\gamma \in \Gamma$. For $u \in E[3]$, let t_u be the translation $e \to e+u$ of E. Then $t_u^* \mathcal{O}_E(o) \cong \mathcal{O}_E(-u)$, so

$$t_u^* \mathcal{O}_E(3o) \cong \mathcal{O}_E(-u)^{\otimes 3} \cong \mathcal{O}_E(3o),$$

where the second isomorphism holds because, under the isomorphism $E \to \mathrm{Pic}^0(E)$ above, the class

$$[\mathcal{O}_E(-u)^{\otimes 3}] - [\mathcal{O}_E(3o)] = 3([\mathcal{O}_E(-u)] - [\mathcal{O}_E(o)])$$

corresponds to $-3u = o$. Moreover, $\tau^* \mathcal{O}_E(3o) \cong \mathcal{O}_E(3o)$ because $\tau(o) = o$. Since Γ is generated by τ and the translations t_u for $u \in E[3]$, $(\gamma^{-1})^* \mathcal{O}_E(3o) \cong \mathcal{O}_E(3o)$. Thus $(\gamma^{-1})^*$ induces an automorphism of $\overline{\mathrm{Pic}}(E)$ commuting with deg, and $\rho \to (\gamma^{-1})^* \circ \rho$ preserves $\mathrm{Hom}_\delta(P, \overline{\mathrm{Pic}}(E))$. Since

$$((\gamma_1 \gamma_2)^{-1})^* = (\gamma_1^{-1})^* \circ (\gamma_2^{-1})^*$$

for any $\gamma_1, \gamma_2 \in \Gamma$, the maps $\rho \to (\gamma^{-1})^* \circ \rho$ for $\gamma \in \Gamma$ form an action of Γ on $\mathrm{Hom}_\delta(P, \overline{\mathrm{Pic}}(E))$. The homomorphism $\Gamma \to H$ of Construction-Proposition 4.3(3) is a group homomorphism and Γ is generated by τ and the translations t_u for $u \in E[3]$, so it suffices to compare the actions of τ and of t_u.

Let $u \in E[3]$. For any $e \in E$, $(t_u^{-1})^* \mathcal{O}_E(e) \cong \mathcal{O}_E(e+u)$. Hence

$$(t_u^{-1})^* \left([\mathcal{O}_E(e)] - [\mathcal{O}_E(o)]\right) = \left([\mathcal{O}_E(e+u)] - [\mathcal{O}_E(o)]\right) - \left([\mathcal{O}_E(u)] - [\mathcal{O}_E(o)]\right),$$

which corresponds to $(e+u) - u = e$. Thus $(t_u^{-1})^*$ is the identity on $\mathrm{Pic}^0(E)$. Moreover, for any $p \in P$,

$$(t_u^{-1})^* \rho_0(p) = \delta(p)[\mathcal{O}_E(u)] = \rho_0(p) + \delta(p)\left([\mathcal{O}_E(u)] - [\mathcal{O}_E(o)]\right) = \rho_0(p) + \iota_u(p).$$

Therefore $(t_u^{-1})^* \circ (\rho_0 + \chi) = \rho_0 + \iota_u + \chi$ for any $\chi \in A$, that is, t_u acts as the translation by ι_u, which is the image of t_u in H.

Since $\tau(o) = o$, we have $(\tau^{-1})^* \mathcal{O}_E(e) \cong \mathcal{O}_E(\tau(e))$ for any $e \in E$. Thus $(\tau^{-1})^*$ fixes ρ_0 and acts on $\mathrm{Pic}^0(E) = E$ by τ. Therefore $(\tau^{-1})^* \circ (\rho_0 + \chi) = \rho_0 + \tau \circ \chi$ for any $\chi \in A$, that is, τ acts as τ_*, which is the image of τ in H. □

We compute the action of H/W on M_2 in the model of [Rai23, p. 45].

Proposition 4.7. *Let $\phi : A/W \to \mathbb{P}(1,1,1,2,2,2,3)$ be an isomorphism (Theorem 4.4), and let M_2 be the space of generators of degree 2 of $\mathbb{P}(1,1,1,2,2,2,3)$ (Definition-Lemma 2.10). Then $\dim M_2 = 3$, and the finite group $\{\phi \circ h \circ \phi^{-1} \mid h \in H/W\}$ acts primitively on M_2, where H/W is viewed as a subgroup of $\mathrm{Aut}(A/W)$ by Proposition 4.5.*

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we recall the model of [Rai23, Section 5.2, (5.7), pp. 44–45] for cubic surfaces, possibly singular, with anti-canonical curve E and $\mathcal{O}_{\mathbb{P}^3}(1)|_E \cong \mathcal{O}_E(3o)$.

Let

$$F := x^3 + y^3 + z^3 - 3(1+\sqrt{3})xyz,$$

so that $E = \{F = 0\} \subset \mathbb{P}^2$ and $\mathcal{O}_E(1) \cong \mathcal{O}_E(3o)$ (Construction-Proposition 4.1(1)). The homogeneous coordinate ring of E is $\mathbb{C}[x,y,z]/(F)$.

Let $\rho \in \mathrm{Hom}_\delta(P, \overline{\mathrm{Pic}}(E))$. By [Rai23, Section 5.2, pp. 43–44], ρ determines a smooth projective rational surface $\widetilde{S}$, an embedding $i : E \to \widetilde{S}$ of E as an anti-canonical curve with $i^* \mathcal{O}_{\widetilde{S}}(-K_{\widetilde{S}}) \cong \mathcal{O}_E(3o)$, and the marking $\mathrm{Pic}(\widetilde{S}) \cong I_{1,6}$ given by the blow-ups below, which sends the class of the pullback of a line to h and the class of the total transform of the j-th exceptional curve to e_j for $1 \le j \le 6$, hence $[K_{\widetilde{S}}]$ to κ, so that ρ is induced by i^*. For any $\rho \in \mathrm{Hom}_\delta(P, \overline{\mathrm{Pic}}(E))$, $\widetilde{S}$ is constructed as follows: the lift of $\rho([h])$ to $\mathrm{Pic}(E)$ of degree 3, which is unique since $\deg \mathcal{O}_E(3o) = 3$, embeds E in $\mathbb{P}^2$ as a plane cubic, the lifts of $\rho([e_1]), \dots, \rho([e_6])$ of degree 1 determine six points of E, and $\widetilde{S}$ is obtained by blowing up the six points successively on the strict transforms of E. The restriction i^* sends κ to $[\mathcal{O}_E(-3o)]$, and the image of $\ell \in I_{1,6}$ has degree $\ell \cdot (-K_{\widetilde{S}}) = -\ell \cdot \kappa$, in accordance with $\deg \circ \rho = \delta$.

The curve $i(E)$ is the zero locus of a section $x_0 \in \Gamma(\widetilde{S}, \omega_{\widetilde{S}}^{-1})$, unique up to a non-zero constant. The quotient of the graded algebra

$$\bigoplus_{k\geq 0} \Gamma(\widetilde{S}, \omega_{\widetilde{S}}^{-k})$$

by $x_0 - 1$ is a filtered algebra whose associated graded algebra is the quotient by x_0, which is the homogeneous coordinate ring $\mathbb{C}[x, y, z]/(F)$ of E [Rai23, Section 5.2, pp. 43–44]. Lifts of x, y, z to the filtered algebra still generate it, and they satisfy a single relation, which is F plus terms of lower degree:

$$F + q_2 + q_1 + q_0 = 0, \tag{4.1}$$

where $q_j \in \mathbb{C}[x, y, z]$ is homogeneous of degree j for $j = 0, 1, 2$. Another choice of lifts changes (4.1) by a translation $(x, y, z) \to (x, y, z) + v$ of the generators, $v \in \mathbb{C}^3$. The corresponding cubic surface, which is the anti-canonical model S of $\widetilde{S}$, that is, Proj of the graded algebra above [Rai23, Section 5.2, (5.3)], is

$$\{F + x_0 q_2 + x_0^2 q_1 + x_0^3 q_0 = 0\} \subset \mathbb{P}^3,$$

whose hyperplane section $\{x_0 = 0\}$ is E. By adjunction, E is an anti-canonical curve on S, and $\mathcal{O}_{\mathbb{P}^3}(1)|_E \cong \mathcal{O}_E(3o)$. The model provides an isomorphism from A/W onto the space of equations (4.1) modulo the translations $(x, y, z) \to (x, y, z) + v$ of the generators, $v \in \mathbb{C}^3$, and modulo the weighted scaling $(q_2, q_1, q_0) \to (cq_2, c^2 q_1, c^3 q_0)$, $c \in \mathbb{C}^*$. The coefficients of q_2, q_1, and q_0 have weights 1, 2, and 3, and v has weight 1. Under the isomorphism, W changes only the marking of $\widetilde{S}$, and the point of A/W given by ρ corresponds to the anti-canonical model S of $(\widetilde{S}, i)$ together with the embedding of E, up to isomorphism.

Step 2. Let $\mathrm{Aut}_{\mathcal{O}_E(3o)}(E)$ be the group of $\gamma \in \mathrm{Aut}(E)$ with $\gamma^* \mathcal{O}_E(3o) \cong \mathcal{O}_E(3o)$. By Definition-Proposition 4.6(2), it contains Γ. In this step we show that the identification of **Step 1** transforms the action of $H/W \cong \Gamma$ on A/W into the restriction to Γ of the action of $\mathrm{Aut}_{\mathcal{O}_E(3o)}(E)$ described in [Rai23, p. 45]: the projective action on the plane cubic $E \subset \mathbb{P}^2$, followed by a translation of the generators.

By Definition-Proposition 4.6(2), $\gamma \in \Gamma$, as an element of H, maps $\rho \in \mathrm{Hom}_\delta(P, \overline{\mathrm{Pic}}(E))$ to $(\gamma^{-1})^* \circ \rho$. Let $\rho \in \mathrm{Hom}_\delta(P, \overline{\mathrm{Pic}}(E))$, and let $\widetilde{S}$, $i : E \to \widetilde{S}$, and the marking $\mathrm{Pic}(\widetilde{S}) \cong I_{1,6}$ be attached to ρ as in **Step 1**. Since $(\gamma^{-1})^* \mathcal{O}_E(3o) \cong \mathcal{O}_E(3o)$, the embedding $i \circ \gamma^{-1} : E \to \widetilde{S}$ satisfies

$$(i \circ \gamma^{-1})^* \mathcal{O}_{\widetilde{S}}(-K_{\widetilde{S}}) \cong \mathcal{O}_E(3o),$$

and with the marking $\mathrm{Pic}(\widetilde{S}) \cong I_{1,6}$ above it induces $(i \circ \gamma^{-1})^* = (\gamma^{-1})^* \circ i^*$, that is, $(\gamma^{-1})^* \circ \rho$. Since the procedure of **Step 1** recovers a surface with an embedding of E as an anti-canonical curve and a marking given by blow-ups of $\mathbb{P}^2$ from the homomorphism they induce [Rai23, Section 5.2, p. 43], ρ and $(\gamma^{-1})^* \circ \rho$ both correspond to the surface $\widetilde{S}$, and the identity of $\widetilde{S}$ maps $i(e)$ to $(i \circ \gamma^{-1})(\gamma(e))$ for any $e \in E$. Hence the identity of $\widetilde{S}$ is an isomorphism from $(\widetilde{S}, i)$ to $(\widetilde{S}, i \circ \gamma^{-1})$ whose restriction to the anti-canonical curve E is γ.

The filtered algebra (4.1) attached in **Step 1** to $(\widetilde{S}, i)$ depends only on $\widetilde{S}$ and i. Hence the filtered algebras of $(\widetilde{S}, i)$ and $(\widetilde{S}, i \circ \gamma^{-1})$ are isomorphic, and the isomorphism induced on the associated graded algebras, both equal to the homogeneous coordinate ring $\mathbb{C}[x, y, z]/(F)$ of E, is induced by γ, that is, by a linear lift $\widehat{\gamma} \in \mathrm{GL}(3, \mathbb{C})$ of the element of $\langle g_1, g_2, g_3 \rangle$ restricting to γ on E (Construction-Proposition 4.1(3)). The scalar ambiguity of $\widehat{\gamma}$ is absorbed by the weighted scaling and by multiplying (4.1) by a constant. Therefore the generators x, y, z of filtration degree one transform by $\widehat{\gamma}$ up to elements of filtration degree zero, and the constants are removed by a translation $(x, y, z) \to (x, y, z) + v$. This is the algorithm of [Rai23, p. 45]. The step uses only the naturality of the surface $\widetilde{S}$ and of the anti-canonical curve E.

Therefore the identification of **Step 1** transforms the action of $H/W \cong \Gamma$ on A/W into the restriction to Γ of the action of $\mathrm{Aut}_{\mathcal{O}_E(3o)}(E)$ in [Rai23, p. 45]: both actions are algebraic, and

since **Step 1** attaches $(\widetilde{S}, i)$ to any $\rho \in \mathrm{Hom}_\delta(P, \overline{\mathrm{Pic}}(E))$, they coincide on the closed points of the reduced variety A/W, so they coincide as morphisms.

Step 3. In this step we bring (4.1) into a normal form preserved by a finite linear lift of Γ, and compute the action of H/W on the generators of degree 2.

For $v = (v_1, v_2, v_3) \in \mathbb{C}^3$ and a polynomial $f \in \mathbb{C}[x, y, z]$, let

$$\partial_v f := v_1 \partial f/\partial x + v_2 \partial f/\partial y + v_3 \partial f/\partial z.$$

Let $T_2 \subset \mathrm{Sym}^2((\mathbb{C}^3)^*)$ be the span of $\partial F/\partial x$, $\partial F/\partial y$, and $\partial F/\partial z$. The linear map $v \to \partial_v F$ is injective: if $\partial_v F = 0$ for some $v \neq 0$, then after a linear change of coordinates with $v = (1, 0, 0)$, F does not depend on x, so F and all partial derivatives of F vanish at $[1 : 0 : 0]$, and E is singular at $[1 : 0 : 0]$, which contradicts Construction-Proposition 4.1(1). Thus $\dim T_2 = 3$, and $v \to \partial_v F$ is an isomorphism $\mathbb{C}^3 \to T_2$.

Let $\widehat{\Gamma} \subset \mathrm{SL}(3, \mathbb{C})$ be the inverse image of $\langle g_1, g_2, g_3 \rangle$. Since the kernel of the natural homomorphism from $\mathrm{SL}(3, \mathbb{C})$ to $\mathrm{PGL}(3, \mathbb{C})$ is μ_3, $\widehat{\Gamma}$ is finite, and the image of $\widehat{\Gamma}$ in $\mathrm{PGL}(3, \mathbb{C})$ is $\langle g_1, g_2, g_3 \rangle$. Let $\mathrm{GL}(3, \mathbb{C})$ act on $\mathbb{C}[x, y, z]$ by $(\widehat{\gamma} \cdot f)(X) := f(\widehat{\gamma}^{-1} X)$, where $X = (x, y, z)^{\mathrm{T}}$. For $\widehat{\gamma} \in \widehat{\Gamma}$, the class of $\widehat{\gamma}$ preserves E (Construction-Proposition 4.1(2)), so $\widehat{\gamma} \cdot F$ is a cubic form vanishing on E. Since E is smooth, F is irreducible, so

$$\widehat{\gamma} \cdot F = \lambda(\widehat{\gamma}) F$$

for some $\lambda(\widehat{\gamma}) \in \mathbb{C}^*$, and $\lambda : \widehat{\Gamma} \to \mathbb{C}^*$ is a homomorphism. By the chain rule,

$$\widehat{\gamma} \cdot \partial_v F = \lambda(\widehat{\gamma}) \partial_{\widehat{\gamma} v} F$$

for any $v \in \mathbb{C}^3$, so T_2 is $\widehat{\Gamma}$-stable. By Maschke's theorem [Ser77, Section 1.3, Theorem 1], there exists a $\widehat{\Gamma}$-stable complement C_2 of T_2 in $\mathrm{Sym}^2((\mathbb{C}^3)^*)$. We fix C_2.

Write $(x_1, x_2, x_3) = (x, y, z)$. For $v \in \mathbb{C}^3$, Taylor's formula reads

$$F(X + v) = F(X) + \partial_v F(X) + \frac{1}{2} \sum_{l,m=1}^{3} v_l v_m \frac{\partial^2 F}{\partial x_l \partial x_m}(X) + F(v),$$

and the third term is linear in X. Hence the translation $(x, y, z) \to (x, y, z) + v$ maps (4.1) to $F + q_2' + q_1' + q_0' = 0$, where

$$\begin{aligned} q_2' &= q_2 + \partial_v F, \\ q_1' &= q_1 + \partial_v q_2 + \frac{1}{2} \sum_{l,m=1}^{3} v_l v_m \frac{\partial^2 F}{\partial x_l \partial x_m}, \\ q_0' &= q_0 + q_2(v) + q_1(v) + F(v). \end{aligned} \tag{4.2}$$

Since $v \to \partial_v F$ is an isomorphism onto T_2, there exists a unique $v \in \mathbb{C}^3$ with $q_2' \in C_2$, namely the negative of the inverse image of the T_2-component of q_2, and v depends linearly on the coefficients of q_2. Thus any equation (4.1) is equivalent under translations to a unique equation with $q_2 \in C_2$, which we call the *normal form*.

Let $\mathcal{S}$ be the graded polynomial ring of functions on the space $C_2 \times (\mathbb{C}^3)^* \times \mathbb{C}$ of normal forms (q_2, q_1, q_0), generated by the three coordinates of q_2 in a basis of C_2 in degree 1, the coefficients a_1, a_2, a_3 of $q_1 = a_1 x + a_2 y + a_3 z$ in degree 2, and q_0 in degree 3. By **Step 1**, the model recalled there provides an isomorphism $\sigma : A/W \to \mathrm{Proj}\, \mathcal{S}$. The weighted projective space $\mathrm{Proj}\, \mathcal{S} \cong \mathbb{P}(1, 1, 1, 2, 2, 2, 3)$ is well-formed, and by Definition-Proposition 2.9(1) we may regard $\mathcal{S}$ as the Cox ring of $\mathrm{Proj}\, \mathcal{S}$. The degree 2 part of $\mathcal{S}$ is spanned by a_1, a_2, a_3, and the products of two coordinates of degree 1, and the products of two coordinates of degree 1 span $(\mathcal{S}_+^2)_2$. Thus the classes $[a_1]$, $[a_2]$, $[a_3]$ form a basis of the space M_2' of generators of degree 2 of $\mathrm{Proj}\, \mathcal{S}$ (Definition-Lemma 2.10). In particular, by Definition-Proposition 2.9(1), $\dim M_2 = 3$.

Let $\widehat{\gamma} \in \widehat{\Gamma}$. The projective action of $\widehat{\gamma}$ maps (4.1) to

$$\lambda(\widehat{\gamma})F + \widehat{\gamma} \cdot q_2 + \widehat{\gamma} \cdot q_1 + q_0 = 0,$$

and dividing by $\lambda(\widehat{\gamma})$, it acts on the parameters by

$$\Phi_{\widehat{\gamma}}(q_2, q_1, q_0) := \left(\lambda(\widehat{\gamma})^{-1}\widehat{\gamma} \cdot q_2,\ \lambda(\widehat{\gamma})^{-1}\widehat{\gamma} \cdot q_1,\ \lambda(\widehat{\gamma})^{-1}q_0\right).$$

Since C_2 is $\widehat{\Gamma}$-stable, $\Phi_{\widehat{\gamma}}$ maps normal forms to normal forms without a translation, and $\Phi_{\widehat{\gamma}}$ is a linear automorphism of $C_2 \times (\mathbb{C}^3)^* \times \mathbb{C}$ commuting with the weighted scaling. Thus $\Phi_{\widehat{\gamma}}$ induces a graded automorphism of $\mathcal{S}$ that is linear on the generators. By **Step 2**, the automorphisms of $\operatorname{Proj}\mathcal{S}$ induced by $\Phi_{\widehat{\gamma}}$ for $\widehat{\gamma} \in \widehat{\Gamma}$ form the group $\{\sigma \circ \eta \circ \sigma^{-1} \mid \eta \in H/W\}$.

We compute the action on the generators of degree 2. Write $q_1 = a^{\mathrm{T}}X$ with $a = (a_1, a_2, a_3)^{\mathrm{T}}$. Then $\widehat{\gamma} \cdot q_1 = a^{\mathrm{T}}\widehat{\gamma}^{-1}X$, so $\Phi_{\widehat{\gamma}}$ maps a to

$$\lambda(\widehat{\gamma})^{-1}(\widehat{\gamma}^{-1})^{\mathrm{T}}a.$$

Therefore the graded automorphism $s \to s \circ \Phi_{\widehat{\gamma}}^{-1}$ of $\mathcal{S}$ maps (a_1, a_2, a_3) to

$$(a_1, a_2, a_3)\lambda(\widehat{\gamma})\widehat{\gamma};$$

it induces the inverse of the automorphism of $\operatorname{Proj}\mathcal{S}$ induced by $\Phi_{\widehat{\gamma}}$. By Definition-Lemma 2.10(1), the class in $\mathrm{PGL}(M_2')$ of the corresponding element of $\{\sigma \circ \eta \circ \sigma^{-1}\}$ is the class of the matrix $\lambda(\widehat{\gamma})\widehat{\gamma}$ in the basis $[a_1], [a_2], [a_3]$. Since $\widehat{\Gamma}$ is a group and λ is a homomorphism, the set of matrices $\lambda(\widehat{\gamma})\widehat{\gamma}$ for $\widehat{\gamma} \in \widehat{\Gamma}$ is closed under taking inverse matrices. Thus

$$\{[\sigma \circ \eta \circ \sigma^{-1}]_2 \mid \eta \in H/W\} = \text{the image in } \mathrm{PGL}(M_2') \text{ of } \{\lambda(\widehat{\gamma})\widehat{\gamma} \mid \widehat{\gamma} \in \widehat{\Gamma}\}. \tag{4.3}$$

Here M_2', spanned by the coefficient functions of q_1, is dual to the space of linear forms q_1, and $\widehat{\Gamma}$ acts on M_2' by the natural representation on $\mathbb{C}^3$ multiplied by λ. If one uses γ^{-1} instead of γ, the group of automorphisms of A/W does not change, since $\{\gamma^{-1} \mid \gamma \in \Gamma\} = \Gamma$. If one uses the action $f(X) \to f(\widehat{\gamma}^{\mathrm{T}}X)$ on polynomials, the matrices obtained are $\lambda'(\widehat{\gamma})(\widehat{\gamma}^{-1})^{\mathrm{T}}$ for a character λ' of $\widehat{\Gamma}$, and they also form a primitive subgroup by Lemma 4.2.

We compare the computation of $\Phi_{\widehat{\gamma}}$ above with the algorithm of [Rai23, p. 45], which restores a normalization by a translation v after the projective action. Since $\Phi_{\widehat{\gamma}}$ preserves normal forms, the computation above uses no translation. For another normalization, v depends linearly on the coordinates of degree 1, and by (4.2) the coefficients of q_1 change by the coefficients of

$$\partial_v q_2 + \frac{1}{2}\sum_{l,m=1}^{3} v_l v_m \frac{\partial^2 F}{\partial x_l \partial x_m},$$

which are quadratic in the coordinates of degree 1, that is, elements of $\mathcal{S}_+^2$. Hence the linear map induced on M_2' does not depend on the choice of normalization.

Step 4. In this step we prove that $\{\phi \circ \eta \circ \phi^{-1} \mid \eta \in H/W\}$ acts primitively on M_2.

The group $\widehat{\Gamma} \subset \mathrm{GL}(3, \mathbb{C})$ is finite with image $\langle g_1, g_2, g_3\rangle$ in $\mathrm{PGL}(3, \mathbb{C})$. By Lemma 4.2, the subgroup of $\mathrm{GL}(3, \mathbb{C})$ consisting of the matrices $\lambda(\widehat{\gamma})\widehat{\gamma}$ for $\widehat{\gamma} \in \widehat{\Gamma}$ is primitive. By (4.3), in the basis $[a_1], [a_2], [a_3]$ of M_2', the matrices $\lambda(\widehat{\gamma})\widehat{\gamma}$ for $\widehat{\gamma} \in \widehat{\Gamma}$ form a subgroup of $\mathrm{GL}(M_2')$ whose image in $\mathrm{PGL}(M_2')$ is $\{[\sigma \circ \eta \circ \sigma^{-1}]_2 \mid \eta \in H/W\}$. Hence $\{\sigma \circ \eta \circ \sigma^{-1} \mid \eta \in H/W\}$ acts primitively on M_2' by the last sentence of Definition-Lemma 2.10.

Now let ϕ be as in the statement, and let $\beta := \phi \circ \sigma^{-1}$, an isomorphism from $\operatorname{Proj}\mathcal{S}$ onto $\mathbb{P}(1, 1, 1, 2, 2, 2, 3)$. By Definition-Lemma 2.10(2),

$$[\phi \circ \eta \circ \phi^{-1}]_2 = [\beta]_2 \circ [\sigma \circ \eta \circ \sigma^{-1}]_2 \circ [\beta]_2^{-1}$$

for any $\eta \in H/W$. Let $b : M_2' \to M_2$ be a linear isomorphism representing $[\beta]_2$. Since conjugation by b preserves determinants, it maps the inverse image in $\mathrm{SL}(M_2')$ of the right-hand group onto the inverse image in $\mathrm{SL}(M_2)$ of the left-hand group, and conjugation by b preserves irreducibility

and maps systems of imprimitivity to systems of imprimitivity. Hence $\{\phi \circ \eta \circ \phi^{-1} \mid \eta \in H/W\}$ acts primitively on M_2. □

We now prove that B is not a finite quotient of $\mathbb{P}^6$.

Theorem 4.8. *The variety B is not isomorphic to $\mathbb{P}^6/G$ for any finite subgroup $G \subset \mathrm{PGL}(7,\mathbb{C})$.*

Proof. Suppose that there exist a finite subgroup $G \subset \mathrm{PGL}(7,\mathbb{C})$ and an isomorphism

$$f : \mathbb{P}^6/G \to B.$$

We apply Set-up 3.1 with $X = A$ and the group H: A is an abelian variety and $H \subset \mathrm{Aut}(A)$ is finite (Construction-Proposition 4.3(3)), and $H_0 = W$ by Proposition 4.5. By Proposition 3.3 applied to $d = 6$ and the isomorphism f, there exist a normal subgroup $J \subset G$ and an isomorphism $\psi : \mathbb{P}^6/J \to A/W$ such that, for any $g \in G$, $\psi \circ \bar{g} \circ \psi^{-1}$ is the automorphism of A/W induced by the image of gJ under the group isomorphism $G/J \to H/W$ of Proposition 3.3, where $\bar{g}$ is the automorphism of $\mathbb{P}^6/J$ induced by g. By Theorem 4.4, there exists an isomorphism $\phi_0 : A/W \to \mathbb{P}(1,1,1,2,2,2,3)$. Since three of the weights of $\mathbb{P}(1,1,1,2,2,2,3)$ are 1, the greatest common divisor of any six of the weights is 1, so $\mathbb{P}(1,1,1,2,2,2,3)$ is well-formed and $\mathbb{P}^6/J$ is isomorphic to a well-formed weighted projective space via $\phi_0 \circ \psi$. By Lemma 3.5 with $d = 6$, there exist a finite complex reflection group $\widehat{J} \subset \mathrm{GL}(7,\mathbb{C})$ whose image in $\mathrm{PGL}(7,\mathbb{C})$ is J, and a finite subgroup $\widehat{G} \subset \mathrm{GL}(7,\mathbb{C})$ whose image in $\mathrm{PGL}(7,\mathbb{C})$ is G, with $\widehat{J}$ normal in $\widehat{G}$.

Let M_2 be the space of generators of degree 2 of $\mathbb{P}(1,1,1,2,2,2,3)$. By Proposition 4.7, $\dim M_2 = 3$. By Lemma 3.7(2) applied to $\widehat{J}$, $\widehat{G}$, the weighted projective space $\mathbb{P}(1,1,1,2,2,2,3)$, the isomorphism $\phi_0 \circ \psi$, and $m = 2$, there exists a decomposition

$$M_2 = M_{2,1} \oplus \cdots \oplus M_{2,s}$$

with $\dim M_{2,j} \in \{1,2\}$ for any j, such that for any $g \in G$ and any $1 \le j \le s$, we have

$$[\phi_0 \circ \psi \circ \bar{g} \circ \psi^{-1} \circ \phi_0^{-1}]_2(M_{2,j}) = M_{2,j'}$$

for some $1 \le j' \le s$. Since the composition $G \to G/J \to H/W$ is surjective, $\psi \circ \bar{g} \circ \psi^{-1}$ runs over H/W as g runs over G. Hence $\{[\phi_0 \circ h \circ \phi_0^{-1}]_2 \mid h \in H/W\}$ permutes $M_{2,1}, \ldots, M_{2,s}$. Since $\dim M_{2,j} \le 2$ for any j and

$$\sum_{j=1}^{s} \dim M_{2,j} = \dim M_2 = 3,$$

we have $s \ge 2$. Scalars preserve each $M_{2,j}$, so the inverse image of $\{[\phi_0 \circ h \circ \phi_0^{-1}]_2 \mid h \in H/W\}$ in $\mathrm{SL}(M_2)$ also permutes $M_{2,1}, \ldots, M_{2,s}$. Thus $M_2 = M_{2,1} \oplus \cdots \oplus M_{2,s}$ is a system of imprimitivity of the inverse image in the sense of Definition 2.7, which contradicts Proposition 4.7 with $\phi = \phi_0$. □

5. Proof of the main theorem

In this section, we prove Theorem 1.2. The variety B of Section 4 is the base of a Lagrangian fibration on a terminalization of a symplectic quotient of dimension 12. The definition of irreducible symplectic varieties involves all finite quasi-étale covers. By the purity of the branch locus, we reduce the irreducibility to the fundamental group of the regular locus of the terminalization (Subsection 5.2) and to invariant holomorphic forms on an abelian variety of dimension 12 (Subsection 5.3).

5.1. **A symplectic quotient of dimension twelve.** In this subsection, we construct a symplectic variety Z of dimension 12 with a surjective morphism onto B, and we fix a terminalization of Z.

As explained in the sketch of the proof in the introduction, H preserves no non-zero holomorphic 2-form on A, so we consider $A \times A$, where the second factor carries the contragredient action of W and τ_* with respect to $(\cdot,\cdot)$, and the translations act only on the first factor.

Construction-Proposition 5.1. Let $Y := A \times A$, and let $\widetilde{H} \subset \mathrm{Aut}(Y)$ be the subgroup generated by the automorphisms

$$(x, y) \to (wx, wy), \quad (x, y) \to (x + \iota_u, y), \quad (x, y) \to (\tau_* x, \tau_*^{-1} y)$$

for $w \in W$ and $u \in E[3]$. Fix a non-zero tangent vector of E at o, which induces an isomorphism

$$T_0 A = \mathrm{Hom}_{\mathbb{Z}}(P, T_o E) \cong Q \otimes_{\mathbb{Z}} \mathbb{C},$$

and let σ_Y be the translation-invariant holomorphic 2-form on Y whose value at the origin is

$$\sigma_Y((v_1, v_2), (v_1', v_2')) = (v_1, v_2') - (v_1', v_2), \quad v_1, v_2, v_1', v_2' \in Q \otimes_{\mathbb{Z}} \mathbb{C}.$$

Let $Z := Y/\widetilde{H}$. Then the following hold.

(1) σ_Y is a symplectic form on Y, $\widetilde{H}$ is finite, and any element of $\widetilde{H}$ preserves σ_Y.
(2) The first projection $Y \to A$ induces a group isomorphism $\widetilde{H} \to H$ and a surjective morphism $g : Z \to B$.
(3) Z is a normal projective variety, and σ_Y descends to a symplectic form σ_Z on Z such that (Z, σ_Z) is a symplectic variety.

Proof. By Construction-Proposition 4.1(2), τ is a group automorphism of E of order 4 with $\tau^2 = -\mathrm{id}_E$. Let $\lambda \in \mathbb{C}$ be the derivative of τ at o. Then $\lambda^2 = -1$, so λ is a primitive fourth root of unity.

We first prove (2). For $h_1, h_2 \in \mathrm{Aut}(A)$, let $h_1 \times h_2 \in \mathrm{Aut}(Y)$ be the automorphism sending (x, y) to $(h_1 x, h_2 y)$, and for $u \in E[3]$, let $t_u \in \mathrm{Aut}(A)$ be the translation by ι_u, which is the image of the translation of E by u under the homomorphism $\Gamma \to H$ of Construction-Proposition 4.3(3). Then $\widetilde{H}$ is generated by $w \times w$, $t_u \times \mathrm{id}_A$, and $\tau_* \times \tau_*^{-1}$ for $w \in W$ and $u \in E[3]$. By Construction-Proposition 4.1(3), $\Gamma \cong E[3] \rtimes \langle \tau \rangle$. Since $\tau_*^4 = \mathrm{id}_A$, there exists a homomorphism $\psi : \Gamma \to \mathrm{Aut}(A)$ that is trivial on the translations of E by $E[3]$ and sends τ to τ_*^{-1}. For any $w \in W$ and $\chi \in A$,

$$w(\tau_* \chi) = \tau \circ \chi \circ w^{-1} = \tau_*(w\chi),$$

so τ_* commutes with W. Thus $(w, \gamma) \to w\psi(\gamma)$ is a homomorphism $W \times \Gamma \to \mathrm{Aut}(A)$. Composing it with the inverse of the isomorphism $W \times \Gamma \to H$ in Construction-Proposition 4.3(3), we obtain a homomorphism $\phi : H \to \mathrm{Aut}(A)$ such that $\phi(w) = w$, $\phi(t_u) = \mathrm{id}_A$, and $\phi(\tau_*) = \tau_*^{-1}$ for any $w \in W$ and $u \in E[3]$. Since H is generated by W, the translations t_u, and τ_*, the subgroup $\{h \times \phi(h) \mid h \in H\}$ of $\mathrm{Aut}(Y)$ is generated by the generators of $\widetilde{H}$. Hence

$$\widetilde{H} = \{h \times \phi(h) \mid h \in H\},$$

and $h \times \phi(h) \to h$ is a group isomorphism $\widetilde{H} \to H$. Let $\mathrm{pr}_1 : Y \to A$ be the first projection. Then for any $h \in H$,

$$\mathrm{pr}_1 \circ (h \times \phi(h)) = h \circ \mathrm{pr}_1,$$

so the composition of pr_1 with the quotient morphism $A \to B$ is $\widetilde{H}$-invariant. Since Z is the topological quotient of Y by $\widetilde{H}$ and $\mathcal{O}_Z$ is the sheaf of $\widetilde{H}$-invariant sections of the pushforward of $\mathcal{O}_Y$ [Mum74, §7, Theorem on p. 66 and Remark on p. 69], any $\widetilde{H}$-invariant morphism from Y to a variety factors uniquely through Z. Thus it factors through a morphism $g : Z \to B$, and g is surjective since pr_1 and $A \to B$ are surjective. This proves (2).

We prove (1). Since H is finite (Construction-Proposition 4.3(3)), $\widetilde{H}$ is finite by (2). Through $T_0 A \cong Q \otimes_{\mathbb{Z}} \mathbb{C}$, the derivative at the origin of $w \in W$ is $w \otimes 1$ (Construction-Proposition 4.3(1)), and we write it as w. Since $T_0 A = \mathrm{Hom}_{\mathbb{Z}}(P, T_o E)$ and $\tau_* \chi = \tau \circ \chi$, the derivative of τ_* at the origin is the multiplication by λ. Hence $w \times w$ and $\tau_* \times \tau_*^{-1}$ are group automorphisms of Y whose derivatives at the origin are $(v_1, v_2) \to (wv_1, wv_2)$ and $(v_1, v_2) \to (\lambda v_1, \lambda^{-1} v_2)$, and $t_u \times \mathrm{id}_A$ is a translation. Since $(\cdot, \cdot)$ is W-invariant, for any $v_1, v_2, v_1', v_2' \in Q \otimes_{\mathbb{Z}} \mathbb{C}$,

$$\sigma_Y((wv_1, wv_2), (wv_1', wv_2')) = \sigma_Y((v_1, v_2), (v_1', v_2')) = \sigma_Y((\lambda v_1, \lambda^{-1} v_2), (\lambda v_1', \lambda^{-1} v_2')).$$

Since σ_Y is translation-invariant, the generators of $\widetilde{H}$ preserve σ_Y, hence any element of $\widetilde{H}$ preserves σ_Y. Let $\alpha_1, \dots, \alpha_6$ be a basis of Q consisting of simple roots, and let $z_1, \dots, z_6, z'_1, \dots, z'_6$ be the linear coordinates on $(Q \otimes_{\mathbb{Z}} \mathbb{C})^{\oplus 2}$ such that $v_1 = \sum_i z_i \alpha_i$ and $v_2 = \sum_j z'_j \alpha_j$. Then

$$\sigma_Y = \sum_{i,j=1}^{6} (\alpha_i, \alpha_j) dz_i \wedge dz'_j.$$

The coefficients are constant, so σ_Y is closed. The matrix $((\alpha_i, \alpha_j))_{i,j}$ is the Gram matrix of the positive definite form $(\cdot, \cdot)$, hence invertible, so σ_Y is non-degenerate at any point of Y. Thus σ_Y is a symplectic form on Y.

We prove (3). Since Y is a normal projective variety, the quotient morphism $Y \to Z$ is finite and surjective, and $\mathcal{O}_Z$ is the sheaf of $\widetilde{H}$-invariant sections of the pushforward of $\mathcal{O}_Y$ [Mum74, §7, Theorem on p. 66 and Remark on p. 69]. Since the ring of invariants of a finite group acting on a normal domain is normal, Z is normal. By [Har92, Lecture 10, pp. 126–127], Z is projective. By (1), σ_Y is $\widetilde{H}$-invariant. Thus σ_Y descends to a symplectic form σ_Z on Z, and (Z, σ_Z) is a symplectic variety [Bea00, Proposition 2.4]. □

By [BGMM25, Proposition 3.3], Z has Gorenstein canonical singularities, so a projective $\mathbb{Q}$-factorial terminalization of Z exists [BCHM10, Corollary 1.4.3]. By Lemma 2.5(3), the pullback of σ_Z to such a terminalization extends to a symplectic form, and the extension is unique, since the inverse image of Z_{reg} is dense in the regular locus of the terminalization.

Set-up 5.2. Let X, μ, σ_X, and f be as follows.

(1) $\mu : X \to Z$ is a projective $\mathbb{Q}$-factorial terminalization of the variety Z in Construction-Proposition 5.1.
(2) σ_X is the symplectic form on X such that $\sigma_X = \mu^* \sigma_Z$ on $\mu^{-1}(Z_{\mathrm{reg}})$.
(3) $f := g \circ \mu : X \to B$.

5.2. **Fundamental groups.** In this subsection, we compare the fundamental group of the locus where $\widetilde{H}$ acts freely with $\pi_1(X_{\mathrm{reg}})$.

Construction 5.3. Notation as in Set-up 5.2. Let $\Lambda_E \subset T_o E$ be the period lattice of E, so that $E = T_o E / \Lambda_E$. Let

$$V := (Q \otimes_{\mathbb{Z}} \mathbb{C})^{\oplus 2}, \quad \Lambda_Y := (Q \otimes_{\mathbb{Z}} \Lambda_E)^{\oplus 2} \subset V,$$

so that $Y = V / \Lambda_Y$ through the isomorphism $T_o E \cong \mathbb{C}$ in Construction-Proposition 5.1. Let $\pi_V : V \to Y$ be the quotient map, and let σ_0 be the value of σ_Y at the origin, an alternating bilinear form on $V = T_0 Y$. Let Π be the group of affine automorphisms of V that lift elements of $\widetilde{H}$, and let $\Pi_0 \subset \Pi$ be the subgroup generated by the translations by elements of Λ_Y and the linear automorphisms $(v_1, v_2) \to (wv_1, wv_2)$ for $w \in W$. Let $Y^\circ \subset Y$ be the open subset of points whose stabilizer in $\widetilde{H}$ is trivial, let $V^\circ \subset V$ be the inverse image of Y°, and let $U := Y^\circ / \widetilde{H}$.

Lemma 5.4. *Notation as in Construction 5.3. Then the following hold.*

(1) $\mathrm{codim}_Y(Y \backslash Y^\circ) \geq 2$, *and* V° *is simply connected.*
(2) U *is an open subset of* Z_{reg}, μ *is an isomorphism over* U, *and the covering* $V^\circ \to U$ *induces an isomorphism* $\pi_1(U) \cong \Pi$.
(3) The homomorphism $\pi_1(U) \to \pi_1(X_{\mathrm{reg}})$ *induced by the open immersion* $\mu^{-1}(U) \to X_{\mathrm{reg}}$ *is surjective.*
(4) Any element of $\widetilde{H}$ *lifts to an element of* Π, *the elements of* Π *lifting* id_Y *are exactly the translations by elements of* Λ_Y, *and* Π *acts properly discontinuously on* V.

Proof. We first prove (4). The generators of $\widetilde{H}$ are compositions of translations with group automorphisms of Y, hence so is any element of $\widetilde{H}$, and any element of $\widetilde{H}$ lifts to an affine automorphism of V. Thus the projection $\Pi \to \widetilde{H}$ is surjective, and the kernel of $\Pi \to \widetilde{H}$ is the group of translations by elements of Λ_Y. In particular, Π acts properly discontinuously on V,

since Λ_Y does and has finite index in Π. For $\tilde{h} \in \Pi$, let $\ell(\tilde{h}) \in \mathrm{GL}(V)$ be the linear part of $\tilde{h}$. Then $\ell : \Pi \to \mathrm{GL}(V)$ is a homomorphism that is trivial on Λ_Y, so $\ell(\tilde{h})$ has finite order. Moreover, $\ell(\tilde{h})$ preserves σ_0 by Construction-Proposition 5.1(1).

(1) Let $\tilde{h} \in \Pi$ be an element that is not a translation by an element of Λ_Y, and assume that $\tilde{h}(v) = v$ for some $v \in V$. In linear coordinates centered at v, $\tilde{h} = \ell(\tilde{h})$, and $\ell(\tilde{h}) \neq \mathrm{id}_V$. Since $\ell(\tilde{h})$ has finite order, it is diagonalizable. Let $V^{\tilde{h}}$ be the fixed subspace of $\ell(\tilde{h})$, and let V' be the sum of the eigenspaces of $\ell(\tilde{h})$ with eigenvalues different from 1. For $x \in V^{\tilde{h}}$ and an eigenvector $x' \in V'$ with eigenvalue $c \neq 1$,

$$\sigma_0(x, x') = \sigma_0(\ell(\tilde{h})x, \ell(\tilde{h})x') = c\sigma_0(x, x'),$$

so $\sigma_0(x, x') = 0$. Since $V = V^{\tilde{h}} \oplus V'$ and σ_0 is non-degenerate by Construction-Proposition 5.1(1), $\sigma_0|_{V'}$ is non-degenerate. Thus $\dim V'$ is even, and $\dim V' \geq 2$ since $\ell(\tilde{h}) \neq \mathrm{id}_V$. Hence the fixed locus $\mathrm{Fix}(\tilde{h}) = v + V^{\tilde{h}}$ is an affine subspace of V of codimension ≥ 2. A point $v \in V$ is not in V° if and only if some non-trivial element of $\widetilde{H}$ fixes $\pi_V(v)$, if and only if $\tilde{h}(v) = v$ for some $\tilde{h} \in \Pi$ that is not a translation by an element of Λ_Y. Thus $V \backslash V^\circ$ is the union of the fixed loci $\mathrm{Fix}(\tilde{h})$ of such elements $\tilde{h}$, which is a locally finite union of affine subspaces of codimension ≥ 2, as Π acts properly discontinuously on V. Since π_V is a local biholomorphism, $\mathrm{codim}_Y(Y \backslash Y^\circ) \geq 2$. Moreover, $V \backslash V^\circ$ is a locally finite union of real submanifolds of real codimension ≥ 4 in $V \cong \mathbb{R}^{24}$. By general position, any path in V connecting two points of V°, and any continuous map from a closed disk to V that maps the boundary circle into V°, can be deformed into V° relative to the boundary, since $4 > 2$. As V is contractible, V° is connected and simply connected.

(2) Since $\widetilde{H}$ acts freely on the $\widetilde{H}$-stable open subset Y°, U is an open subset of Z and the quotient morphism $Y^\circ \to U$ is étale [Mum74, §7, Theorem on p. 66 and Remark on p. 69]. Hence U is smooth, so $U \subset Z_{\mathrm{reg}}$, and μ is an isomorphism over U by Lemma 2.5(1). The group Π preserves V°. If $\tilde{h} \in \Pi$ fixes a point $v \in V^\circ$, then the image of $\tilde{h}$ in $\widetilde{H}$ fixes $\pi_V(v) \in Y^\circ$, so $\tilde{h}$ is a translation by an element of Λ_Y fixing v, and $\tilde{h} = \mathrm{id}_V$. Thus Π acts freely and properly discontinuously on V°, and $V^\circ/\Pi = Y^\circ/\widetilde{H} = U$. Therefore $V^\circ \to U$ is a covering with deck transformation group Π, and by (1), it is a universal covering. Thus $\pi_1(U) \cong \Pi$ after the choice of a base point of V°.

(3) Since X is an irreducible normal variety, X_{reg} is a connected complex manifold [Gro71, Exposé XII, Proposition 2.4], and $\mu^{-1}(U)$ is a non-empty Zariski open subset of X_{reg}. By [Kol95, Proposition 2.10.1], the homomorphism $\pi_1(\mu^{-1}(U)) \to \pi_1(X_{\mathrm{reg}})$ is surjective. Since μ restricts to an isomorphism $\mu^{-1}(U) \to U$ by (2), (3) follows. □

Since Y is not simply connected, [BGMM25, Proposition 8.1] does not apply to Z, and we show directly that the terminalization kills the period lattice $\Lambda_Y \cong \mathbb{Z}^{24}$ in $\pi_1(U) \cong \Pi$ (Lemma 5.4) as well as the diagonal W.

Proposition 5.5. *Notation as in Construction 5.3. Then the kernel of the homomorphism $\Pi \cong \pi_1(U) \to \pi_1(X_{\mathrm{reg}})$, induced by the covering $V^\circ \to U$ and the open immersion $\mu^{-1}(U) \to X_{\mathrm{reg}}$, contains Π_0.*

Proof. Since V° is connected, the kernel does not depend on the base point of V° in Lemma 5.4(2): for a path γ in V° from a base point p to a base point p', the change of base point along the image of γ in U commutes with the isomorphisms $\pi_1(U) \cong \Pi$ given by p and p', and the change of base point along the image of γ in X_{reg} commutes with the homomorphisms $\pi_1(U) \to \pi_1(X_{\mathrm{reg}})$ for the base points given by p and p'.

Let $\alpha \in Q$ be an element with $(\alpha, \alpha) = 2$, and let $\omega \in \Lambda_E$. We regard ω as a complex number through $T_o E \cong \mathbb{C}$, so that

$$\alpha \otimes \omega \in Q \otimes_{\mathbb{Z}} \Lambda_E \subset Q \otimes_{\mathbb{Z}} \mathbb{C}.$$

On $Q \otimes_{\mathbb{Z}} \mathbb{C}$, $s_\alpha(v) = v - (v,\alpha)\alpha$, so $s_\alpha(\alpha \otimes \omega) = -\alpha \otimes \omega$. Let $h_\alpha \in \widetilde{H}$ be the automorphism $(x,y) \to (s_\alpha x, s_\alpha y)$ of Y, and let r_0, r_1, r_2 be the affine automorphisms of V given by

$$r_0(v_1,v_2) = (s_\alpha v_1, s_\alpha v_2), \quad r_1(v_1,v_2) = (s_\alpha v_1 + \alpha \otimes \omega, s_\alpha v_2), \quad r_2(v_1,v_2) = (s_\alpha v_1, s_\alpha v_2 + \alpha \otimes \omega).$$

By Construction-Proposition 4.3(1), r_0, r_1, and r_2 lift h_α, so they belong to Π. Since $s_\alpha(\alpha \otimes \omega) = -\alpha \otimes \omega$, they are involutions. Let

$$V_\alpha := \{(v_1,v_2) \in V \mid (\alpha, v_1) = (\alpha, v_2) = 0\}.$$

Then V_α is the fixed subspace of r_0, and

$$\mathrm{Fix}(r_1) = \{(\alpha,v_1) = \omega,\ (\alpha,v_2) = 0\}, \quad \mathrm{Fix}(r_2) = \{(\alpha,v_1) = 0,\ (\alpha,v_2) = \omega\}.$$

Thus $\mathrm{Fix}(r_i)$ is a translate of V_α for $i = 0,1,2$, and $\mathrm{codim}_V \mathrm{Fix}(r_i) = 2$.

Fix $i \in \{0,1,2\}$. We show that r_i belongs to the kernel. Any lift of h_α to V has linear part r_0. If $h_\alpha(y') = y'$ for some $y' \in Y$, and $\tilde{h} \in \Pi$ is the lift of h_α fixing a point $v' \in \pi_V^{-1}(y')$, then near y', $\mathrm{Fix}(h_\alpha)$ is the image of $\mathrm{Fix}(\tilde{h}) = v' + V_\alpha$, because $\tilde{h}(x) - x \in \Lambda_Y$ implies $\tilde{h}(x) = x$ for x close to v'. Hence $\mathrm{Fix}(h_\alpha)$ is a closed subvariety of Y of pure dimension 10. Since $\mathrm{Fix}(r_i)$ is irreducible and $\mathrm{Fix}(h_\alpha)$ has finitely many irreducible components, there exists an irreducible component F of $\mathrm{Fix}(h_\alpha)$ such that $\pi_V(\mathrm{Fix}(r_i)) \subset F$. Let $D \subset Y$ be the union of the irreducible components of $\mathrm{Fix}(h)$, for $h \in \widetilde{H} \backslash \{\mathrm{id}_Y\}$, that do not contain F. Then $\dim(D \cap F) \le 9$. Since π_V is a local biholomorphism, $\pi_V^{-1}(D \cap F)$ is an analytic subset of V of dimension ≤ 9, and $\mathrm{Fix}(r_i) \cap \pi_V^{-1}(D) = \mathrm{Fix}(r_i) \cap \pi_V^{-1}(D \cap F)$. Since $\dim \mathrm{Fix}(r_i) = 10$, there exists a point $v \in \mathrm{Fix}(r_i)$ such that $y := \pi_V(v) \notin D$. Let z be the image of y in Z, let $G_y \subset \widetilde{H}$ be the stabilizer of y, and let $\Pi_v \subset \Pi$ be the stabilizer of v. Then $r_i \in \Pi_v$, and the projection $\Pi \to \widetilde{H}$ restricts to an isomorphism $\Pi_v \to G_y$, since any lift of an element of G_y maps v into $v + \Lambda_Y$, and for $\tilde{h}_1, \tilde{h}_2 \in \Pi_v$ whose images in G_y are equal, $\tilde{h}_1^{-1}\tilde{h}_2$ is a translation by an element of Λ_Y fixing v, hence $\tilde{h}_1 = \tilde{h}_2$.

We describe the action of Π_v near v. Let $\tilde{h} \in \Pi_v$, and let $h \in G_y$ be the image of $\tilde{h}$. Since $y \in \mathrm{Fix}(h)$ and $y \notin D$, any irreducible component of $\mathrm{Fix}(h)$ containing y contains F, so $F \subset \mathrm{Fix}(h)$. For $x \in \mathrm{Fix}(r_i)$, $\pi_V(\tilde{h}(x)) = h(\pi_V(x)) = \pi_V(x)$, so $\tilde{h}(x) - x \in \Lambda_Y$, and $\tilde{h}(x) = x$ for x close to v. Since $\tilde{h}$ is affine, $\tilde{h}$ fixes $\mathrm{Fix}(r_i) = v + V_\alpha$ pointwise. Hence, in linear coordinates centered at v, Π_v is a finite group of linear automorphisms of V that fix V_α pointwise and preserve σ_0. Let

$$V_\alpha^\perp := \{(a\alpha, b\alpha) \mid a,b \in \mathbb{C}\} \subset V.$$

For $(v_1,v_2) \in V$ and $a,b \in \mathbb{C}$,

$$\sigma_0((v_1,v_2),(a\alpha,b\alpha)) = b(v_1,\alpha) - a(\alpha,v_2),$$

so $V_\alpha^\perp$ is the σ_0-orthogonal complement of V_α, $V = V_\alpha \oplus V_\alpha^\perp$, and $\sigma_0|_{V_\alpha^\perp}$ is non-degenerate. Thus Π_v preserves $V_\alpha^\perp$ and acts on $V_\alpha^\perp$ faithfully, preserving $\sigma_0|_{V_\alpha^\perp}$. In particular, Π_v is isomorphic to a finite subgroup of $\mathrm{SL}(V_\alpha^\perp)$, which is isomorphic to $\mathrm{SL}(2,\mathbb{C})$, and a non-trivial element of Π_v has eigenvalues c and c^{-1} on $V_\alpha^\perp$ with $c \ne 1$, so it fixes only 0 in $V_\alpha^\perp$.

Let $\bar{B} \subset V$ be a closed ball centered at v. Since Π acts properly discontinuously on V (Lemma 5.4(4)), the set of $\tilde{h} \in \Pi$ with $\tilde{h}(\bar{B}) \cap \bar{B} \ne \emptyset$ is finite, and $\tilde{h}(v) \ne v$ for any such $\tilde{h} \notin \Pi_v$. Thus for any sufficiently small open neighborhood Ω_v of v of the form $\Omega_v = (v + \Omega_1) + \Omega_2$, where $\Omega_1 \subset V_\alpha$ and $\Omega_2 \subset V_\alpha^\perp$ are open balls centered at 0 with respect to Π_v-invariant Hermitian norms, if $\tilde{h} \in \Pi$ and $\tilde{h}(\Omega_v) \cap \Omega_v \ne \emptyset$, then $\tilde{h} \in \Pi_v$. Moreover, Ω_v is Π_v-stable, since Π_v fixes V_α pointwise and preserves $V_\alpha^\perp$ in linear coordinates centered at v.

The quotient morphism $Y \to Z$ is finite, and the fibers of $Y \to Z$ are the $\widetilde{H}$-orbits [Mum74, §7, Theorem on p. 66 and Remark on p. 69], so the natural map $V/\Pi \to Z$ is a homeomorphism for the classical topologies, and it is biholomorphic over U, since $Y^\circ \to U$ is étale. Since Z is normal, the associated analytic space is normal [Gro71, Exposé XII, Proposition 2.1], so a function on an open subset N of Z whose pullback to V is holomorphic is continuous on N and holomorphic on $N \cap U$, hence holomorphic on N by the Riemann extension theorem on normal

complex spaces [GR84, Chapter 7, §4.2], since $N\backslash U$ is a nowhere dense analytic subset of N and the function is locally bounded. Thus the analytic space associated with Z is the quotient V/Π of [Car57, Théorème 4], and the image of Ω_v in Z is an open neighborhood of z isomorphic to Ω_v/Π_v. Hence the analytic germ $(Z \ni z)$ is isomorphic to $(\mathbb{C}^{10} \times (\mathbb{C}^2/\Pi_v) \ni (0, \bar{0}))$, where Π_v acts on $\mathbb{C}^2 \cong V_\alpha^\perp$ through the linear action of Π_v on $V_\alpha^\perp$ in linear coordinates centered at v. By Lemma 2.5(2), there exists an analytic open neighborhood N_z of z such that $\mu^{-1}(N_z)$ is a simply connected complex manifold. In particular, $\mu^{-1}(N_z) \subset X_{\mathrm{reg}}$. Possibly shrinking Ω_1 and Ω_2, we may assume that the image of Ω_v in Z is contained in N_z.

Let $\Omega_v^\circ := \Omega_v \cap V^\circ$. By the choice of Ω_v, a point of Ω_v has non-trivial stabilizer in Π if and only if it has non-trivial stabilizer in Π_v, and by the previous paragraph,

$$\Omega_v^\circ = (v + \Omega_1) + (\Omega_2\backslash\{0\}).$$

Since $\Omega_2\backslash\{0\}$ is a punctured ball in $\mathbb{C}^2$, Ω_v° is simply connected. Let $U_v \subset U$ be the image of Ω_v°. Then $U_v = \Omega_v^\circ/\Pi_v$, and Ω_v° is a connected component of the inverse image of U_v in V°. Taking the base point of V° in Ω_v°, the image of $\pi_1(U_v) \to \pi_1(U) \cong \Pi$ is Π_v. Since $\mu^{-1}(U_v) \subset \mu^{-1}(N_z)$ and $\mu^{-1}(N_z)$ is simply connected, the homomorphism $\pi_1(\mu^{-1}(U_v)) \to \pi_1(X_{\mathrm{reg}})$ is trivial. Thus Π_v is contained in the kernel. In particular, r_i belongs to the kernel.

Hence r_0, r_1, and r_2 belong to the kernel for any $\alpha \in Q$ with $(\alpha, \alpha) = 2$ and any $\omega \in \Lambda_E$. Since $s_\alpha^2 = \mathrm{id}_{Q \otimes_{\mathbb{Z}} \mathbb{C}}$, $r_1 r_0$ and $r_2 r_0$ are the translations by $(\alpha \otimes \omega, 0)$ and $(0, \alpha \otimes \omega)$, respectively. By Construction-Proposition 4.3(1), Q is the root lattice of type E_6, so Q is generated by the elements α with $(\alpha, \alpha) = 2$. Thus the elements $\alpha \otimes \omega$ generate $Q \otimes_{\mathbb{Z}} \Lambda_E$, and the translations by $(\alpha \otimes \omega, 0)$ and $(0, \alpha \otimes \omega)$ generate the group of translations by elements of Λ_Y. By definition, W is generated by the reflections s_α, so the elements r_0 for all α generate the group of linear automorphisms $(v_1, v_2) \to (wv_1, wv_2)$ for $w \in W$. Therefore Π_0 is contained in the kernel. □

5.3. **Reflexive forms on quasi-étale covers.** In this subsection, we prove that (X, σ_X) is an irreducible symplectic variety. As explained at the end of the sketch of the proof in the introduction, by Proposition 5.5 the proof reduces to determining the holomorphic forms on Y invariant under $(x, y) \to (wx, wy)$ for any $w \in W$ (Lemma 5.6).

Lemma 5.6. *Notation as in Construction-Proposition 5.1. Let W act on Y by $(x, y) \to (wx, wy)$. Then*

$$\bigoplus_{k \geq 0} H^0(Y, \Omega_Y^k)^W = \mathbb{C}[\sigma_Y],$$

and $\sigma_Y^i \neq 0$ for any $0 \leq i \leq 6$.

Proof. Since Y is an abelian variety, Ω_Y^1 is freely generated by translation-invariant 1-forms [Mum74, §4, (iii) on p. 42], so for any $k \geq 0$, $\Omega_Y^k = \bigwedge^k \Omega_Y^1$ is freely generated by translation-invariant k-forms. Since regular functions on Y are constant [Har77, Chapter I, Theorem 3.4(a)], any global k-form on Y is translation-invariant. Thus evaluation at the origin 0 of Y is an isomorphism

$$H^0(Y, \Omega_Y^k) \to \bigwedge^k T_0^* Y.$$

Since W acts on Y by group automorphisms, the isomorphism is W-equivariant, where $w \in W$ acts on forms by pullback along w^{-1} and on $T_0^* Y$ by the contragredient of the tangent map of w at 0.

Let $Q_{\mathbb{C}} := Q \otimes_{\mathbb{Z}} \mathbb{C}$. By Construction-Proposition 4.3(1), under the isomorphism $T_0 A \cong Q_{\mathbb{C}}$ in Construction-Proposition 5.1, the tangent map at 0 of $w \in W$ acting on A is the action of w on $Q_{\mathbb{C}}$. Thus $T_0 Y = Q_{\mathbb{C}} \oplus Q_{\mathbb{C}}$, with W acting diagonally. The form $(\cdot, \cdot)$ on $Q_{\mathbb{C}}$ is non-degenerate and W-invariant, so $Q_{\mathbb{C}} \cong Q_{\mathbb{C}}^*$ as W-modules. Moreover, for any $0 \leq i \leq 6$, the bilinear form

$$(v_1 \wedge \cdots \wedge v_i, v_1' \wedge \cdots \wedge v_i') \to \det\left((v_a, v_b')\right)_{1 \leq a, b \leq i}$$

on $\bigwedge^i Q_{\mathbb{C}}$ is non-degenerate and W-invariant, so $\bigwedge^i Q_{\mathbb{C}} \cong (\bigwedge^i Q_{\mathbb{C}})^*$ as W-modules. Therefore, for any $k \geq 0$,

$$H^0(Y, \Omega_Y^k)^W \cong \Big(\bigwedge^k (Q_{\mathbb{C}} \oplus Q_{\mathbb{C}})\Big)^W \cong \bigoplus_{\substack{i+j=k \\ 0\leq i,j\leq 6}} \Big(\bigwedge^i Q_{\mathbb{C}} \otimes \bigwedge^j Q_{\mathbb{C}}\Big)^W$$
$$\cong \bigoplus_{\substack{i+j=k \\ 0\leq i,j\leq 6}} \operatorname{Hom}_W\Big(\bigwedge^i Q_{\mathbb{C}}, \bigwedge^j Q_{\mathbb{C}}\Big).$$

By Construction-Proposition 4.3(1), $(Q, (\cdot,\cdot))$ is the root lattice of type E_6 with Weyl group W. Since the root system E_6 is irreducible, $Q \otimes_{\mathbb{Z}} \mathbb{R}$ is an absolutely simple W-module [Bou02, Chapter VI, §1, no. 2, Corollary of Proposition 5], that is, $Q_{\mathbb{C}}$ is a simple W-module. Thus, by a theorem of Steinberg [Ste68, Section 14] (see also [Bou02, Chapter V, §2, Exercise 3(d)] and [Hu25, Theorem 1.2]), the W-modules $\bigwedge^i Q_{\mathbb{C}}$, $0 \leq i \leq 6$, are simple and pairwise non-isomorphic. By Schur's lemma, for any $0 \leq i, j \leq 6$,

$$\dim \operatorname{Hom}_W\Big(\bigwedge^i Q_{\mathbb{C}}, \bigwedge^j Q_{\mathbb{C}}\Big) = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{if } i \neq j. \end{cases}$$

Hence

$$\dim H^0(Y, \Omega_Y^{2i})^W = 1 \text{ for } 0 \leq i \leq 6, \quad \text{and} \quad H^0(Y, \Omega_Y^k)^W = 0 \text{ for odd } k \text{ or } k > 12.$$

By Construction-Proposition 5.1(1), σ_Y is a symplectic form on Y preserved by $\widetilde{H}$. Since the automorphisms $(x, y) \to (wx, wy)$ belong to $\widetilde{H}$, $\sigma_Y \in H^0(Y, \Omega_Y^2)^W$. Since σ_Y is non-degenerate at 0 and $\dim Y = 12$, $\sigma_Y^6 \neq 0$. Since $\sigma_Y^6 = \sigma_Y^i \wedge \sigma_Y^{6-i}$, $\sigma_Y^i \neq 0$ for any $0 \leq i \leq 6$. Thus σ_Y^i spans $H^0(Y, \Omega_Y^{2i})^W$ for any $0 \leq i \leq 6$, and Lemma 5.6 follows. □

Proposition 5.7. *Notation as in Set-up 5.2. Then (X, σ_X) is a projective irreducible symplectic variety of dimension 12.*

Proof. By Lemma 2.5(3), (X, σ_X) is a symplectic variety. Since μ and $Z \to \operatorname{Spec} \mathbb{C}$ are projective (Construction-Proposition 5.1(3)), X is projective [Har77, Chapter II, Exercise 4.9]. Since μ is birational and $\widetilde{H}$ is finite, $\dim X = \dim Z = \dim Y = 12$.

Let $q : X' \to X$ be a finite quasi-étale morphism from a normal variety X'. We show that

$$\bigoplus_{k\geq 0} H^0(X', \Omega_{X'}^{[k]}) = \mathbb{C}[q^{[*]}\sigma_X].$$

Let $T \subset X$ be a closed subset with $\operatorname{codim}_X T \geq 2$ such that q is étale over $X \backslash T$. Since q is finite, the set of points of $q^{-1}(X_{\mathrm{reg}})$ at which q is not étale is contained in $q^{-1}(T)$, so it is of codimension at least 2 in X'. Since X' is normal and X_{reg} is smooth, by purity of the branch locus [Gro71, Exposé X, Théorème 3.1], q is étale over X_{reg}. In particular, $q^{-1}(X_{\mathrm{reg}})$ is smooth, and $q^{-1}(X_{\mathrm{reg}}) \to X_{\mathrm{reg}}$ is a finite topological covering. Since $q^{-1}(X_{\mathrm{reg}})$ is a non-empty open subset of the variety X', it is connected [Gro71, Exposé XII, Proposition 2.4].

We use the notation of Construction 5.3. Let $U' := q^{-1}(\mu^{-1}(U))$. By Lemma 5.4(2), $\mu^{-1}(U) \subset X_{\mathrm{reg}}$ and μ induces an isomorphism $\mu^{-1}(U) \to U$, so $\mu \circ q$ induces a finite étale morphism $U' \to U$. By Lemma 5.4(4), the homomorphism $\Pi \to \widetilde{H}$ is surjective, and the kernel consists of the translations by elements of Λ_Y.

Take a base point $v_0 \in V^\circ$, take the image of v_0 as the base point of U, and take base points of X_{reg} and $q^{-1}(X_{\mathrm{reg}})$ compatibly. The connected covering $q^{-1}(X_{\mathrm{reg}}) \to X_{\mathrm{reg}}$ corresponds to a subgroup $\Sigma \subset \pi_1(X_{\mathrm{reg}})$ of finite index. Consider the isomorphism $\Pi \cong \pi_1(U)$ of Lemma 5.4(2) and the homomorphism $\pi_1(U) \to \pi_1(X_{\mathrm{reg}})$ of Proposition 5.5, and let $\Pi' \subset \Pi$ be the inverse image of Σ under their composition. Since $\pi_1(U) \to \pi_1(X_{\mathrm{reg}})$ is surjective (Lemma 5.4(3)), the inverse image U' of $\mu^{-1}(U)$ in $q^{-1}(X_{\mathrm{reg}})$ is connected, and $U' \to U$ is the connected covering of

U corresponding to Π'. Since $V^\circ \to U$ is a universal covering (Lemma 5.4(1)(2)), the covering $U' \to U$ is isomorphic to $V^\circ/\Pi' \to V^\circ/\Pi = U$ [Hat02, Theorem 1.38 and Proposition 1.40]. By Proposition 5.5,

$$\Pi_0 \subset \mathrm{Ker}\left(\Pi \to \pi_1(X_{\mathrm{reg}})\right) \subset \Pi'.$$

In particular, Π' contains the translations by elements of Λ_Y, and $H' := \Pi'/\Lambda_Y$ is a subgroup of $\Pi/\Lambda_Y = \widetilde{H}$. Since $V^\circ/\Lambda_Y = Y^\circ$, we obtain an isomorphism of coverings of U

$$U' \cong V^\circ/\Pi' = Y^\circ/H',$$

and, since $\Pi' \supset \Pi_0$, H' contains the automorphisms $(x, y) \to (wx, wy)$ for $w \in W$. Since $\widetilde{H}$ acts freely on Y°, both $U' \to U$ and $Y^\circ/H' \to U$ are finite étale, so the isomorphism $U' \cong Y^\circ/H'$ is an isomorphism of varieties over U [Gro71, Exposé XII, Théorème 5.1]. Thus there exists a finite étale surjective morphism $r : Y^\circ \to U'$ such that $r \circ \varphi = r$ for any $\varphi \in H'$ and $\mu \circ q \circ r$ is the restriction to Y° of the quotient morphism $Y \to Z$.

Let η be a global reflexive k-form on X'. Since $U' \subset q^{-1}(X_{\mathrm{reg}})$ is smooth, $\eta|_{U'}$ is a k-form on U', and $r^*(\eta|_{U'})$ is a k-form on Y° invariant under $(x, y) \to (wx, wy)$ for any $w \in W$. Since Y is smooth and $\mathrm{codim}_Y(Y \backslash Y^\circ) \geq 2$ (Lemma 5.4(1)), $r^*(\eta|_{U'})$ extends uniquely to a k-form η_Y on Y [Har77, Chapter II, Proposition 6.3A], and by uniqueness, $\eta_Y \in H^0(Y, \Omega_Y^k)^W$. The linear map

$$\bigoplus_{k\geq 0} H^0(X', \Omega_{X'}^{[k]}) \to \bigoplus_{k\geq 0} H^0(Y, \Omega_Y^k)^W, \quad \eta \to \eta_Y,$$

is compatible with exterior products. It is injective: since $\Omega_{X'}^{[k]}$ is torsion free and U' is dense in X', η is determined by $\eta|_{U'}$, and since r is étale and surjective, $\eta|_{U'}$ is determined by $r^*(\eta|_{U'})$. On U', $q^{[*]}\sigma_X = q^*\sigma_X$, and $\sigma_X = \mu^*\sigma_Z$ on $\mu^{-1}(U)$. Since the pullback of σ_Z to Y° is $\sigma_Y|_{Y^\circ}$ (Construction-Proposition 5.1(3)),

$$(q^{[*]}\sigma_X)_Y = \sigma_Y.$$

By Lemma 5.6 and the injectivity of $\eta \to \eta_Y$, $H^0(X', \Omega_{X'}^{[k]}) = 0$ for odd k or $k > 12$, and $\dim H^0(X', \Omega_{X'}^{[2i]}) \leq 1$ for any $0 \leq i \leq 6$. On the other hand, for any $0 \leq i \leq 6$, $((q^{[*]}\sigma_X)^i)_Y = \sigma_Y^i \neq 0$ (Lemma 5.6), so $(q^{[*]}\sigma_X)^i \neq 0$ spans $H^0(X', \Omega_{X'}^{[2i]})$. Thus

$$\bigoplus_{k\geq 0} H^0(X', \Omega_{X'}^{[k]}) = \mathbb{C}[q^{[*]}\sigma_X]$$

as graded algebras. Since q is arbitrary, (X, σ_X) is an irreducible symplectic variety. □

5.4. **The Lagrangian fibration.** In this subsection, we prove that f is a Lagrangian fibration and complete the proof of Theorem 1.2.

Proposition 5.8. *Notation as in Set-up 5.2. Then $f : X \to B$ is a Lagrangian fibration of (X, σ_X), and general fibers of f are abelian varieties of dimension 6.*

Proof. By Lemma 2.5(3), (X, σ_X) is a symplectic variety. By Construction-Proposition 4.3(3), B is a normal projective variety. Since Z is projective and B is separated, g is projective, so $f = g \circ \mu$ is projective [Har77, Chapter II, Exercises 4.8(e) and 4.9]. Since μ is projective and birational, μ is surjective. Since g is surjective (Construction-Proposition 5.1(2)), f is surjective.

Let $p : Y \to Z$ and $p_A : A \to B$ be the quotient morphisms, and let $\mathrm{pr}_1 : Y \to A$ be the first projection. By Construction-Proposition 5.1(2), pr_1 induces the morphism g and a group isomorphism $\widetilde{H} \to H$, which we write as $\varphi \to \bar{\varphi}$. That is, $g \circ p = p_A \circ \mathrm{pr}_1$ and

$$\mathrm{pr}_1 \circ \varphi = \bar{\varphi} \circ \mathrm{pr}_1 \quad \text{for any } \varphi \in \widetilde{H}.$$

In particular, $\varphi(\{a\} \times A) = \{\bar{\varphi}(a)\} \times A$ for any $\varphi \in \widetilde{H}$ and $a \in A$.

We show that f has connected fibers. Let $b \in B$ and $a \in p_A^{-1}(b)$. Then $p_A^{-1}(b) = \{\bar{\varphi}(a) \mid \varphi \in \widetilde{H}\}$, so

$$p^{-1}(g^{-1}(b)) = \bigcup_{\varphi \in \widetilde{H}} \{\bar{\varphi}(a)\} \times A = \bigcup_{\varphi \in \widetilde{H}} \varphi(\{a\} \times A).$$

Thus $g^{-1}(b) = p(\{a\} \times A)$, which is connected since A is connected. Since μ is projective and birational and Z is normal, the fibers of μ are connected [Har77, Chapter III, Corollary 11.4]. Since $f^{-1}(b) = \mu^{-1}(g^{-1}(b))$, $g^{-1}(b)$ is connected, and μ is a closed surjective morphism with connected fibers, $f^{-1}(b)$ is connected.

We show that general fibers of f are Lagrangian abelian varieties of dimension 6. We use the notation of Construction 5.3. Let $A^\circ \subset A$ be the open subset of points whose stabilizer in H is trivial. Since $H \subset \mathrm{Aut}(A)$ is finite, A° is a non-empty H-invariant open subset of A, so $B^\circ := p_A(A^\circ)$ is a non-empty open subset of B, $A^\circ = p_A^{-1}(B^\circ)$, and $A^\circ \to B^\circ$ is finite étale. If $\varphi \in \widetilde{H}$ fixes a point $(x, y) \in A^\circ \times A$, then $\bar{\varphi}$ fixes x, so $\bar{\varphi} = \mathrm{id}_A$, hence $\varphi = \mathrm{id}_Y$. Thus $A^\circ \times A \subset Y^\circ$, and

$$g^{-1}(B^\circ) = p(A^\circ \times A) \subset U \subset Z_{\mathrm{reg}}$$

by Lemma 5.4(2). By Lemma 2.5(1), μ is an isomorphism over $g^{-1}(B^\circ)$. Consider the morphism

$$A^\circ \times A \to A^\circ \times_{B^\circ} g^{-1}(B^\circ), \quad (x, y) \to (x, p(x, y)).$$

Both $A^\circ \times A$ and $A^\circ \times_{B^\circ} g^{-1}(B^\circ)$ are finite étale over the variety $g^{-1}(B^\circ)$ of degree $|\widetilde{H}| = |H|$. If $(x, p(x, y)) = (x, p(x, y'))$, then $\varphi(x, y) = (x, y')$ for some $\varphi \in \widetilde{H}$, so $\bar{\varphi}$ fixes $x \in A^\circ$, hence $\varphi = \mathrm{id}_Y$ and $y = y'$. Thus the morphism is finite étale and injective on points, hence an open and closed immersion (its image is open and closed, and over its image it is finite étale with one-point fibers over closed points, hence of degree one) [Gro67, Théorème 17.9.1], and it is surjective since both sides are finite étale of degree $|H|$ over $g^{-1}(B^\circ)$ and $g^{-1}(B^\circ)$ is connected, being a non-empty open subset of the irreducible variety Z. Hence it is an isomorphism. Therefore, after the étale base change $A^\circ \to B^\circ$, $f^{-1}(B^\circ) \cong g^{-1}(B^\circ) \to B^\circ$ becomes the first projection $A^\circ \times A \to A^\circ$.

Let $b \in B^\circ$, $a \in p_A^{-1}(b)$, and $F := f^{-1}(b)$. Taking the fibers over a of the isomorphism $(x, y) \to (x, p(x, y))$ above, the morphism $A \to g^{-1}(b)$, $y \to p(a, y)$, is an isomorphism. Since $f^{-1}(B^\circ) = \mu^{-1}(g^{-1}(B^\circ))$ and μ is an isomorphism over $g^{-1}(B^\circ)$, the morphism

$$j : A \to X, \quad y \to \left(\mu|_{f^{-1}(B^\circ)}\right)^{-1}(p(a, y)),$$

is an isomorphism onto F. In particular, F is an abelian variety, $\dim F = 6 = \frac{1}{2} \dim X$, and $F \subset \mu^{-1}(U) \subset X_{\mathrm{reg}}$, so $F \not\subset X_{\mathrm{sing}}$. Since $\sigma_X = \mu^* \sigma_Z$ on $\mu^{-1}(U)$ and the pullback of σ_Z to Y° is $\sigma_Y|_{Y^\circ}$, $j^* \sigma_X$ is the pullback of σ_Y along $A \to Y$, $y \to (a, y)$. The tangent vectors of $\{a\} \times A$ are of the form $(0, v_2)$ with $v_2 \in Q \otimes_{\mathbb{Z}} \mathbb{C}$. By the formula for σ_Y at the origin in Construction-Proposition 5.1, $\sigma_Y((v_1, v_2), (v_1', v_2')) = 0$ for any $v_2, v_2' \in Q \otimes_{\mathbb{Z}} \mathbb{C}$ if $v_1 = v_1' = 0$. Since σ_Y is translation-invariant, $j^* \sigma_X = 0$. Thus F is a Lagrangian subvariety of X. Since B° is a non-empty open subset of B, f is a Lagrangian fibration of (X, σ_X), and general fibers of f are abelian varieties of dimension 6. □

Proof of Theorem 1.2. Consider the abelian variety Y, the symplectic form σ_Y, the group $\widetilde{H}$, and the morphism $g : Y/\widetilde{H} = Z \to B$ of Construction-Proposition 5.1. By Construction-Proposition 4.3(1)(3), $Y = A \times A$ is an abelian variety of dimension 12 and B is a normal projective variety. By Construction-Proposition 5.1(1)(2), σ_Y is a symplectic form on Y, $\widetilde{H}$ is a finite group of automorphisms of Y preserving σ_Y, and g is surjective.

(1) Let $\mu : X \to Y/\widetilde{H}$ be a projective $\mathbb{Q}$-factorial terminalization. The symplectic form on $Y/\widetilde{H}$ induced by σ_Y is σ_Z (Construction-Proposition 5.1(3)). By Lemma 2.5(3), $\mu^* \sigma_Z$ extends to a symplectic form σ_X on X, and σ_X is the symplectic form in Set-up 5.2(2). Thus (1) follows from Proposition 5.7 and Proposition 5.8, applied to X, μ, σ_X, and $f := g \circ \mu$.

(2) This is Theorem 4.8. □

References

[AD09] M. Artebani and I. V. Dolgachev, *The Hesse pencil of plane cubic curves*, Enseign. Math. (2) **55** (2009), no. 3–4, 235–273.

[BGL22] B. Bakker, H. Guenancia, and C. Lehn, *Algebraic approximation and the decomposition theorem for Kähler Calabi–Yau varieties*, Invent. Math. **228** (2022), no. 3, 1255–1308.

[BS26] B. Bakker and C. Schnell, *A Hodge-theoretic proof of Hwang's theorem on base manifolds of Lagrangian fibrations*, Ann. Inst. Fourier (Grenoble) (2026), DOI 10.5802/aif.3786; arXiv:2311.08977.

[BHPV04] W. P. Barth, K. Hulek, C. A. M. Peters, and A. Van de Ven, *Compact complex surfaces*, second edition, Ergeb. Math. Grenzgeb. (3) **4**, Springer-Verlag, Berlin (2004).

[BFS21] E. Bayer-Fluckiger and J.-P. Serre, *Lines on cubic surfaces, Witt invariants and Stiefel–Whitney classes*, Indag. Math. (N.S.) **32** (2021), no. 5, 920–938.

[Bea00] A. Beauville, *Symplectic singularities*, Invent. Math. **139** (2000), no. 3, 541–549.

[BS06] J. Bernstein and O. Schwarzman, *Chevalley's theorem for the complex crystallographic groups*, J. Nonlinear Math. Phys. **13** (2006), no. 3, 323–351.

[BGMM25] V. Bertini, A. Grossi, M. Mauri, and E. Mazzon, *Terminalizations of quotients of compact hyperkähler manifolds by induced symplectic automorphisms*, Épijournal Géom. Algébrique **9** (2025), Art. 14, 53 pp.

[BCHM10] C. Birkar, P. Cascini, C. D. Hacon, and J. McKernan, *Existence of minimal models for varieties of log general type*, J. Amer. Math. Soc. **23** (2010), no. 2, 405–468.

[Bou02] N. Bourbaki, *Lie groups and Lie algebras. Chapters 4–6*, Elements of Mathematics, Springer-Verlag, Berlin (2002).

[Car26] R. Carini, *Irreducible symplectic varieties via K3-del Pezzo double covers*, arXiv:2606.18357.

[Car57] H. Cartan, *Quotient d'un espace analytique par un groupe d'automorphismes*, in: *Algebraic geometry and topology. A symposium in honor of S. Lefschetz*, Princeton University Press, Princeton, NJ (1957), 90–102.

[DHMV24] O. Debarre, D. Huybrechts, E. Macrì, and C. Voisin, *Computing Riemann–Roch polynomials and classifying hyper-Kähler fourfolds*, J. Amer. Math. Soc. **37** (2024), no. 1, 151–185.

[Dol82] I. V. Dolgachev, *Weighted projective varieties*, in: *Group actions and vector fields (Vancouver, 1981)*, Lecture Notes in Math. **956**, Springer, Berlin (1982), 34–71.

[Dol12] I. V. Dolgachev, *Classical algebraic geometry: a modern view*, Cambridge University Press, Cambridge (2012).

[FSY22] C. Felisetti, J. Shen, and Q. Yin, *On intersection cohomology and Lagrangian fibrations of irreducible symplectic varieties*, Trans. Amer. Math. Soc. **375** (2022), no. 4, 2987–3001.

[Fuj83] A. Fujiki, *On primitively symplectic compact Kähler V-manifolds of dimension four*, in: *Classification of algebraic and analytic manifolds (Katata, 1982)*, Progr. Math. **39**, Birkhäuser, Boston (1983), 71–250.

[GR84] H. Grauert and R. Remmert, *Coherent analytic sheaves*, Grundlehren der Mathematischen Wissenschaften **265**, Springer-Verlag, Berlin (1984).

[GGK19] D. Greb, H. Guenancia, and S. Kebekus, *Klt varieties with trivial canonical class: holonomy, differential forms, and fundamental groups*, Geom. Topol. **23** (2019), no. 4, 2051–2124.

[GL14] D. Greb and C. Lehn, *Base manifolds for Lagrangian fibrations on hyperkähler manifolds*, Int. Math. Res. Not. IMRN (2014), no. 19, 5483–5487.

[Gro61] A. Grothendieck, *Éléments de géométrie algébrique. II. Étude globale élémentaire de quelques classes de morphismes*, Inst. Hautes Études Sci. Publ. Math. **8** (1961), 5–222.

[Gro67] A. Grothendieck, *Éléments de géométrie algébrique. IV. Étude locale des schémas et des morphismes de schémas, Quatrième partie*, Inst. Hautes Études Sci. Publ. Math. **32** (1967), 5–361.

[Gro71] A. Grothendieck, *Revêtements étales et groupe fondamental (SGA 1)*, Lecture Notes in Math. **224**, Springer-Verlag, Berlin (1971).

[Har92] J. Harris, *Algebraic geometry. A first course*, Graduate Texts in Mathematics **133**, Springer-Verlag, New York (1992).

[Har77] R. Hartshorne, *Algebraic geometry*, Graduate Texts in Mathematics **52**, Springer-Verlag, New York-Heidelberg (1977).

[Hat02] A. Hatcher, *Algebraic topology*, Cambridge University Press, Cambridge (2002).

[HP19] A. Höring and T. Peternell, *Algebraic integrability of foliations with numerically trivial canonical bundle*, Invent. Math. **216** (2019), no. 2, 395–419.

[Hu25] H. Hu, *On exterior powers of reflection representations, II*, Canad. J. Math. (2025), published online.

[HX22] D. Huybrechts and C. Xu, *Lagrangian fibrations of hyperkähler fourfolds*, J. Inst. Math. Jussieu **21** (2022), no. 3, 921–932.

[Hwa08] J.-M. Hwang, *Base manifolds for fibrations of projective irreducible symplectic manifolds*, Invent. Math. **174** (2008), no. 3, 625–644.

[Kal06] D. Kaledin, *Symplectic singularities from the Poisson point of view*, J. Reine Angew. Math. **600** (2006), 135–156.

[Kol95] J. Kollár, *Shafarevich maps and automorphic forms*, M. B. Porter Lectures, Princeton University Press, Princeton, NJ (1995).
[KM98] J. Kollár and S. Mori, *Birational geometry of algebraic varieties*, Cambridge Tracts in Math. **134**, Cambridge Univ. Press, Cambridge (1998).
[LLX24] Y. Liu, Z. Liu, and C. Xu, *Irreducible symplectic varieties with a large second Betti number*, arXiv:2410.01566.
[LLX25] Y. Liu, Z. Liu, and C. Xu, *Irreducible symplectic varieties with a large second Betti number*, J. Reine Angew. Math. **825** (2025), 1–31.
[Loo76] E. Looijenga, *Root systems and elliptic curves*, Invent. Math. **38** (1976/77), no. 1, 17–32.
[Mat99] D. Matsushita, *On fibre space structures of a projective irreducible symplectic manifold*, Topology **38** (1999), no. 1, 79–83.
[Mat00] D. Matsushita, *Equidimensionality of Lagrangian fibrations on holomorphic symplectic manifolds*, Math. Res. Lett. **7** (2000), no. 4, 389–391.
[Mat01] D. Matsushita, *Addendum: On fibre space structures of a projective irreducible symplectic manifold*, Topology **40** (2001), no. 2, 431–432.
[Mat15] D. Matsushita, *On base manifolds of Lagrangian fibrations*, Sci. China Math. **58** (2015), no. 3, 531–542.
[Men22] G. Menet, *Thirty-three deformation classes of compact hyperkähler orbifolds*, arXiv:2211.14524.
[MX26] N. Müller and Z. Xu, *Lagrangian fibrations onto varieties with isolated quotient singularities*, J. Reine Angew. Math. **832** (2026), 297–307; arXiv:2508.04135.
[Mum74] D. Mumford, *Abelian varieties*, second edition, Tata Inst. Fund. Res. Stud. Math. **5**, Oxford University Press, London (1974).
[OAI26] OpenAI, *Projective-space bases of Lagrangian fibrations*, Paper No. 041 of https://github.com/openai/math.
[Ou19] W. Ou, *Lagrangian fibrations on symplectic fourfolds*, J. Reine Angew. Math. **746** (2019), 117–147.
[PS17] Yu. Prokhorov and C. Shramov, *Jordan constant for Cremona group of rank 3*, Mosc. Math. J. **17** (2017), no. 3, 457–509; arXiv:1608.00709.
[Rai23] E. M. Rains, *Quotients of abelian varieties by reflection groups*, arXiv:2303.04786.
[Ser56] J.-P. Serre, *Géométrie algébrique et géométrie analytique*, Ann. Inst. Fourier (Grenoble) **6** (1956), 1–42.
[Ser77] J.-P. Serre, *Linear representations of finite groups*, Graduate Texts in Mathematics **42**, Springer-Verlag, New York-Heidelberg (1977).
[SY22] J. Shen and Q. Yin, *Topology of Lagrangian fibrations and Hodge theory of hyper-Kähler manifolds*, Duke Math. J. **171** (2022), no. 1, 209–241.
[ST54] G. C. Shephard and J. A. Todd, *Finite unitary reflection groups*, Canad. J. Math. **6** (1954), 274–304.
[Sho92] V. V. Shokurov, *Threefold log flips*, with an appendix in English by Y. Kawamata, Izv. Ross. Akad. Nauk Ser. Mat. **56** (1992), no. 1, 105–203.
[Ste68] R. Steinberg, *Endomorphisms of linear algebraic groups*, Mem. Amer. Math. Soc. **80**, Amer. Math. Soc., Providence, RI (1968).

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